

# Notes on Varieties of Semiring Solvers

Amaury Mazoyer

June 5, 2026

## CONTENTS

|          |                                                                             |           |
|----------|-----------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Preliminaries</b>                                                        | <b>2</b>  |
| 1.1      | Algebraic Structures . . . . .                                              | 2         |
| 1.2      | Categories . . . . .                                                        | 3         |
| 1.3      | Monads . . . . .                                                            | 10        |
| 1.4      | 2-categories . . . . .                                                      | 14        |
| 1.5      | Multicategories . . . . .                                                   | 17        |
| 1.6      | Operads . . . . .                                                           | 19        |
| 1.7      | Free algebras . . . . .                                                     | 21        |
| 1.8      | Multi-homomorphisms . . . . .                                               | 24        |
| 1.9      | Free extensions . . . . .                                                   | 26        |
| 1.10     | Distributive combination of theories . . . . .                              | 26        |
| <b>2</b> | <b>Constructing free algebras</b>                                           | <b>27</b> |
| 2.1      | Semigroups . . . . .                                                        | 27        |
| 2.2      | Combining semigroups and commutative semigroups . . . . .                   | 32        |
| 2.3      | Monoids . . . . .                                                           | 39        |
| 2.4      | Distributive combination of semigroups and commutative semigroups . . . . . | 41        |
| 2.5      | Free algebras for the combination . . . . .                                 | 45        |
| 2.6      | Counter example of Lemma 95 . . . . .                                       | 48        |
| 2.7      | General case . . . . .                                                      | 49        |
| 2.8      | Link with distributive laws . . . . .                                       | 55        |
| <b>3</b> | <b>Constructing free extensions</b>                                         | <b>56</b> |
| 3.1      | Semigroups . . . . .                                                        | 56        |
| 3.2      | Sem-rgs . . . . .                                                           | 58        |
| 3.3      | Distributive combination of free extensions . . . . .                       | 60        |
| 3.4      | Practical combination . . . . .                                             | 61        |
| <b>A</b> | <b>Daily recap</b>                                                          | <b>65</b> |
| <b>B</b> | <b>TODO</b>                                                                 | <b>67</b> |
| <b>C</b> | <b>Questions for Ohad</b>                                                   | <b>67</b> |



**Definition 3 (Group)**

A *group* is a set  $G$  equipped with an associative binary operation  $\cdot : G \times G \rightarrow G$ , an identity element  $e \in G$ , and an inverse operation such that for all  $a, b, c \in G$ :

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$  (associativity)
- $e \cdot a = a \cdot e = a$  (identity element)
- For each  $a \in G$ , there exists an element  $a^{-1} \in G$  such that  $a \cdot a^{-1} = a^{-1} \cdot a = e$  (inverse element)

A group is called *abelian* if its operation is commutative, i.e., for all  $a, b \in G$ ,  $a \cdot b = b \cdot a$ .

**Definition 4 (Semiring)**

A *semiring* is a set  $R$  equipped with two binary operations: addition  $+$  and multiplication  $\cdot$ , such that:

- $(R, +)$  is a commutative monoid with identity element 0.
- $(R, \cdot)$  is a monoid with identity element 1.
- Multiplication distributes over addition from both sides: for all  $a, b, c \in R$ ,

$$\begin{aligned} a \cdot (b + c) &= a \cdot b + a \cdot c, \\ (a + b) \cdot c &= a \cdot c + b \cdot c. \end{aligned}$$

- The additive identity 0 is an absorbing element for multiplication: for all  $a \in R$ ,  $0 \cdot a = a \cdot 0 = 0$ .

## 1.2 Categories

**Definition 5 (Category)**

A *category*  $\mathcal{C}$  consists of:

- A class of objects  $\text{Ob}(\mathcal{C})$ ,
- For each pair of objects  $a, b \in \text{Ob}(\mathcal{C})$ , a set of morphisms  $\text{Hom}_{\mathcal{C}}(a, b)$ ,
- For each object  $a \in \text{Ob}(\mathcal{C})$ , an identity morphism  $\text{id}_a \in \text{Hom}_{\mathcal{C}}(a, a)$ ,
- A composition operation  $\circ$  such that for morphisms  $f : a \rightarrow b$  and  $g : b \rightarrow c$ , there is a morphism  $g \circ f : a \rightarrow c$ ,

satisfying the following axioms:

- Associativity: For morphisms  $f : a \rightarrow b$ ,  $g : b \rightarrow c$ , and  $h : c \rightarrow d$ , we have  $h \circ (g \circ f) = (h \circ g) \circ f$ .
- Identity: For any morphism  $f : a \rightarrow b$ , we have  $\text{id}_b \circ f = f = f \circ \text{id}_a$ .

**Example 6 (Set Category)**

The category **Set** has sets as objects and functions as morphisms. The identity morphism for a set  $a$  is the identity function  $\text{id}_a : a \rightarrow a$  defined by  $\text{id}_a(x) = x$  for all  $x \in a$ . Composition of morphisms is given by function composition.

**Example 7 (Walking Morphism)**

The *walking morphism* category **2** or  $I$  has two objects, say 0 and 1, and a single non-identity morphism  $f : 0 \rightarrow 1$ . The identity morphisms are  $\text{id}_0 : 0 \rightarrow 0$  and  $\text{id}_1 : 1 \rightarrow 1$ . The composition is defined by the identities and the morphism  $f$ .

**Definition 8 (Product)**

Given two objects  $a$  and  $b$  in a category  $\mathcal{C}$ , a *product* of  $a$  and  $b$  is an object  $a \times b$  together with two morphisms  $\pi_a : a \times b \rightarrow a$  and  $\pi_b : a \times b \rightarrow b$  such that for any object  $c$  and morphisms  $f : c \rightarrow a$  and  $g : c \rightarrow b$ , there exists a unique morphism  $\langle f, g \rangle : c \rightarrow a \times b$  making the following diagram commute:

$$\begin{array}{ccccc} & & c & & \\ & f \swarrow & \downarrow \langle f, g \rangle & \searrow g & \\ a & \xleftarrow{\pi_a} & a \times b & \xrightarrow{\pi_b} & b \end{array}$$

**Definition 9 (Coproduct)**

Given two objects  $a$  and  $b$  in a category  $\mathcal{C}$ , a *coproduct* of  $a$  and  $b$  is an object  $a + b$  together with two morphisms  $\iota_a : a \rightarrow a + b$  and  $\iota_b : b \rightarrow a + b$  such that for any object  $c$  and morphisms  $f : a \rightarrow c$  and  $g : b \rightarrow c$ , there exists a unique morphism  $[f, g] : a + b \rightarrow c$  making the following diagram commute:

$$\begin{array}{ccccc} a & \xrightarrow{\iota_a} & a + b & \xleftarrow{\iota_b} & b \\ & \searrow f & \downarrow [f, g] & \swarrow g & \\ & & c & & \end{array}$$

**Definition 10 (Monoidal Category)**

A *monoidal category* is a category  $\mathcal{C}$  equipped with a bifunctor  $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ , an object  $I$  called the unit object, and natural isomorphisms (called associators and unitors) that satisfy certain coherence conditions. Specifically, for all objects  $a, b, c$  in  $\mathcal{C}$ , there are natural isomorphisms:

- Associator:  $\alpha_{a,b,c} : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c)$ ,
- Left unitor:  $\lambda_a : I \otimes a \rightarrow a$ ,
- Right unitor:  $\rho_a : a \otimes I \rightarrow a$ ,

such that the following diagrams commute:

$$\begin{array}{ccc} ((a \otimes b) \otimes c) \otimes d & \xrightarrow{\alpha_{a,b,c} \otimes 1_d} & (a \otimes (b \otimes c)) \otimes d \xrightarrow{\alpha_{a,b \otimes c,d}} a \otimes ((b \otimes c) \otimes d) \\ \alpha_{a,b,c \otimes d} \uparrow & & 1_a \otimes \alpha_{b,c,d} \uparrow \\ (a \otimes b) \otimes (c \otimes d) & & a \otimes (b \otimes (c \otimes d)) \end{array}$$

$$(I \otimes a) \otimes b \xrightarrow{\alpha_{I,a,b}} I \otimes (a \otimes b) \xleftarrow{\rho_a} a = a \otimes I \xrightarrow{\lambda_a} a \otimes I \xrightarrow{\alpha_{a,I,I}} (a \otimes I) \otimes I$$

**Definition 11 (Functor)**

Given two categories  $\mathcal{C}$  and  $\mathcal{D}$ , a *functor*  $F : \mathcal{C} \rightarrow \mathcal{D}$  consists of:

- A mapping  $F : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$  that assigns to each object  $a$  in  $\mathcal{C}$  an object  $F(a)$  in  $\mathcal{D}$ ,
- For each pair of objects  $a, b \in \text{Ob}(\mathcal{C})$ , a mapping  $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$  that assigns to each morphism  $f : a \rightarrow b$  in  $\mathcal{C}$  a morphism  $F(f) : F(a) \rightarrow F(b)$  in  $\mathcal{D}$ ,

such that for all objects  $a, b, c$  in  $\mathcal{C}$  and morphisms  $f : a \rightarrow b$ ,  $g : b \rightarrow c$ , the following conditions hold:

- $F(g \circ f) = F(g) \circ F(f)$  (preservation of composition),
- $F(\text{id}_a) = \text{id}_{F(a)}$  (preservation of identity).

For a category  $\mathcal{C}$ , the *identity functor*  $\text{Id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$  is defined by  $\text{Id}_{\mathcal{C}}(a) = a$  for all objects  $a$  and  $\text{Id}_{\mathcal{C}}(f) = f$  for all morphisms  $f$ .

- A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is *faithful* if, for every pair of objects  $a, b \in \text{Ob}(\mathcal{C})$ , the map  $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$  is injective.
- A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is *full* if, for every pair of objects  $a, b \in \text{Ob}(\mathcal{C})$ , the map  $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$  is surjective.
- A functor that is both full and faithful is called *fully faithful*.

**Notation:** We often denote the image of an object  $a$  under a functor  $F$  as  $Fa$  instead of  $F(a)$  and the image of a morphism  $f : a \rightarrow b$  as  $Ff : Fa \rightarrow Fb$ .

**Example 12 (Diagonal functor)**

Given a category  $\mathcal{C}$ , the *diagonal functor*  $\Delta : \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$  is defined by  $\Delta(a) = (a, a)$  for each object  $a$  in  $\mathcal{C}$  and  $\Delta(f) = (f, f)$  for each morphism  $f : a \rightarrow b$  in  $\mathcal{C}$ .

**Definition 13 (Product preserving functor)**

A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is said to *preserve products* if for every collection of objects  $\{a_i\}_{i \in I}$  in  $\mathcal{C}$ , the natural morphism from  $F(\prod_{i \in I} a_i)$  to  $\prod_{i \in I} F(a_i)$  is an isomorphism in  $\mathcal{D}$ . In other words,  $F$  preserves the structure of products up to isomorphism.

**Definition 14 (Natural Transformation)**

Given two functors  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  between categories  $\mathcal{C}$  and  $\mathcal{D}$ , a *natural transformation*  $\eta : F \rightarrow G$  consists of a family of morphisms  $\{\eta_a : Fa \rightarrow Ga\}_{a \in \text{Ob}(\mathcal{C})}$  such that for every morphism  $f : a \rightarrow b$  in  $\mathcal{C}$ , the following diagram commutes:

$$\begin{array}{ccc} Fa & \xrightarrow{Ff} & Fb \\ \eta_a \downarrow & & \downarrow \eta_b \\ Ga & \xrightarrow{Gf} & Gb \end{array}$$



**Definition 15 (Adjunction)**

Given two categories  $\mathcal{C}$  and  $\mathcal{D}$ , a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a *left adjoint* to a functor  $G : \mathcal{D} \rightarrow \mathcal{C}$  (and  $G$  is a *right adjoint* to  $F$ ) if there is a natural isomorphism:

$$\text{Hom}_{\mathcal{D}}(Fa, b) \cong \text{Hom}_{\mathcal{C}}(a, Gb)$$

for all objects  $a$  in  $\mathcal{C}$  and  $b$  in  $\mathcal{D}$ . This means that for every morphism  $f : Fa \rightarrow b$  in  $\mathcal{D}$ , there is a unique morphism  $g : a \rightarrow Gb$  in  $\mathcal{C}$  such that the following diagram commutes:

$$\begin{array}{ccc} Fa & \xrightarrow{f} & b \\ \uparrow F & & \downarrow G \\ a & \xrightarrow{g} & Gb \end{array}$$

We often denote this adjunction by  $F \dashv G$ .

There are actually several different but equivalent ways to define adjunctions. They are linked thanks to the Yoneda lemma.

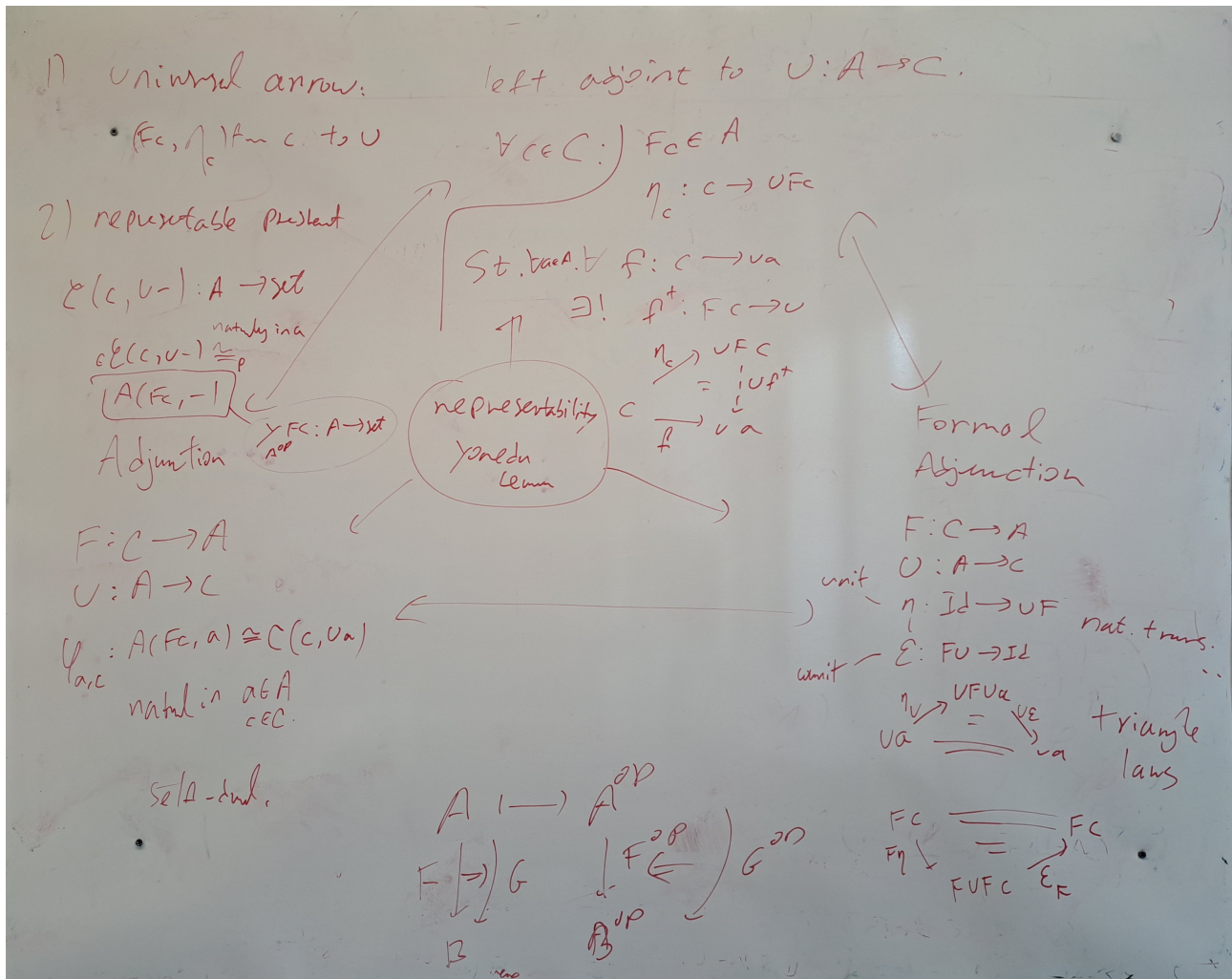


Figure 2: Different equivalent definitions of adjunctions.

One important definition is that of the universal arrow from MacLane's *Categories for the Working Mathematician* [Mac71].

**Definition 16 (Universal arrow)**

Let  $U : A \rightarrow C$  be a functor, and let  $c \in C$ .

An *universal arrow* from  $c$  to  $U$  is a pair  $(Fc, \eta_c)$  where:

- $Fc \in A$
- $\eta_c : c \rightarrow UFc$  is a morphism in  $C$

such that for every  $a \in A$ , and every  $f : c \rightarrow Ua$  in  $C$ , there exists a unique morphism  $f^\dagger : Fc \rightarrow a$  in  $A$  such that the following diagram commutes:

$$\begin{array}{ccc} c & \xrightarrow{\eta_c} & UFc \\ & \searrow f & \downarrow Uf^\dagger \\ & & Ua \end{array} \quad \begin{array}{ccc} Fc & & \\ \downarrow f^\dagger & & \\ a & & \end{array}$$

We can dualize this definition to get the notion of a universal arrow from  $U$  to  $c$ .

A universal arrow from  $U$  to  $c$  is a pair  $(Fc, \varepsilon_c)$  where:

- $Fc \in A$
- $\varepsilon_c : UFc \rightarrow c$  is a morphism in  $C$

such that for every  $a \in A$ , and every  $f : Ua \rightarrow c$  in  $C$ , there exists a unique morphism  $f^\dagger : a \rightarrow Fc$  in  $A$  such that the following diagram commutes:

$$\begin{array}{ccc} c & \xleftarrow{\varepsilon_c} & UFc \\ & \swarrow f & \uparrow Uf^\dagger \\ & & Ua \end{array} \quad \begin{array}{ccc} Fc & & \\ \uparrow f^\dagger & & \\ a & & \end{array}$$

**Example 17 (Adjunction from Universal Arrows)**

Given a functor  $U : A \rightarrow C$  such that for every object  $c \in C$ , there exists a universal arrow from  $c$  to  $U$ , we can construct an adjunction  $F \dashv U$  where:

- The functor  $F : C \rightarrow A$  is defined by sending each object  $c$  in  $C$  to the object  $Fc$  in  $A$  that appears in the universal arrow from  $c$  to  $U$ , and sending each morphism  $f : c \rightarrow c'$  in  $C$  to the unique morphism  $Ff : Fc \rightarrow Fc'$  in  $A$  that makes the corresponding diagram commute.
- The unit  $\eta : \text{Id}_C \rightarrow UF$  is given by the family of morphisms  $\{\eta_c : c \rightarrow UFc\}_{c \in C}$  that are part of the universal arrows.

Universal arrows have more uses than just defining adjunctions. Here are other examples.

**Example 18 (Defining the coproduct with an universal arrow)**

Let  $A$  be a category, and let  $a, b \in A$ . A *coproduct* of  $a$  and  $b$  is a universal arrow from  $(a, b)$  to the diagonal functor  $\Delta : A \rightarrow A \times A$ , where  $\Delta(c) = (c, c)$  for any object  $c \in A$ .

$$\begin{array}{ccc} a & \xrightarrow{\iota_a} & a + b \xleftarrow{\iota_b} b \\ & \searrow f & \downarrow [f, g] \\ & & c \end{array} \quad \sim \quad \begin{array}{ccc} (a, b) & \xrightarrow{\eta_{(a,b)}} & \Delta(a + b) \\ & \searrow (f, g) & \downarrow \Delta[f, g] \\ & & \Delta c \end{array}$$

We have  $\eta_{(a,b)} = (\iota_a, \iota_b)$ .

**Example 19 (Defining the product with an universal arrow)**

Let  $A$  be a category, and let  $a, b \in A$ . A *product* of  $a$  and  $b$  is a universal arrow from the diagonal functor  $\Delta : A \rightarrow A \times A$  to  $(a, b)$ .

$$\begin{array}{ccc}
 & c & \\
 f \swarrow & \downarrow (f,g) & \searrow g \\
 a & \xleftarrow{\pi_a} a \times b \xrightarrow{\pi_b} & b
 \end{array}
 \quad \sim \quad
 \begin{array}{ccc}
 \Delta c & \xrightarrow{\varepsilon_{(a,b)}} & \Delta(a \times b) \\
 (f,g) \searrow & & \downarrow (\pi_a, \pi_b) \\
 & & (a, b)
 \end{array}$$

We have  $\varepsilon_{(a,b)} = \langle f, g \rangle$ .

**Definition 20 (Pullback of two morphisms)**

Let  $f : X \rightarrow Z$  and  $g : Y \rightarrow Z$  be two morphisms in a category  $\mathcal{C}$ . A *pullback* of  $f$  and  $g$  consists of an object  $P$  and two morphisms  $p_1 : P \rightarrow X$  and  $p_2 : P \rightarrow Y$  such that the following diagram commutes:

$$\begin{array}{ccc}
 P & \xrightarrow{p_2} & Y \\
 p_1 \downarrow & & \downarrow g \\
 X & \xrightarrow{f} & Z
 \end{array}$$

Moreover, for any triple  $(Q, q_1, q_2)$  where  $q_1 : Q \rightarrow X$  and  $q_2 : Q \rightarrow Y$  are morphisms with  $f q_1 = g q_2$ , there must exist  $u : Q \rightarrow P$  such that  $p_1 \circ u = q_1$  and  $p_2 \circ u = q_2$ .

The situation is illustrated in the following commutative diagram:

$$\begin{array}{ccccc}
 Q & & & & \\
 & \searrow u & & \searrow q_2 & \\
 & & P & \xrightarrow{p_2} & Y \\
 & & p_1 \downarrow & & \downarrow g \\
 & & X & \xrightarrow{f} & Z \\
 & \searrow q_1 & & & 
 \end{array}$$

This condition can be reformulated with universal arrows as follows: a pullback of  $f$  and  $g$  is a universal arrow from the functor  $\Delta : \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$  to the object  $(f, g)$  in  $\mathcal{C} \times \mathcal{C}$ .

**Example 21 (Pullback in the category of categories)**

Let  $\mathcal{A}, \mathcal{B}, \mathcal{Z} \in \mathbf{Cat}$  such that we have functors  $F : \mathcal{A} \rightarrow \mathcal{Z}$  and  $G : \mathcal{B} \rightarrow \mathcal{Z}$ .

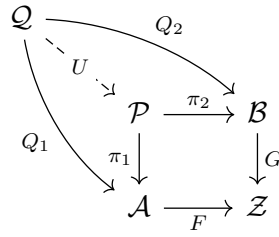
The pullback of  $F$  and  $G$  is the category  $\mathcal{P}$  defined as follows:

- The objects of  $\mathcal{P}$  are pairs  $(a, b)$  where  $a$  is an object of  $\mathcal{A}$ ,  $b$  is an object of  $\mathcal{B}$ , and  $Fa = Gb$ .
- The morphisms of  $\mathcal{P}$  from  $(a, b)$  to  $(a', b')$  are pairs  $(f, g)$  where  $f : a \rightarrow a'$  is a morphism in  $\mathcal{A}$ ,  $g : b \rightarrow b'$  is a morphism in  $\mathcal{B}$ , and  $Ff = Gg$ .
- The composition of morphisms in  $\mathcal{P}$  is defined componentwise: if  $(f, g) : (a, b) \rightarrow (a', b')$  and  $(f', g') : (a', b') \rightarrow (a'', b'')$  are morphisms in  $\mathcal{P}$ , then their composition is given by  $(f' \circ f, g' \circ g) : (a, b) \rightarrow (a'', b'')$ .
- The identity morphism on an object  $(a, b)$  in  $\mathcal{P}$  is given by  $(\text{id}_a, \text{id}_b)$ , where  $\text{id}_a$  is the identity morphism on  $a$  in  $\mathcal{A}$  and  $\text{id}_b$  is the identity morphism on  $b$  in  $\mathcal{B}$ .



**Proof**

Consider the following commuting diagram:



We need to find  $U$  and to prove that it is unique up to isomorphism.

We define  $U$  as follows: for any object  $q$  in  $\mathcal{Q}$ , we have  $Q_1q \in \mathcal{A}$  and  $Q_2q \in \mathcal{B}$ . Since  $FQ_1q = GQ_2q$ , we can define  $Uq = (Q_1q, Q_2q)$ . For any morphism  $m : q \rightarrow q'$  in  $\mathcal{Q}$ , we have  $Q_1m : Q_1q \rightarrow Q_1q'$  and  $Q_2m : Q_2q \rightarrow Q_2q'$ . Since  $FQ_1m = GQ_2m$ , we can define  $Um = (Q_1m, Q_2m)$ .

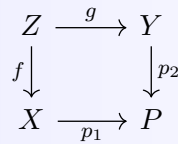
It is straightforward to check that  $U$  is a functor and that the diagram commutes.

Moreover, if  $U' : \mathcal{Q} \rightarrow \mathcal{P}$  is another functor such that the diagram commutes, then for any object  $q$  in  $\mathcal{Q}$ , we have  $U'q = (Q_1q, Q_2q) = Uq$ . For any morphism  $m : q \rightarrow q'$  in  $\mathcal{Q}$ , we have  $U'm = (Q_1m, Q_2m) = Um$ . Hence,  $U$  is unique up to isomorphism.  $\square$

**Definition 22 (Pushout of two morphisms)**

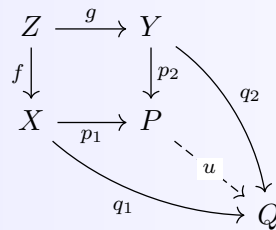
Let  $f : Z \rightarrow X$  and  $g : Z \rightarrow Y$  be two morphisms in a category  $\mathcal{C}$ .

A *pushout* of  $f$  and  $g$  consists of an object  $P$  and two morphisms  $p_1 : X \rightarrow P$  and  $p_2 : Y \rightarrow P$  such that the following diagram commutes:



Moreover, for any triple  $(Q, q_1, q_2)$  where  $q_1 : X \rightarrow Q$  and  $q_2 : Y \rightarrow Q$  are morphisms with  $q_1f = q_2g$ , there must exist  $u : P \rightarrow Q$  such that  $u \circ p_1 = q_1$  and  $u \circ p_2 = q_2$ .

The situation is illustrated in the following commutative diagram:



This condition can be reformulated with universal arrows as follows: a pushout of  $f$  and  $g$  is a universal arrow from the object  $(f, g)$  in  $\mathcal{C} \times \mathcal{C}$  to the functor  $\Delta : \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$ .

**Definition 23 (Coequalizer)**

Let  $f, g : X \rightarrow Y$  be two morphisms in a category  $\mathcal{C}$ .

A *coequalizer* of  $f$  and  $g$  consists of an object  $Q$  and a morphism  $q : Y \rightarrow Q$  such that the following diagram commutes:

$$X \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} Y \xrightarrow{q} Q$$

Moreover, for any pair  $(R, r)$  where  $r : Y \rightarrow R$  is a morphism with  $r \circ f = r \circ g$ , there must exist a unique morphism  $u : Q \rightarrow R$  such that  $u \circ q = r$ .

The situation is illustrated in the following commutative diagram:

$$\begin{array}{ccc} X & \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} & Y \\ & \searrow r & \downarrow \text{---} u \\ & & R \end{array}$$

**Lemma 24 (Pushout as coproducts and coequalizers)**

Let  $f : Z \rightarrow X$  and  $g : Z \rightarrow Y$  be two morphisms in a category  $\mathcal{C}$ . The pushout of  $f$  and  $g$  can be constructed as the coequalizer of the morphisms  $\iota_X f : Z \rightarrow X \sqcup Y$  and  $\iota_Y g : Z \rightarrow X \sqcup Y$ , where  $X \sqcup Y$  is the coproduct of  $X$  and  $Y$ , and  $\iota_X : X \rightarrow X \sqcup Y$  and  $\iota_Y : Y \rightarrow X \sqcup Y$  are the canonical injections:

$$\begin{array}{ccccc} Z & \xrightarrow{g} & Y & & \\ \downarrow f & \searrow \tilde{g} & \swarrow \iota_Y & & \\ & X \sqcup Y & & & \\ \uparrow \iota_X & \swarrow q & \downarrow \iota_Y \circ q & & \\ X & \xrightarrow{\iota_X \circ q} & P & & \end{array}$$

**1.3 Monads****Definition 25 (Monad)**

A *monad* on a category  $\mathcal{C}$  is a triple  $(T, \eta, \mu)$  where:

- $T : \mathcal{C} \rightarrow \mathcal{C}$  is a functor,
- $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$  is a natural transformation called the unit,
- $\mu : T^2 \rightarrow T$  is a natural transformation called the multiplication,

such that the following diagrams commute:

$$\begin{array}{ccc} T & \xrightarrow{T\eta} & T^2 \\ \eta T \downarrow & \searrow & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array} \quad \begin{array}{ccc} T^3 & \xrightarrow{T\mu} & T^2 \\ \mu T \downarrow & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array}$$

**Example 26 (Monad from an Adjunction)**

Given an adjunction  $F \dashv G$  between categories  $\mathcal{C}$  and  $\mathcal{D}$ , we can construct a monad on  $\mathcal{C}$  as follows:

- The functor  $T : \mathcal{C} \rightarrow \mathcal{C}$  is defined by  $T = GF$ .
- The unit  $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$  is given by the natural transformation that assigns to each object  $a$  in  $\mathcal{C}$  the morphism  $\eta_a : a \rightarrow GFa$  corresponding to the identity morphism on  $Fa$  under the adjunction.
- The multiplication  $\mu : T^2 \rightarrow T$  is given by the natural transformation that assigns to each object  $a$  in  $\mathcal{C}$  the morphism  $\mu_a : GF GFa \rightarrow GFa$  corresponding to the composition of morphisms in  $\mathcal{D}$  under the adjunction.

### Definition 27 (Distributive law between monads, [Bec69])

Given two monads  $(T, \eta^T, \mu^T)$  and  $(S, \eta^S, \mu^S)$  on a category  $\mathcal{C}$ , a *distributive law* of  $T$  over  $S$  is a natural transformation  $\lambda : TS \rightarrow ST$  such that the following diagrams commute:

$$\begin{array}{ccc}
 & T & \\
 T\eta^S \swarrow & & \searrow \eta^{ST} \\
 TS & \xrightarrow{\lambda} & ST
 \end{array}
 \qquad
 \begin{array}{ccc}
 & S & \\
 \eta^{TS} \swarrow & & \searrow S\eta^T \\
 TS & \xrightarrow{\lambda} & ST
 \end{array}$$
  

$$\begin{array}{ccccc}
 T^2S & \xrightarrow{T\lambda} & TST & \xrightarrow{\lambda T} & ST^2 \\
 \mu^T S \downarrow & & & & \downarrow S\mu^T \\
 TS & \xrightarrow{\lambda} & & & ST
 \end{array}
 \qquad
 \begin{array}{ccccc}
 TS^2 & \xrightarrow{\lambda S} & STS & \xrightarrow{S\lambda} & S^2T \\
 T\mu^S \downarrow & & & & \downarrow \mu^{ST} \\
 TS & \xrightarrow{\lambda} & & & ST
 \end{array}$$

This law induces a composite monad  $ST$  on  $\mathcal{C}$  with

- unit  $\eta^{ST} = \eta^S \circ \eta^T$
- multiplication  $STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu^S \mu^T} ST$

### Definition 28 (Algebra over a monad)

Let  $(T, \eta, \mu)$  be a monad on a category  $\mathcal{C}$ .

An *algebra over the monad  $T$*  (or simply a  $T$ -algebra) consists of:

- An object  $A$  in  $\mathcal{C}$ ,
- A morphism  $a : TA \rightarrow A$  in  $\mathcal{C}$ , called the structure map, such that the following diagrams commute:

$$\begin{array}{ccc}
 A & \xrightarrow{\eta_A} & TA \\
 \searrow \text{id}_A & & \downarrow a \\
 & & A
 \end{array}
 \qquad
 \begin{array}{ccc}
 T^2A & \xrightarrow{Ta} & TA \\
 \mu_A \downarrow & & \downarrow a \\
 TA & \xrightarrow{a} & A
 \end{array}$$

The left one is called the unit property, and the right one is called the action property.

We'll now reformulate the definition of monads in terms of Kleisli categories and extension operators.

**Definition 29 (Kleisli category)**

Given a monad  $(T, \eta, \mu)$  on a category  $\mathcal{C}$ , the *Kleisli category*  $\mathcal{C}_T$  has:

- The same objects as  $\mathcal{C}$ ,
- For each pair of objects  $a, b$ , the morphisms from  $a$  to  $b$  in  $\mathcal{C}_T$  are given by  $\text{Hom}_{\mathcal{C}_T}(a, b) = \text{Hom}_{\mathcal{C}}(a, Tb)$ ,
- The identity morphism on an object  $a$  is given by  $\eta_a : a \rightarrow Ta$ ,
- The composition of morphisms  $f : a \rightarrow Tb$  and  $g : b \rightarrow Tc$  in  $\mathcal{C}_T$  is given by the morphism  $g \circ_T f : a \rightarrow Tc$  defined as the composition:

$$a \xrightarrow{f} Tb \xrightarrow{Tg} T^2c \xrightarrow{\mu_c} Tc$$

**Definition 30 (Extension operator)**

Given a monad  $(T, \eta, \mu)$  on a category  $\mathcal{C}$ , the *extension operator*  $(-)^{\sharp}$  is defined for each morphism  $f : a \rightarrow Tb$  in  $\mathcal{C}$  as the morphism  $f^{\sharp} : Ta \rightarrow Tb$  given by the composition:

$$Ta \xrightarrow{Tf} T^2b \xrightarrow{\mu_b} Tb$$

In other words,  $f^{\sharp} = \mu_b \circ Tf$ .

We have the following properties of the extension operator:

- $\eta_a^{\sharp} = \text{id}_{Ta}$  for all objects  $a$  in  $\mathcal{C}$ .
- For any morphism  $f : a \rightarrow Tb$ , we have  $f^{\sharp} \circ \eta_a = f$ .
- For any morphisms  $f : a \rightarrow Tb$  and  $g : b \rightarrow Tc$ , we have  $(g^{\sharp} \circ f)^{\sharp} = g^{\sharp} \circ f^{\sharp}$ .

**Definition 31 (Kleisli triple)**

A *Kleisli triple* on a category  $\mathcal{C}$  consists of:

- A mapping  $T : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{C})$ ,
- For each object  $a$  in  $\mathcal{C}$ , a morphism  $\eta_a : a \rightarrow Ta$ ,
- For each morphism  $f : a \rightarrow Tb$  in  $\mathcal{C}$ , a morphism  $f^{\sharp} : Ta \rightarrow Tb$ ,

such that the following conditions hold:

- $\eta_a^{\sharp} = \text{id}_{Ta}$  for all objects  $a$  in  $\mathcal{C}$ .
- the following diagrams commute

$$\begin{array}{ccc} a & \xrightarrow{\eta_a} & Ta \\ & \searrow f & \downarrow f^{\sharp} \\ & & Tb \end{array} \quad \begin{array}{ccc} Ta & \xrightarrow{f^{\sharp}} & Tb \\ & \searrow (g^{\sharp} \cdot f)^{\sharp} & \downarrow g^{\sharp} \\ & & Tc \end{array}$$

**Proposition 32**

Given a category  $\mathcal{C}$ , there is a one-to-one correspondence between monads on  $\mathcal{C}$  and Kleisli triples on  $\mathcal{C}$ .

**Proof**

Given a monad  $(T, \eta, \mu)$  on  $\mathcal{C}$ , we can construct a Kleisli triple by defining the extension operator as  $f^\sharp = \mu_b \circ T f$  for each morphism  $f : a \rightarrow Tb$ .

Conversely, given a Kleisli triple  $(T, \eta, (-)^\sharp)$  on  $\mathcal{C}$ , we can construct a monad by defining the multiplication  $\mu_a : T^2 a \rightarrow Ta$  as  $\mu_a = (\text{id}_{Ta})^\sharp$  for each object  $a$  in  $\mathcal{C}$ .

The conditions of the Kleisli triple ensure that the monad axioms are satisfied, and the construction is reversible, establishing the one-to-one correspondence.  $\square$

**Remark :** Kleisli triples are sometimes referred to as *monads in extensive form*.

**Example 33 (Monads in terms of binds and returns)**

In the context of programming languages, a monad can be described in terms of a type constructor  $T$  and two operations: **return** (or **unit**) and **bind**. The **return** operation corresponds to the unit  $\eta$  of the monad, while the **bind** operation corresponds to the extension operator  $(-)^\sharp$ . Specifically, for a monad  $(T, \eta, \mu)$  on a category  $\mathcal{C}$ , we can define:

- $\text{return}_a = \eta_a : a \rightarrow Ta$  for each object  $a$  in  $\mathcal{C}$ ,
- $\text{bind}_{a,b}(f) = f^\sharp : Ta \rightarrow Tb$  for each morphism  $f : a \rightarrow Tb$  in  $\mathcal{C}$ .

We use the notion of  $\mathbb{S}$ -algebra of [MW10] to define the distributive laws for monads in extensive form.

**Definition 34 ( $\mathbb{S}$ -algebra, [MW10])**

Given a monad in extensive form  $\mathbb{S} = (S, \eta, (-)^\mathbb{S})$  on a category  $\mathcal{C}$ , an  $\mathbb{S}$ -algebra is a pair  $\mathbb{B} = (B, (-)^\mathbb{B})$  where:

- $B \in \mathcal{C}$
- $(-)_X^\mathbb{B} : \mathcal{C}(X, B) \rightarrow \mathcal{C}(SX, B)$

such that the following diagrams commute for every  $h : X \rightarrow B$  and  $k : Y \rightarrow SX$ :

$$\begin{array}{ccc} X & \xrightarrow{\eta^X} & SX \\ & \searrow h & \downarrow h^\mathbb{B} \\ & & B \end{array} \quad \begin{array}{ccc} SY & \xrightarrow{k^\mathbb{S}} & SX \\ & \searrow (h^\mathbb{B} k)^\mathbb{B} & \downarrow h^\mathbb{B} \\ & & B \end{array}$$

A morphism of  $\mathbb{S}$ -algebras  $(B, (-)^\mathbb{B})$  to  $(A, (-)^\mathbb{A})$  can be defined as an arrow  $l : B \rightarrow A$  in  $\mathcal{C}$  such that the following diagram commutes for every  $h : X \rightarrow B$ :

$$\begin{array}{ccc} SX & \xrightarrow{h^\mathbb{B}} & B \\ & \searrow (l \cdot h)^\mathbb{A} & \downarrow l \\ & & A \end{array}$$

**Theorem 35 (Distributive law between monads in extensive form, [MW10])**

Let  $\mathbb{T} = (T, \eta_T, (-)^\mathbb{T})$  and  $\mathbb{S} = (S, \eta_S, (-)^\mathbb{S})$  be monads on  $\mathcal{C}$ . A distributive law of  $\mathbb{T}$  over  $\mathbb{S}$  can equivalently be given by an  $\mathbb{T}$ -algebra  $(STA, (-)^\lambda)$  for every  $A \in \mathcal{C}$  such that

- $(S\eta_T A \cdot \eta_S A)^\lambda = \eta_S T A$
- for every  $f : B \rightarrow TSA$ ,  $(f^\lambda)^\mathbb{S} : (STB, (-)^\lambda) \rightarrow (STA, (-)^\lambda)$  is a morphism of  $\mathbb{T}$ -algebras.

Unfolding the conditions for the  $\mathbb{T}$ -algebra and for the morphism, the following diagrams must commute for every  $h : X \rightarrow STA$ ,  $k : Y \rightarrow TX$ ,  $f : B \rightarrow STA$  and  $g : X \rightarrow STB$ :

$$\begin{array}{ccc}
 X & \xrightarrow{\eta_T X} & TX \\
 & \searrow h & \downarrow h^\lambda \\
 & & STA
 \end{array}
 \quad
 \begin{array}{ccc}
 TY & \xrightarrow{k^\mathbb{T}} & TX \\
 & \searrow (h^\lambda k)^\lambda & \downarrow h^\lambda \\
 & & STA
 \end{array}
 \quad
 \begin{array}{ccc}
 TB & \xrightarrow{g^\lambda} & STB \\
 & \searrow ((f^\lambda)^\mathbb{S} \cdot g)^\lambda & \downarrow (f^\lambda)^\mathbb{S} \\
 & & STA
 \end{array}$$

**Proposition 36 ([MW10])**

Given a distributive law  $\lambda$  of  $\mathbb{T}$  over  $\mathbb{S}$ , the composite monad is given by  $\mathbb{S} \circ_\lambda \mathbb{T} = (ST, S\eta_T \cdot \eta_S, ((-)^\lambda)^\mathbb{S})$

**Proposition 37 ([MW10])**

Let  $\mathbb{S}$  and  $\mathbb{T}$  be monads on  $\mathcal{C}$ . There is a bijection between distributive laws of  $\mathbb{T}$  over  $\mathbb{S}$  and monad structures  $(ST, S\eta_T \cdot \eta_S, (-)^{(\mathbb{S}\mathbb{T})})$  on  $ST$  such that:

- for every  $k : Y \rightarrow TX$ ,  $S(k^\mathbb{T}) = (\eta_S TX \cdot k)^{(\mathbb{S}\mathbb{T})}$
- for every  $m : Y \rightarrow STX$  and  $h : X \rightarrow STU$ :
  - $h^{(\mathbb{S}\mathbb{T})} = (h^{(\mathbb{S}\mathbb{T})} \cdot \eta_S TX)^\mathbb{T}$
  - the following diagram commutes

$$\begin{array}{ccc}
 SY & \xrightarrow{m^\mathbb{S}} & STX \\
 & \searrow (h^{(\mathbb{S}\mathbb{T})} \cdot m)^\mathbb{S} & \downarrow h^{(\mathbb{S}\mathbb{T})} \\
 & & STU
 \end{array}$$

**1.4 2-categories**



**Definition 38 (2-category)**

A 2-category  $C$  consists of:

- Cellular structure:
  - A collection  $C_0$  of 0-cells  $a, b, c, \dots$  (objects),
  - For each pair of 0-cells  $a, b$ , a collection  $C_1(a, b)$  of 1-cells (morphisms) from  $a$  to  $b$ ,
  - For each pair of 1-cells  $f, g \in C_1(a, b)$ , a collection  $C_2(f, g)$  of 2-cells (2-morphisms) from  $f$  to  $g$
- Identities:
  - A 1-cell  $\text{id}_a : a \rightarrow a$  for each 0-cell  $a \in C_0$  (identity 1-cell),
  - A 2-cell  $\text{id}_f : f \Rightarrow f$  for each 1-cell  $f \in C_1(a, b)$  (identity 2-cell)
- Composition:
  - A composition operation on 1-cells  $g \circ f$  for every pair of composable 1-cells  $f : a \rightarrow b$  and  $g : b \rightarrow c$ ,
  - A vertical composition operation on 2-cells  $\beta \circ \alpha : f \Rightarrow h$  for every pair of composable 2-cells  $\alpha : f \Rightarrow g$  and  $\beta : g \Rightarrow h$ ,
  - A horizontal composition operation on 2-cells  $\beta \bullet \alpha : g \circ f \Rightarrow g' \circ f'$  for every pair of composable 2-cells  $\alpha : f \Rightarrow f'$  and  $\beta : g \Rightarrow g'$

such that

- 1-cell compositions are associative and identities are neutral
- 2-cell vertical compositions are associative and identities are neutral
- 2-cell horizontal compositions are associative and identities are neutral, and composition of 2-cell identities preserves 1-cell composition
- the interchange law holds: for any composable 2-cells  $\alpha, \alpha', \beta, \beta'$ , we have

$$(\beta' \circ \beta) \bullet (\alpha' \circ \alpha) = (\beta' \bullet \alpha') \circ (\beta \bullet \alpha)$$

**Example 39 (2-category of Categories)**

The 2-category **Cat** has:

- 0-cells: categories,
- 1-cells: functors between categories,
- 2-cells: natural transformations between functors.
- pointwise composition of functors as 1-cell composition,
- pointwise composition of natural transformations as vertical composition,
- horizontal composition of natural transformations defined by the composition of functors.

**Definition 40 (Adjunction in a 2-category)**

Given a 2-category  $C$ , an *adjunction* between 0-cells  $a$  and  $b$  in  $C$  consists of:

- Two 1-cells  $f : a \rightarrow b$  and  $g : b \rightarrow a$ ,
- Two 2-cells  $\eta : \text{id}_a \Rightarrow g \circ f$  (unit) and  $\varepsilon : f \circ g \Rightarrow \text{id}_b$  (counit),

such that the following triangle identities hold:

$$(f \bullet \eta) \circ (\varepsilon \bullet f) = \text{id}_f \quad (\eta \bullet g) \circ (g \bullet \varepsilon) = \text{id}_g$$

**Definition 41 (2-functor)**

Given two 2-categories  $C$  and  $D$ , a *2-functor*  $F : C \rightarrow D$  consists of three mappings:

- $F : C_0 \rightarrow D_0$  that assigns to each 0-cell  $a$  in  $C$  a 0-cell  $Fa$  in  $D$ ,
- $F : C_1(a, b) \rightarrow D_1(Fa, Fb)$  for each pair of 0-cells  $a, b$  in  $C$ , that assigns to each 1-cell  $f : a \rightarrow b$  in  $C$  a 1-cell  $Ff : Fa \rightarrow Fb$  in  $D$ ,
- $F : C_2(f, g) \rightarrow D_2(Ff, Fg)$  for each pair of 1-cells  $f, g : a \rightarrow b$  in  $C$ , that assigns to each 2-cell  $\alpha : f \Rightarrow g$  in  $C$  a 2-cell  $F\alpha : Ff \Rightarrow Fg$  in  $D$ ,

such that  $F$  preserves identities and compositions of 1-cells and 2-cells.

**Lemma 42**

Let  $F : C \rightarrow D$  be a 2-functor.

If  $\langle f : a \rightarrow b, u, \eta, \varepsilon \rangle$  is an adjunction in  $C$ , then  $\langle Ff : Fa \rightarrow Fb, Fu, F\eta, F\varepsilon \rangle$  is an adjunction in  $D$ .

**Proof**

The candidate adjunction has the right type, and a 2-functor preserves pasting diagrams and identities, so the triangle identities are preserved.  $\square$

**Definition 43 (natural 2-transformation)**

Let  $F, G : C \rightarrow D$  be two 2-functors between 2-categories.

A (strict) natural 2-transformation  $C \begin{matrix} \xrightarrow{F} \\ \Downarrow \Xi \\ \xrightarrow{G} \end{matrix} D$  consists of a 1-cell  $\Xi_a : F_0 a \rightarrow G_0 a$  for each 0-cell  $a$  such that:

- for every 1-cell  $f : a \rightarrow b$  in  $C$ :

$$\begin{array}{ccc} F_0 a & \xrightarrow{F_1 f} & F_0 b \\ \Xi_a \downarrow & & \downarrow \Xi_b \\ G_0 a & \xrightarrow{G_1 f} & G_0 b \end{array}$$

- for every 2-cell  $a \begin{matrix} \xrightarrow{f} \\ \Downarrow \alpha \\ \xrightarrow{g} \end{matrix} b$ :

$$\begin{array}{ccccc} & & Ga & \xrightarrow{Gf} & \\ \Xi_a \nearrow & & \Downarrow G\alpha & & \\ Fa & \xrightarrow{Fg} & Fb & \xrightarrow{\Xi_b} & Gb \end{array} = \begin{array}{ccccc} & & Ga & \xrightarrow{Gf} & \\ \Xi_a \nearrow & & \Downarrow F\alpha & & \\ Fa & \xrightarrow{Ff} & Fb & \xrightarrow{\Xi_b} & Gb \end{array}$$

**1.5 Multicategories****Definition 44 (Multicategory)**

A multicategory  $\mathcal{U}$  consists of:

- A class of objects  $\text{Ob}(\mathcal{U})$ ,
- For each finite (possibly empty) sequence of objects  $(a_1, a_2, \dots, a_n)$  and an object  $b$ , a set of morphisms  $\mathcal{U}(a_1, a_2, \dots, a_n; b)$ ,
- For each object  $a$ , an identity morphism  $\text{id}_a : (a) \rightarrow a$ ,
- A composition operation that allows for the composition of morphisms in a way that generalizes the composition in a category, satisfying appropriate associativity and identity axioms.

**Example 45 (Category with Finite Products as a Multicategory)**

Given a category  $\mathcal{C}$  with finite products, we can construct a multicategory  $\mathcal{U}$  where:

- The objects of  $\mathcal{U}$  are the same as the objects of  $\mathcal{C}$ ,
- The morphisms  $\mathcal{U}(a_1, a_2, \dots, a_n; b)$  are given by the morphisms in  $\mathcal{C}$  from the product  $a_1 \times a_2 \times \dots \times a_n$  to  $b$ ,
- The identities rely on the canonical isomorphism

$$\text{id}_a := \prod_{i=1}^n a \cong a$$

- The composition is defined using the universal property of products and the composition in  $\mathcal{C}$ .

#### Example 46 (Monoidal Category as a Multicategory)

A monoidal category  $(\mathcal{C}, \otimes, I)$  can be viewed as a multicategory where:

- The objects of the multicategory are the same as the objects of  $\mathcal{C}$ ,
- The morphisms  $\mathcal{U}(a_1, a_2, \dots, a_n; b)$  are given by the morphisms in  $\mathcal{C}$  from  $a_1 \otimes a_2 \otimes \dots \otimes a_n$  to  $b$  (by nesting the monoidal product)

#### Definition 47 (Multifunctor)

A *multi-functor*  $F : \mathcal{U} \rightarrow \mathcal{V}$  between multicategories  $\mathcal{U}$  and  $\mathcal{V}$  consists of two mappings:

- A mapping  $F : \text{Ob}(\mathcal{U}) \rightarrow \text{Ob}(\mathcal{V})$  that assigns to each object  $a$  in  $\mathcal{U}$  an object  $Fa$  in  $\mathcal{V}$ ,
- A mapping  $F : \mathcal{U}(a_1, a_2, \dots, a_n; b) \rightarrow \mathcal{V}(Fa_1, Fa_2, \dots, Fa_n; Fb)$  that assigns to each morphism  $f : (a_1, a_2, \dots, a_n) \rightarrow b$  in  $\mathcal{U}$  a morphism  $Ff : (Fa_1, Fa_2, \dots, Fa_n) \rightarrow Fb$  in  $\mathcal{V}$ ,

such that  $F$  preserves:

- Identities: For each object  $a$  in  $\mathcal{U}$ ,  $F\text{id}_a = \text{id}_{Fa}$ ,
- Composition: For morphisms  $f : (a_1, a_2, \dots, a_n) \rightarrow b$  and  $g_i : (a_{i1}, a_{i2}, \dots, a_{ik_i}) \rightarrow a_i$  in  $\mathcal{U}$ , we have

$$F(f \circ (g_1, g_2, \dots, g_n)) = Ff \circ (Fg_1, Fg_2, \dots, Fg_n).$$

#### Definition 48 (Multi-natural transformation)

Given two multi-functors  $F, G : \mathcal{U} \rightarrow \mathcal{V}$  between multicategories  $\mathcal{U}$  and  $\mathcal{V}$ , a *multi-natural transformation*  $\eta : F \rightarrow G$  consists of a family of morphisms  $\{\eta_a : Fa \rightarrow Ga\}_{a \in \text{Ob}(\mathcal{U})}$  such that for every morphism  $f : (a_1, a_2, \dots, a_n) \rightarrow b$  in  $\mathcal{U}$ , the following diagram commutes:

$$\begin{array}{ccc} (Fa_1, Fa_2, \dots, Fa_n) & \xrightarrow{Ff} & Fb \\ (\eta_{a_1}, \eta_{a_2}, \dots, \eta_{a_n}) \downarrow & & \downarrow \eta_b \\ (Ga_1, Ga_2, \dots, Ga_n) & \xrightarrow{Gf} & Gb \end{array}$$

#### Example 49 (Category of multicategories)

We have a 2-category **Multicat** where:

- 0-cells are multi-categories
- 1-cells are multi-functors
- 2-cells are multi-natural transformations

with multi-functor composition and identities:

$$\text{Id}_C a := a \quad \text{Id}_C f := f$$

vertical, horizontal composition of multi-natural transformations, with identities:

$$(\text{id}_F)_a := \text{id}_{Fa} : (a) \rightarrow a$$

### Example 50 (Ordinary 2-functor)

We have a 2-functor structure  $-_o : \mathbf{Multicat} \rightarrow \mathbf{Cat}$  sending:

- each multi-category  $\mathcal{U}$  to its (ordinary) category of morphisms:
  - $\text{Ob}(\mathcal{U}_o) := \text{Ob}(\mathcal{U})$
  - $\mathcal{U}_o(a, b) := \mathcal{U}(a; b)$ , i.e the morphisms in  $\mathcal{U}_o$  are the multi-morphisms with a length-1 source
  - identities and composition are given by identity morphisms and composition in the multi-category
- each multi-functor  $F : \mathcal{U} \rightarrow \mathcal{V}$  to the functor sending:
  - $a \in \text{Ob}(\mathcal{U})$  to  $Fa \in \text{Ob}(\mathcal{V})$
  - $f : (a) \rightarrow b$  to  $F_o f := Ff : (Fa) \rightarrow Fb$
- each multi-natural transformation  $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$  to  $(\alpha_o)_a := \alpha_a$

It is 2-faithful.

## 1.6 Operads

### Definition 51 (Operad)

An *operad*  $\mathcal{O}$  consists of:

- A collection of sets  $\{\mathcal{O}_n\}_{n \geq 0}$ , where  $\mathcal{O}_n$  is the set of  $n$ -ary operations,
- A composition function  $\gamma : \mathcal{O}_n \times \mathcal{O}_{k_1} \times \cdots \times \mathcal{O}_{k_n} \rightarrow \mathcal{O}_{k_1 + \cdots + k_n}$ ,
- An identity element  $e \in \mathcal{O}_1$ ,

satisfying the following axioms:

- Associativity: For all  $f \in \mathcal{O}_n$ ,  $g_i \in \mathcal{O}_{k_i}$ , and  $h_{ij} \in \mathcal{O}_{m_{ij}}$ , we have

$$\gamma(f; g_1, \dots, g_n) = f(g_1(h_{11}, \dots, h_{1k_1}), \dots, g_n(h_{n1}, \dots, h_{nk_n})).$$

- Identity: For all  $f \in \mathcal{O}_n$ , we have

$$f(e, \dots, e) = f.$$

**Definition 52 (Algebraic Theory)**

An *algebraic theory*  $\mathcal{T}$  consists of:

- A collection of sets  $\{\mathcal{T}_n\}_{n \geq 0}$ , where  $\mathcal{T}_n$  is the set of  $n$ -ary operations,
- A collection of equations between terms built from these operations,

such that the operations and equations define a class of algebras (models) that satisfy the given equations.

**Example 53 (Operad from an Algebraic Theory)**

Given an algebraic theory  $\mathcal{T}$ , we can construct an operad  $\mathcal{O}^{\mathcal{T}}$  where:

- $\mathcal{O}_n^{\mathcal{T}}$  consists of the  $n$ -ary operations in  $\mathcal{T}$ ,
- The composition function  $\gamma$  is defined by substituting operations according to the equations of the theory,
- The identity element  $e$  corresponds to the identity operation in the theory.

**Definition 54 (Renaming)**

A *renaming*  $\rho : n \rightarrow m$  is any function from the set of  $n$  variables to the set of  $m$  variables.

**Example 55 (of renaming)**

Let  $\rho : 3 \rightarrow 2$  be the renaming defined by  $\rho(x_1) = y_1$ ,  $\rho(x_2) = y_2$ , and  $\rho(x_3) = y_1$ . Then for any term  $t \in \mathcal{T}_3$ , we have  $\rho(t) \in \mathcal{T}_2$  obtained by substituting  $y_1$  for  $x_1$  and  $x_3$ , and  $y_2$  for  $x_2$  in  $t$ .

**Definition 56 (Translation)**

Let  $\mathcal{O}$  be an operad and  $\mathcal{T}$  be a theory.

A *translation*  $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{T}$  consists of an assignment  $\mathfrak{T}f \in \mathcal{T}_n$  for every  $f \in \mathcal{O}_n$  such that:

- the identity  $\text{id} : \mathcal{O}_1$  is sent to the unique variable projection:  $\mathfrak{T}\text{id} = x \in \mathcal{T}_1$
- the translation respects composition, for all  $f \in \mathcal{O}_n$  composable with  $g_i \in \mathcal{O}_{m_i}$ :

$$\mathfrak{T}(f \circ (g_1, \dots, g_n)) = \mathfrak{T}f[x_i \mapsto \mathfrak{T}g_i]_{i=1}^n$$

A translation is *surjective*, written  $\mathfrak{T} : \mathcal{O} \twoheadrightarrow \mathcal{T}$ , if for every term  $t \in \mathcal{T}_n$ , there exists an operator  $f \in \mathcal{O}_m$  and a renaming  $\rho : m \rightarrow n$  such that  $t = \mathfrak{T}f[\rho]$ .

**Definition 57 (Model)**

Let  $\mathcal{O}$  be an operad, and  $\mathcal{U}$  be a multicategory. A *model* of  $\mathcal{O}$  in  $\mathcal{U}$  is a multi-functor  $\mathcal{A} : \mathcal{O} \rightarrow \mathcal{U}$  that preserves the operadic structure, i.e., it respects the composition and identity of the operad.

**Definition 58**

Homomorphism of Models A *homomorphism*  $h : \mathcal{A} \rightarrow \mathcal{B}$  between  $\mathcal{O}$ -models in  $\mathcal{U}$  consists of a morphism  $\underline{h} : (\underline{\mathcal{A}}) \rightarrow (\underline{\mathcal{B}})$  such that for all  $f \in \mathcal{O}_n$ ,

$$\underline{h} \circ (\mathcal{A} \llbracket f \rrbracket) = \mathcal{B} \llbracket f \rrbracket (\underline{h}, \dots, \underline{h})$$

**Example 59 (Category of Models)**

Given an operad  $\mathcal{O}$  and a multicategory  $\mathcal{U}$ , we can construct a category  $\mathbf{Mod}(\mathcal{O}, \mathcal{U})$  where:

- The objects are the models of  $\mathcal{O}$  in  $\mathcal{U}$ ,
- The morphisms are the homomorphisms between these models.

**Proposition 60**

Let  $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{T}$  be a translation, and let  $\mathcal{C}$  be a category with chosen finite product  $\prod$ .

There is a carrier-preserving, fully-faithful functor  $\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) : \mathbf{Mod}(\mathcal{T}, \mathcal{C}) \rightarrow \mathbf{Mod}(\mathcal{O}, \langle \mathcal{C}, \prod \rangle)$  given by:

$$(\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle)A) \llbracket f \rrbracket := A \llbracket \mathfrak{T}f \rrbracket \quad \mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle)h := h$$

**Proposition 61**

The compatibility with composition and identities of the operad follows from their preservation by  $\mathfrak{T}$ . Preservation of identities and composition, and fully-faithfulness are evident.

**Proposition 62**

We define a 2-functor  $\mathbf{Mod}(\mathcal{O}, -) : \mathbf{Multicat} \rightarrow \mathbf{Cat}$  such that:

- for every multi-category  $\mathcal{U}$ ,  $\mathbf{Mod}(\mathcal{O}, \mathcal{U})$  is the category of  $\mathcal{O}$ -models in  $\mathcal{U}$ .
- for every multi-functor  $F : \mathcal{U} \rightarrow \mathcal{V}$ , the functor  $\mathbf{Mod}(\mathcal{O}, F) : \mathbf{Mod}(\mathcal{O}, \mathcal{U}) \rightarrow \mathbf{Mod}(\mathcal{O}, \mathcal{V})$  sends:

– each  $\mathcal{O}$ -model  $\mathcal{A}$  in  $\mathcal{U}$  to the  $\mathcal{O}$ -model:

$$* \quad F\mathcal{A} := F\mathcal{A}$$

$$* \quad (F\mathcal{A}) \llbracket f \rrbracket : (F\mathcal{A}, \dots, F\mathcal{A}) \xrightarrow{F(\mathcal{A} \llbracket f \rrbracket)} F\mathcal{A}$$

– each  $\mathcal{O}$ -homomorphism  $h : \mathcal{A} \rightarrow \mathcal{B}$  to the  $\mathcal{O}$ -homomorphism over  $Fh : F\mathcal{A} \rightarrow F\mathcal{B}$

- for every multi-natural transformation  $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$  we have a natural transformation

$$\mathbf{Mod}(\mathcal{O}, \mathcal{U}) \begin{array}{c} \xrightarrow{\mathbf{Mod}(\mathcal{O}, F)} \\ \Downarrow \mathbf{Mod}(\mathcal{O}, \alpha) \\ \xrightarrow{\mathbf{Mod}(\mathcal{O}, G)} \end{array} \mathbf{Mod}(\mathcal{O}, \mathcal{V}) \quad \text{given, for every model } \mathcal{A}, \text{ by}$$

$$\mathbf{Mod}(\mathcal{O}, \alpha)_{\mathcal{A}} : (F\mathcal{A}) \xrightarrow{\alpha_{\mathcal{A}}} G\mathcal{A}$$

**1.7 Free algebras**

We'll use the definitions from [All+25].

**Definition 63 (Finitary signature)**

A *finitary signature*  $\Sigma = (\text{op } \Sigma, \text{arity})$  consists of a set  $\text{op } \Sigma$  of *operators*, and an assignment  $\text{arity} : \text{op } \Sigma \rightarrow \mathbb{N}$  associating to each operator in  $\text{op } \Sigma$  its *arity*.



**Definition 64 (Algebra)**

An *algebra*  $A = (\underline{A}, A \llbracket - \rrbracket)$  over a finitary signature  $\Sigma$  consists of:

- A set  $\underline{A}$  called the *carrier* of  $A$ ,
- For each operator  $f \in \text{op } \Sigma$  of arity  $n$ , a function  $A \llbracket f \rrbracket : \underline{A}^n \rightarrow \underline{A}$  called the *interpretation* of  $f$  in  $A$ .

**Definition 65 (Terms)**

Given a set  $X$  of variables and a finitary signature  $\Sigma$ , the set of *terms*  $\text{Term}_\Sigma(X)$  is defined inductively as follows:

- Every variable  $x \in X$  is a term,
- If  $f \in \text{op } \Sigma$  is an operator of arity  $n$  and  $t_1, t_2, \dots, t_n$  are terms, then  $f(t_1, t_2, \dots, t_n)$  is a term.

**Remark :** The definition of algebraic theory seen before can be reformulated as a finitary signature together with a set of equations between terms built from the operators in the signature.

**Definition 66 (Environment and Interpretation)**

An *environment*  $\rho$  for a context  $X$  in an algebra  $A$  is a function  $\rho : X \rightarrow \underline{A}$  that assigns to each variable in  $X$  an element of the carrier of  $A$ .

Given a term  $t \in \text{Term}_\Sigma(X)$ , denoted  $X \vdash t$ , the algebra  $A$  gives an *interpretation* function  $A \llbracket t \rrbracket : \underline{A}^X \rightarrow \underline{A}$ . Given an environment  $\rho : X \rightarrow \underline{A}$ , we can evaluate the term  $t$  in  $A$  under the environment  $\rho$ , denoted  $A \llbracket t \rrbracket(\rho)$ , by recursively substituting the variables in  $t$  with their corresponding values in  $\underline{A}$  according to  $\rho$  and applying the interpretations of the operators as defined in  $A$ .

**Definition 67 (Valid equations)**

An equation  $t = t'$  between terms in  $\text{Term}_\Sigma(X)$  is *valid* in an algebra  $A$  if for every environment  $\rho : X \rightarrow \underline{A}$ , we have  $A \llbracket t \rrbracket(\rho) = A \llbracket t' \rrbracket(\rho)$ .

**Definition 68 (Presentation and Models)**

A *presentation*  $\mathcal{T} = (\Sigma_{\mathcal{T}}, \mathcal{T}_{ax})$  consists of a finitary signature  $\Sigma_{\mathcal{T}}$  and a set  $\mathcal{T}_{ax}$  of *axioms*, which are equations between terms in  $\text{Term}_{\Sigma_{\mathcal{T}}}(X)$  for some set of variables  $X$ .

A  $\mathcal{T}$ -model  $A$  is a  $\Sigma_{\mathcal{T}}$ -algebra such that all equations in  $\mathcal{T}_{ax}$  are valid in  $A$ .

A  $\mathcal{T}$ -model  $A$  over a context  $X$  is a  $\mathcal{T}$ -model  $A$  together with an environment  $\rho : X \rightarrow \underline{A}$ .

**Definition 69 (Homomorphisms)**

Let  $A = (\underline{A}, A \llbracket - \rrbracket)$  and  $B = (\underline{B}, B \llbracket - \rrbracket)$  be algebras over the same signature  $\Sigma$ .

A *homomorphism* of  $\Sigma$ -algebras from  $A$  to  $B$  is a function  $h : \underline{A} \rightarrow \underline{B}$  such that for every operator  $f \in \text{op } \Sigma$  of arity  $n$ , we have:

$$B \llbracket f \rrbracket(h(a_1), \dots, h(a_n)) = h(A \llbracket f \rrbracket(a_1, \dots, a_n))$$

for all  $a_1, \dots, a_n \in \underline{A}$ .

A *homomorphism* of  $\mathcal{T}$ -models is a homomorphism of  $\Sigma_{\mathcal{T}}$ -algebras between the underlying algebras of the models.

**Definition 70 (Morphism)**

A *morphism*  $h : A \rightarrow B$  of  $\mathcal{T}$ -models over  $X$  is a  $\mathcal{T}$ -model homomorphism that makes the diagram commute:

$$\begin{array}{ccc} X & \xrightarrow{\rho_A} & \underline{A} \\ & \searrow \rho_B & \downarrow h \\ & & \underline{B} \end{array}$$

**Definition 71 (Free model (or free algebra))**

Let  $\mathcal{T}$  be a presentation and  $X$  be a set of variables.

A *free  $\mathcal{T}$ -model* over  $X$  is a  $\mathcal{T}$ -model  $F(X)$  such that for any  $\mathcal{T}$ -model  $A$  over  $X$ , there exists a unique morphism of  $\mathcal{T}$ -models from  $F(X)$  to  $A$ .

The existence-and-uniqueness property is called the *universal property* of free  $\mathcal{T}$ -models over  $X$ .

Informally:

- A signature gives defines the operators and constants (arity 0 operators).
- An algebra gives a concrete interpretation of the signature.
- A presentation defines axioms on top of the signature, for instance the axioms of a group, or of a semiring, etc.
- A model of a presentation is an algebra that satisfies the axioms of the presentation.
- A free algebra is the most general model of a presentation.

We can give an equivalent definition for free algebras using universal arrows.

**Proposition 72**

Let  $\mathcal{T}$  be a presentation, and  $X$  be a set. Let  $\mathcal{T}\text{-Mod}$  be the category of  $\mathcal{T}$ -models over  $X$ . Let  $U : \mathcal{T}\text{-Mod} \rightarrow \mathbf{Set}$  be the forgetful functor that sends a  $\mathcal{T}$ -model to its underlying set.

Then, a free  $\mathcal{T}$ -model over  $X$  is a universal arrow from  $X$  to  $U$ .

**Proof**

Let  $F X$  be a free  $\mathcal{T}$ -model over  $X$ . We want to show that  $(F X, \eta_X)$  is a universal arrow from  $X$  to  $U$ , where  $\eta_X : X \rightarrow U F X$  is the function that sends each variable in  $X$  to its interpretation in  $F X$ .

Let  $A$  be a  $\mathcal{T}$ -model over  $X$ , and let  $\rho_A : X \rightarrow U A$  be the environment of  $A$ . Since  $F X$  is a free  $\mathcal{T}$ -model over  $X$ , there exists a unique morphism of  $\mathcal{T}$ -models  $h : F X \rightarrow A$  such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & U F X \\ & \searrow \rho_A & \downarrow U h \\ & & U A \end{array}$$

This shows that  $(F X, \eta_X)$  satisfies the universal property of being a universal arrow from  $X$  to  $U$ .

Conversely, if  $(F X, \eta_X)$  is a universal arrow from  $X$  to  $U$ , then for any  $\mathcal{T}$ -model  $A$  over  $X$ , there exists a unique morphism of  $\mathcal{T}$ -models from  $F X$  to  $A$ , which means that  $F X$  is a free  $\mathcal{T}$ -model over  $X$ . Therefore, the two definitions are equivalent.  $\square$

## 1.8 Multi-homomorphisms

This section will be a subset of Ohad's note. In particular, all proofs will be omitted.

In all of this section, we fix an algebraic theory  $\mathcal{T}$  and  $\langle \mathcal{C}, \Pi \rangle$  a category with chosen finite products.

### Definition 73 (Strength)

For all  $k \in \mathbb{N}^*$  and  $X, Y \in \mathcal{C}$ , we define the *strength* (for the reader monad  $X^k$ ) by

$$\text{str} : X^k \times Y \xrightarrow{\langle \pi_i \times \text{id} \rangle_{i=1}^k} (X \times Y)^k$$

### Definition 74 (Multi- $\mathcal{T}$ -homomorphism)

Let  $A_1, A_2, \dots, A_k$  and  $B$  be  $\mathcal{T}$ -models in  $\mathcal{C}$ .

A *multi- $\mathcal{T}$ -homomorphism* from  $A_1, A_2, \dots, A_k$  to  $B$  is a morphism  $h : \prod_{i=1}^k \underline{A}_i \rightarrow \underline{B}$  such that for every term  $t \in \mathcal{T}_k$  and  $1 \leq i_0 \leq n$ , the following diagram commutes:

$$\begin{array}{ccccc} A_{i_0}^k \times \prod_{\substack{1 \leq i \leq n \\ i \neq i_0}} A_i & \xrightarrow{\text{str}} & \left( A_{i_0} \times \prod_{\substack{1 \leq i \leq n \\ i \neq i_0}} A_i \right)^k & \xrightarrow{\cong} & \left( \prod_{i=1}^n A_i \right)^k \\ \downarrow A_{i_0}[[t]] \times \text{id} & & & & \downarrow f^k \\ A_{i_0} \times \prod_{\substack{1 \leq i \leq n \\ i \neq i_0}} A_i & \xrightarrow{\cong} & \prod_{i=1}^n A_i & \xrightarrow{f} & B \\ & & & & \downarrow B[[t]] \\ & & & & B \end{array}$$

▷ We can then define a multi-category structure  $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$  over the finite-product multi-category:

- the objects are  $\mathcal{T}$ -algebra
- the multi-morphisms are the multi- $\mathcal{T}$ -homomorphisms
- the identities are the identity morphisms
- composition is inherited from  $\langle \mathcal{C}, \Pi, \rangle$

▷ We also define a multi-functor  $\underline{\phantom{x}} : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \rightarrow \langle \mathcal{C}, \Pi \rangle$  which sends an algebra to its carrier and a multi- $\mathcal{T}$ -homomorphism to its underlying morphism.

▷ We define another 2-category  $\mathbf{Cat}_\times$  consisting of:

- 0-cells are  $\langle \mathcal{C}, \Pi \rangle$ , categories with chosen finite products
- 1-cells are finite product preserving functors, not necessarily preserving the chosen products
- 2-cells are the natural transformations between these functors

Therefore, this is a sub-category of  $\mathbf{Cat}$ .

▷ We define the data of a 2-functor  $\mathbf{Multi}(\mathcal{T}, -) : \mathbf{Cat}_\times \rightarrow \mathbf{Multicat}$  sending:

- each category with chosen products (0-cells)  $\mathcal{C}$  to the multi-category (0-cell)  $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$
- each finite-product preserving functor (1-cell)  $H : \mathcal{C} \rightarrow \mathcal{D}$  to the following multi-functor  $\mathbf{Multi}(\mathcal{T}, H) : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \rightarrow \mathbf{Multi}(\mathcal{T}, \mathcal{D})$ , sending:

- each  $\mathcal{T}$ -algebra  $\mathcal{A}$  to the  $\mathcal{T}$ -algebra  $\mathbf{Multi}(\mathcal{T}, H) := H\mathcal{A}$ , given by:
  - \* its carrier  $H\mathcal{A}$  is  $H\mathcal{A}$
  - \* the interpretation of each  $t \in \mathcal{T}_n$  is  $H\mathcal{A}[\![t]\!] : (H\mathcal{A}^n) \xrightarrow{\cong} H(\mathcal{A}^n) \xrightarrow{\mathcal{A}[\![t]\!] } H\mathcal{A}$
- each multi- $\mathcal{T}$ -homomorphism  $h : (\mathcal{A}_1, \dots, \mathcal{A}_n) \rightarrow \mathcal{B}$  to the multi- $\mathcal{T}$ -homomorphism  $\mathbf{Multi}(\mathcal{T}, H)h : (H\mathcal{A}_1, \dots, H\mathcal{A}_n) \rightarrow H\mathcal{B}$  given by:

$$\mathbf{Multi}(\mathcal{T}, H)h : \prod_{i=1}^n H\mathcal{A}_i \xrightarrow{\cong} H\left(\prod_{i=1}^n \mathcal{A}_i\right) \xrightarrow{Hh} H\mathcal{B}$$

- each natural transformation  $\mathcal{C} \begin{array}{c} \xrightarrow{H} \\ \Downarrow \alpha \\ \xrightarrow{K} \end{array} \mathcal{D}$  (2-cell) to the multi-natural transformation

$$\mathbf{Multi}(\mathcal{T}, \mathcal{C}) \begin{array}{c} \xrightarrow{\mathbf{Multi}(\mathcal{T}, H)} \\ \Downarrow \mathbf{Multi}(\mathcal{T}, \alpha) \\ \xrightarrow{\mathbf{Multi}(\mathcal{T}, K)} \end{array} \mathbf{Multi}(\mathcal{T}, \mathcal{D})$$

, given, at an algebra  $\mathcal{A} \in \mathbf{Mod}(\mathcal{T}, \mathcal{C})$ , by the unary multi- $\mathcal{T}$ -homomorphism corresponding to the following  $\mathcal{T}$ -homomorphism:

$$\mathbf{Multi}(\mathcal{T}, \alpha)'_{\mathcal{A}} : H\mathcal{A} \xrightarrow{\alpha_{\mathcal{A}}} K\mathcal{A}$$

▷ We define another 2-functor  $\langle - \rangle : \mathbf{Cat}_\times \rightarrow \mathbf{Multicat}$ , sending:

- 0-cells: categories with specified products  $\langle \mathcal{C}, \Pi \rangle$  to their associated multi-category  $\langle \mathcal{C}, \Pi \rangle$  (cf. Example 45)
- 1-cells: finite-product preserving functors  $H : \mathcal{C} \rightarrow \mathcal{D}$  to the multi-functor sending:
  - objects  $a \in \text{Ob}(\langle \mathcal{C}, \Pi \rangle) = \text{Ob } \mathcal{C}$  to  $\langle H \rangle a \in \text{Ob } \mathcal{D}$
  - multi-morphisms  $f : (a_1, \dots, a_n) \rightarrow b$  to the multi-morphism:

$$\langle H \rangle f : \prod_{i=1}^n H a_i \xrightarrow{\cong} H\left(\prod_{i=1}^n a_i\right) \xrightarrow{Hf} Hb$$

- 2-cells: natural transformations  $\mathcal{C} \begin{array}{c} \xrightarrow{H} \\ \Downarrow \alpha \\ \xrightarrow{K} \end{array} \mathcal{D}$  to the multi-natural transformation given for each  $a \in \text{Ob}(\mathcal{C})$ , by:

$$\langle \alpha \rangle_a : \prod_{i=1}^1 H a \xrightarrow{\cong} H a \xrightarrow{\alpha_a} K a$$

▷ Finally, we have a natural 2-transformation  $\mathbf{Cat}_\times \begin{array}{c} \xrightarrow{\mathbf{Multi}(\mathcal{T}, -)} \\ \Downarrow \langle - \rangle \\ \xrightarrow{\langle - \rangle} \end{array} \mathbf{Multicat}$  whose components, for each category  $\langle \mathcal{C}, \Pi \rangle$  with chosen finite products, consists of the forgetful multi-functors sending:

- each  $\mathcal{T}$ -algebra  $\mathcal{A}$  to its carrier  $\underline{\mathcal{A}}$
- each multi- $\mathcal{T}$ -homomorphism  $h : (\mathcal{A}_1, \dots, \mathcal{A}_n) \rightarrow \mathcal{B}$  to its underlying morphism  $\underline{h} : \prod_{i=1}^n \underline{\mathcal{A}}_i \rightarrow \underline{\mathcal{B}}$

All of the proofs that the following definitions are indeed of the nature they were described of are in Ohad's notes.

## 1.9 Free extensions

Free extensions were introduced in [YGK18]. They are a unifying theory for specifying (generalized) partially-static data-structures by recourse to universal algebra.

In this section, let  $\mathcal{T}$  be a presentation,  $X$  be a set of variables and  $A$  be a  $\mathcal{T}$ -algebra.

### Definition 75 (Extension)

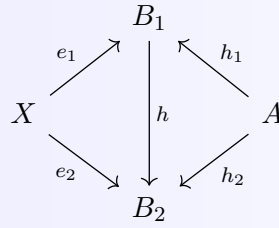
An *extension* of  $A$  by  $X$  is a triple  $\langle B, h, e \rangle$  where:

- $B$  is a  $\mathcal{T}$ -algebra
- $h : A \rightarrow B$  is a  $\mathcal{T}$ -homomorphism
- $e : X \rightarrow B$  is a function

### Definition 76 (Morphism of extensions)

Let  $b_1 = \langle B_1, h_1, e_1 \rangle, b_2 = \langle B_2, h_2, e_2 \rangle$  be two extensions of  $A$  by  $X$ .

A *morphism of extensions*  $h : b_1 \rightarrow b_2$  is a  $\mathcal{T}$ -homomorphism  $h : B_1 \rightarrow B_2$  that makes the following diagram commute:



### Definition 77 (Free extension)

A *free extension* of  $A$  by  $X$  is an extension for which there exists a unique morphism of extensions to every other extension of  $A$  by  $X$ .

## 1.10 Distributive combination of theories

### Definition 78 (Coproduct of signatures)

Given two signatures  $\Sigma_1$  and  $\Sigma_2$ , we define their coproduct  $\Sigma_1 + \Sigma_2$  as the signature with operators the disjoint union of the operators, but retaining their arity:

$$\Sigma_1 + \Sigma_2 = \{ \iota_i f : n \mid (f : n) \in \Sigma_i, i = 1, 2 \}$$

Given a  $\Sigma_i$ -term-in-context  $n \vdash_{\Sigma_i} t$ , we denote by  $n \vdash_{\Sigma_1 + \Sigma_2} \iota_i[t]$  the  $\Sigma_1 + \Sigma_2$ -term-in-context obtained by applying the injection  $\iota_i$  to all the operators of  $t$ .

Similarly, given a  $\Sigma_i$ -equation in context  $eq := (n \vdash_{\Sigma_i} t = t')$ , we denote by  $\iota_i[eq]$  the  $\Sigma_1 + \Sigma_2$ -equation in context  $n \vdash_{\Sigma_1 + \Sigma_2} \iota_i[t] = \iota_i[t']$ .

**Definition 79 (Distributive combination of theories)**

Let  $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$  be a translation from an operad  $\mathcal{O}$  to a theory  $\mathcal{M}$  and  $\mathcal{A}$  be any theory.

We define the *distributive combination of  $\mathfrak{T}$  over  $\mathcal{A}$*  to be the theory  $\mathfrak{T} \triangleleft \mathcal{A}$  given by

$$\begin{aligned} \Sigma_{\mathfrak{T} \triangleleft \mathcal{A}} &:= \Sigma_M + \Sigma_A \\ \text{Ax}_{\mathfrak{T} \triangleleft \mathcal{A}} &:= \iota_1[\text{Ax}_M] \cup \iota_2[\text{Ax}_A] \\ &\cup \left\{ \begin{array}{l} \langle x_i \rangle_{\substack{i=1 \\ i \neq i_0}}^n, \mathbf{y} \mid \mathfrak{T}f(x_1, \dots, x_{i_0-1}, s\mathbf{y}, x_{i_0+1}, \dots, x_n) \\ \qquad \qquad \qquad = s \langle \mathfrak{T}f(x_1, \dots, x_{i_0-1}, \mathbf{y}_j, x_{i_0+1}, \dots, x_n) \rangle_{j=1}^{|\mathbf{y}|} \end{array} \mid \begin{array}{l} (f : n) \in O_n \\ (s : |\mathbf{y}|) \in \Sigma_A \end{array} \right\} \end{aligned}$$

## 2 CONSTRUCTING FREE ALGEBRAS

### 2.1 Semigroups

#### 2.1.1 Free commutative semigroups

Let's try to construct the free algebra for the presentation of commutative semigroups. We cannot rely on the order of the terms, nor the order of applying the operations.

Let  $X$  be a set. We consider the theory of commutative semigroups **CSG**, with a signature  $\Sigma$  with a single operator  $(+)$  of arity 2, and with the two axioms

$$\begin{aligned} a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \end{aligned}$$

We pose

$$SX = \{f : X \rightarrow \mathbb{N} \mid f \text{ has finite non-empty support}\}$$

For  $f, g \in SX$ , we pose

$$f + g = \lambda x. f \ x + g \ x \ x \in SX$$

We define  $\eta_{\text{CSG}} : X \rightarrow SX$  for all  $x \in X$  by

$$\eta_{\text{CSG}}(x) = \lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$$

We pose  $F_{\text{CSG}}X = (SX, (+))$  where  $+$  is the operator defined above.

**Proposition 80**

$F_{\text{CSG}}X$  is a free **CSG**-model over  $X$ .

**Proof**

- ▷ Let's first check that  $F_{\text{CSG}}X$  is indeed a **CSG**-model. We need to check that the axioms of **CSG** are valid in  $F_{\text{CSG}}X$ . This is straightforward and comes down to the commutativity and associativity of addition in  $\mathbb{N}$ .
- ▷ Let  $A$  be a **CSG**-model over  $X$  with environment  $\varphi_A : X \rightarrow U A$ . We want to show that there exists a unique morphism  $h$  of **CSG**-models from  $F_{\text{CSG}}X$  to  $A$  such that the following

diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\mathbf{CSG}}} & SX \\ & \searrow \varphi_A & \downarrow h \\ & & U A \end{array}$$

with  $U$  being the forgetful functor from **CSG**-models to **Set**.

We define  $\Sigma\varphi_A : SX \rightarrow \underline{A}$  for all  $f \in SX$  by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f \ x \cdot \varphi_A(x)$$

We want to show that  $\Sigma\varphi_A$  is the unique morphism  $h$  of **CSG**-models from  $F_{\mathbf{CSG}}X$  to  $A$  above. We need to check that  $\Sigma\varphi_A$  preserves the operator  $+$ , and that the diagram above commutes.

- Preservation of the operator  $+$ : let  $f, g \in SX$ . We have:

$$\begin{aligned} \Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda x. f \ x + g \ x) \\ &= \sum_{x \in X} (f \ x + g \ x) \cdot \varphi_A(x) \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) + \sum_{x \in X} g \ x \cdot \varphi_A(x) \\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g) \end{aligned}$$

- Commutation of the diagram: let  $x \in X$ . We have:

$$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{CSG}}(x)) &= \Sigma\varphi_A \left( \lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \sum_{y \in X} \begin{cases} 1 \cdot \varphi_A(x) & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \\ &= \varphi_A(x) \end{aligned}$$

Therefore, the diagram commutes.

- Uniqueness: let  $h : F_{\mathbf{CSG}}X \rightarrow A$  be a morphism of **CSG**-models such that the diagram above commutes. We want to show that  $h = \Sigma\varphi_A$ . Let  $f \in SX$ . We have:

$$\begin{aligned} h(f) &= h \left( \sum_{x \in X} f \ x \cdot \eta_{\mathbf{CSG}}(x) \right) \\ &= \sum_{x \in X} f \ x \cdot h(\eta_{\mathbf{CSG}}(x)) \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(f) \end{aligned}$$

Therefore, we proved that  $F_{\mathbf{CSG}}X$  is a free **CSG**-model over  $X$ .

□

Finally we'll define a multiplication  $\mu_{\mathbf{CSG}}$  so that we have a monad structure  $(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$  on **Set**. For all  $f \in SSX$ , we pose

$$\mu_{\mathbf{CSG}}(f) = \lambda x. \sum_{g \in SX} f \ g \cdot g \ x$$

**Proposition 81**

$(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$  is a monad on **Set**.

**Proof**

We need to check the monad axioms:

- Left unit: let  $f \in SX$ . We have:

$$\begin{aligned} \mu_{\mathbf{CSG}}(\eta_{\mathbf{CSG}}S(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \begin{cases} 1 & \text{if } g = f \\ 0 & \text{otherwise} \end{cases}\right) \\ &= \lambda x. \sum_{g \in SX} \begin{cases} 1 \cdot g \cdot x & \text{if } g = f \\ 0 & \text{otherwise} \end{cases} \\ &= \lambda x. f \cdot x \\ &= f \end{aligned}$$

- Right unit: let  $f \in SX$ . We have:

$$\begin{aligned} \mu_{\mathbf{CSG}}(S\eta_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \begin{cases} f \cdot x & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise} \end{cases}\right) \\ &= \lambda x. \sum_{g \in SX} \begin{cases} f \cdot x \cdot g \cdot x & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise} \end{cases} \\ &= \lambda x. f(x) \cdot \eta_{\mathbf{CSG}}(x)(x) \\ &= \lambda x. f \cdot x \\ &= f \end{aligned}$$

- Associativity: let  $f \in SSSX$ . We have:

$$\begin{aligned} \mu_{\mathbf{CSG}}(S\mu_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \sum_{h \in SSSX} f \cdot h \cdot h \cdot g\right) \\ &= \lambda x. \sum_{g \in SX} \left(\sum_{h \in SSSX} f \cdot h \cdot h \cdot g\right) \cdot g \cdot x \\ &= \lambda x. \sum_{h \in SSSX} f \cdot h \cdot \left(\sum_{g \in SX} h \cdot g \cdot g \cdot x\right) \\ &= \lambda x. \sum_{h \in SSSX} f \cdot h \cdot \mu_{\mathbf{CSG}}(h) \\ &= \mu_{\mathbf{CSG}}(\mu_{\mathbf{CSG}}S(f)) \end{aligned}$$

□

**2.1.2 Free semigroups**

Compared to the free commutative semigroup, we can rely on the order of the terms. We'll use lists to represent the terms of the free semigroup.

Let  $X$  be a set. We consider the theory of semigroups **SG**, with a signature  $\Sigma$  with a single operator  $(\cdot)$  of arity 2, and with the axiom

$$a, b, c \vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c$$



We pose

$$TX = \{l \in X^* \mid l \text{ is finite and non-empty}\}$$

For  $f, g \in TX$ , we pose

$$f \cdot g = f \mathbin{++} g$$

where  $++$  is the concatenation of lists.

We define  $\eta_{\mathbf{SG}} : X \rightarrow TX$  for all  $x \in X$  by

$$\eta_{\mathbf{SG}}(x) = [x]$$

We pose  $F_{\mathbf{SG}}X = (TX, (\cdot))$  where  $\cdot$  is the operator defined above.

### Proposition 82

$F_{\mathbf{SG}}X$  is a free  $\mathbf{SG}$ -model over  $X$ .

#### Proof

▷  $F_{\mathbf{SG}}$  is a  $\mathbf{SG}$ -model: the associativity of  $\cdot$  in  $F_{\mathbf{SG}}$  comes from the associativity of list concatenation.

▷ Let  $A$  be a  $\mathbf{SG}$ -model over  $X$  with environment  $\varphi_A : X \rightarrow U A$ .

We define  $\Sigma\varphi_A : TX \rightarrow \underline{A}$  for all  $l \in TX$  by

$$\Sigma\varphi_A(l) = \prod_{x \in l} \varphi_A(x)$$

where the product<sup>a</sup> is taken in the order of the list  $l$ , which is well defined because the list is finite, non-empty, and the product is associative.

Let's show that  $\Sigma\varphi_A$  is the unique morphism  $h$  of  $\mathbf{SG}$ -models from  $F_{\mathbf{SG}}X$  to  $A$  such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\mathbf{SG}}} & TX \\ & \searrow \varphi_A & \downarrow h \\ & & U A \end{array}$$

– Preservation of the operator  $\cdot$ : let  $f, g \in TX$ . We have:

$$\begin{aligned} \Sigma\varphi_A(f \cdot g) &= \Sigma\varphi_A(f \mathbin{++} g) \\ &= \prod_{x \in f \mathbin{++} g} \varphi_A(x) \\ &= \prod_{x \in f} \varphi_A(x) \cdot \prod_{x \in g} \varphi_A(x) && \text{by associativity of the product} \\ &= \Sigma\varphi_A(f) \cdot \Sigma\varphi_A(g) \end{aligned}$$

– Commutation of the diagram: let  $x \in X$ . We have:

$$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{SG}}(x)) &= \Sigma\varphi_A([x]) \\ &= \varphi_A(x) \end{aligned}$$

– Uniqueness: let  $h : F_{\mathbf{SG}}X \rightarrow A$  be a morphism of  $\mathbf{SG}$ -models such that the diagram

above commutes. We want to show that  $h = \Sigma\varphi_A$ . Let  $l \in TX$ . We have:

$$\begin{aligned}
 h(l) &= h\left(\prod_{x \in l} \eta_{\mathbf{SG}}(x)\right) \\
 &= \prod_{x \in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism} \\
 &= \prod_{x \in l} \varphi_A(x) && \text{since the diagram commutes} \\
 &= \Sigma\varphi_A(l)
 \end{aligned}$$

Therefore we proved that  $F_{\mathbf{SG}}X$  is a free  $\mathbf{SG}$ -model over  $X$ .

□

<sup>a</sup>product with the operator  $\cdot$ .

We'll also define a multiplication  $\mu_{\mathbf{SG}}$  so that we have a monad structure  $(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$  on  $\mathbf{Set}$ . For all  $l \in TTX$ , we pose

$$\mu_{\mathbf{SG}}(l) = \#_{x \in l} x$$

where the concatenation is taken in the order of the list  $l$ , which is well defined because the list  $l$  is finite, non-empty, and the concatenation is associative.

### Proposition 83

$(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$  is a monad on  $\mathbf{Set}$ .

#### Proof

We need to check the monad axioms:

- Left unit: let  $l \in TX$ . We have:

$$\begin{aligned}
 \mu_{\mathbf{SG}}(\eta_{\mathbf{SG}}T(l)) &= \mu_{\mathbf{SG}}([l]) \\
 &= \#_{x \in [l]} x \\
 &= l
 \end{aligned}$$

- Right unit: let  $l \in TX$ . We have:

$$\begin{aligned}
 \mu_{\mathbf{SG}}(T\eta_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}([ [x_i] \mid x_i \in l ]) \\
 &= \#_{x \in [ [x_i] \mid x_i \in l ]} x \\
 &= \#_{x_i \in l} [x_i] \\
 &= l
 \end{aligned}$$

- Associativity: let  $l \in TTTX$ . We have:

$$\begin{aligned}
 \mu_{\mathbf{SG}}(T\mu_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}([ \#_{x \in l_i} x \mid l_i \in l ]) \\
 &= \#_{x \in [ \#_{x \in l_i} x \mid l_i \in l ]} x \\
 &= \#_{l_i \in l} \#_{x \in l_i} x \\
 &= \#_{x \in \#_{l_j \in l} l_j} x \\
 &= \#_{x \in l} x \\
 &= \mu_{\mathbf{SG}}(\mu_{\mathbf{SG}}T(l))
 \end{aligned}$$

□

## 2.2 Combining semigroups and commutative semigroups

We will call  $\mathbf{SG} \triangleleft \mathbf{CSG}$  the distributive combination of  $\mathbf{SG}$  and  $\mathbf{CSG}$ .

The signature of  $\mathbf{SG} \triangleleft \mathbf{CSG}$  has two operators:  $(\cdot)$  of arity 2, and  $(+)$  of arity 2, and the axioms are the union of the axioms of  $\mathbf{SG}$  and  $\mathbf{CSG}$ :

$$\begin{aligned} a, b, c &\vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c \\ a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \end{aligned}$$

We also have the distributivity axioms:

$$\begin{aligned} a, b, c &\vdash a \cdot (b + c) = a \cdot b + a \cdot c && \text{left distributivity} \\ a, b, c &\vdash (a + b) \cdot c = a \cdot c + b \cdot c && \text{right distributivity} \end{aligned}$$

We'll try to combine the two previous examples to hopefully get the free algebra for the presentation.

The carrier set of our candidate for the free  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model over  $X$  is  $STX$ .

We will keep the same definition for the operator  $+$  as in the free commutative semigroup, but we'll need to change the definition of the operator  $\cdot$  to take into account the fact that we are now working with  $STX$  instead of  $TX$ .

If we write  $(p : n) \in f$  when  $f(p) = n$ , then what we want is

$$f \cdot g = \sum_{(p:n) \in f} \sum_{(q:m) \in g} (p \cdot_{\mathbf{SG}} q : n \cdot m)$$

Formally, this can be defined for all  $f, g \in STX$  as

$$f \cdot g = \lambda l : TX. \sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} f(l_1) \cdot g(l_2)$$

We define  $\eta_{\mathbf{SG} \triangleleft \mathbf{CSG}} : X \rightarrow STX = \eta_{\mathbf{CSG}} \circ \eta_{\mathbf{SG}}$ . Let's first prove that the following diagram commutes:

$$\begin{array}{ccccc} X & \xrightarrow{\eta_{\mathbf{SG}}} & TX & \xrightarrow{\eta_{\mathbf{CSG}}^T} & STX \\ & \searrow \eta_{\mathbf{CSG}} & & \nearrow S\eta_{\mathbf{SG}} & \\ & & SX & & \end{array}$$

Let  $x \in X$ . We have:

$$\begin{aligned} (S\eta_{\mathbf{SG}})(\eta_{\mathbf{CSG}}(x)) &= S\eta_{\mathbf{SG}} \left( \lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \\ &= \eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x)) \end{aligned}$$

We pose  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X = (STX, (\cdot), (+))$ , where  $(\cdot)$  and  $(+)$  are defined above.

### Proposition 84

$F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  is a free  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model over  $X$ .

**Proof**

▷ Let's first check that  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  is indeed a  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model.

The associativity and commutativity of  $(+)$  was already proven for  $F_{\mathbf{CSG}} X$ .

For the associativity of  $(\cdot)$ : let  $f, g, h \in STX$ . We have:

$$\begin{aligned}
 (f \cdot g) \cdot h &= \lambda l : TX. \sum_{l=l_1 \cdot \mathbf{SG} l_2} (f \cdot g)(l_1) \cdot h(l_2) \\
 &= \lambda l : TX. \sum_{l=l_1 \cdot \mathbf{SG} l_2} \sum_{l_1=l_{11} \cdot \mathbf{SG} l_{12}} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) \\
 &= \lambda l : TX. \sum_{l=l_{11} \cdot \mathbf{SG} (l_{12} \cdot \mathbf{SG} l_2)} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) && \text{by associativity of } \cdot \mathbf{SG} \\
 &= \lambda l : TX. \sum_{l=l_{11} \cdot \mathbf{SG} l_3} \sum_{l_3=l_{12} \cdot \mathbf{SG} l_2} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) \\
 &= \lambda l : TX. \sum_{l=l_{11} \cdot \mathbf{SG} l_3} f(l_{11}) \cdot (g \cdot h)(l_3) \\
 &= f \cdot (g \cdot h)
 \end{aligned}$$

The distributivity of  $(\cdot)$  over  $(+)$  comes from the distributivity of the product in  $\mathbb{N}$  over addition, and from the definition of  $(+)$ .

Therefore,  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  is a  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model.

▷ Let  $A$  be a  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model over  $X$  with environment  $\varphi_A : X \rightarrow UA$ . We want to show that there exists a unique morphism  $h$  of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  to  $A$  such that the following diagram commutes:

$$\begin{array}{ccc}
 X & \xrightarrow{\eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}} & STX \\
 & \searrow \varphi_A & \downarrow h \\
 & & UA
 \end{array}$$

We define  $\Sigma\varphi_A : STX \rightarrow \underline{A}$  for all  $f \in STX$  by

$$\Sigma\varphi_A(f) = \sum_{l \in TX} f \cdot l \cdot \prod_{x \in l} \varphi_A(x)$$

Let's show that  $\Sigma\varphi_A$  is the unique morphism  $h$  of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  to  $A$  above. We need to check that  $\Sigma\varphi_A$  preserves the operators  $(\cdot)$  and  $(+)$ , and that the diagram above commutes.

– Preservation of the operator  $(\cdot)$ : let  $f, g \in STX$ . We have:

$$\begin{aligned}
 \Sigma\varphi_A(f \cdot g) &= \Sigma\varphi_A(\lambda l : TX. \sum_{l=l_1 \cdot \mathbf{SG} l_2} f(l_1) \cdot g(l_2)) \\
 &= \sum_{l \in TX} \left( \sum_{l=l_1 \cdot \mathbf{SG} l_2} f(l_1) \cdot g(l_2) \right) \cdot \prod_{x \in l} \varphi_A(x) \\
 &= \sum_{l_1, l_2 \in TX} f(l_1) \cdot g(l_2) \cdot \prod_{x \in l_1 \cdot \mathbf{SG} l_2} \varphi_A(x) \\
 &= \sum_{l_1, l_2 \in TX} f(l_1) \cdot g(l_2) \cdot \prod_{x \in l_1} \varphi_A(x) \cdot \prod_{x \in l_2} \varphi_A(x) && \text{by associativity of the product} \\
 &= \sum_{l_1 \in TX} f(l_1) \cdot \prod_{x \in l_1} \varphi_A(x) \cdot \sum_{l_2 \in TX} g(l_2) \cdot \prod_{x \in l_2} \varphi_A(x) \\
 &= \Sigma\varphi_A(f) \cdot \Sigma\varphi_A(g)
 \end{aligned}$$

- Preservation of the operator (+): let  $f, g \in STX$ . We have:

$$\begin{aligned}
\Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda l : TX.f \ l + g \ l) \\
&= \sum_{l \in TX} (f \ l + g \ l) \cdot \prod_{x \in l} \varphi_A(x) \\
&= \sum_{l \in TX} f \ l \cdot \prod_{x \in l} \varphi_A(x) + \sum_{l \in TX} g \ l \cdot \prod_{x \in l} \varphi_A(x) \\
&= \Sigma\varphi_A(f) + \Sigma\varphi_A(g)
\end{aligned}$$

- Commutation of the diagram: let  $x \in X$ . We have:

$$\begin{aligned}
(\Sigma\varphi_A)(\eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}(x)) &= \Sigma\varphi_A(\eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x))) \\
&= \Sigma\varphi_A\left(\lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases}\right) \\
&= \sum_{y \in TX} \begin{cases} 1 \cdot \prod_{x' \in y} \varphi_A(x') & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \\
&= \prod_{x' \in \eta_{\mathbf{SG}}(x)} \varphi_A(x') \\
&= \varphi_A(x)
\end{aligned}$$

- Uniqueness: let  $h : F_{\mathbf{SG} \triangleleft \mathbf{CSG}}X \rightarrow A$  be a morphism of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models such that the diagram above commutes. We want to show that  $h = \Sigma\varphi_A$ . Let  $f \in STX$ . We have:

$$\begin{aligned}
h(f) &= h\left(\sum_{l \in TX} f \ l \cdot \prod_{x \in l} \eta_{\mathbf{SG}}(x)\right) \\
&= \sum_{l \in TX} f \ l \cdot \prod_{x \in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism} \\
&= \sum_{l \in TX} f \ l \cdot \prod_{x \in l} \varphi_A(x) && \text{since the diagram commutes} \\
&= \Sigma\varphi_A(f)
\end{aligned}$$

Therefore  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}}X$  is a free  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model over  $X$ .

□

We also define a distributive law  $\lambda : TSX \rightarrow STX$  for all  $l \in TSX$  by

$$\lambda([f_1, \dots, f_n]) = \sum_{(x_1 : n_1) \in f_1} \cdots \sum_{(x_n : n_n) \in f_n} \left( [x_1, \dots, x_n] : \prod_{i=1}^n n_i \right)$$

Formaly, this can be defined for all  $l \in TSX$  as

$$\lambda(l) = \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y, f) \in x * l} f(y) & \text{otherwise} \end{cases}$$

where  $x * l$  is the elementwise pairing of the list  $x$  and the list  $l$ , which is well defined because  $x$  and  $l$  have the same length.

### Proposition 85

$\lambda$  is a distributive law of the monad  $T$  over the monad  $S$ .

#### Proof

We need to check the following axioms:

- $\lambda \circ T\eta_S = \eta_S T$ : let  $l \in TX$ . We have:

$$\begin{aligned}
(\lambda \circ T\eta_S)(l) &= \lambda(T\eta_S(l)) \\
&= \lambda([\eta_S(x) \mid x \in l]) \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,f) \in x * [\eta_S(x) \mid x \in l]} f(y) & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,z) \in x * l} \eta_S(z)(y) & \text{otherwise} \end{cases} \\
&= \eta_S T(l)
\end{aligned}$$

- $\lambda \circ \eta_T S = S\eta_T$ : let  $f \in SX$ . We have:

$$\begin{aligned}
(\lambda \circ \eta_T S)(f) &= \lambda(\eta_T S(f)) \\
&= \lambda([f]) \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq 1 \\ \prod_{(y,g) \in x * [f]} g(y) & \text{otherwise} \end{cases} \\
&= S\eta_T(f)
\end{aligned}$$

- $\lambda \circ T\mu_S = \mu_S T \circ S\lambda \circ \lambda S$ : let  $l = [f_1, \dots, f_n] \in TSSX$ . We have:

$$\begin{aligned}
(\lambda \circ T\mu_S)(l) &= \lambda(T\mu_S(l)) \\
&= \lambda([\mu_S(f_i) \mid i = 1, \dots, n]) \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^n \mu_S(f_i)(x_i) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^n \sum_{g \in SX} f_i(g) \cdot g(x_i) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} \sum_{g_1, \dots, g_n \in SX} \prod_{i=1}^n f_i(g_i) \cdot g_i(x_i) & x = [x_1, \dots, x_m] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

$$\begin{aligned}
(\mu_S T \circ S\lambda \circ \lambda S)(l) &= \mu_S T(S\lambda(\lambda S(l))) \\
&= \mu_S T \left( S\lambda \left( \lambda y : T S X. \begin{cases} \prod_{i=1}^n \sum_{g \in S X} f_i(g) \cdot g(y_i) & y = [y_1, \dots, y_n] \\ 0 & \text{otherwise} \end{cases} \right) \right) \\
&= \mu_S T \left( \lambda h : S T X. \sum_{g_1, \dots, g_n \in S X \wedge \lambda([g_1, \dots, g_n]) = h} \prod_{i=1}^n f_i(g_i) \cdot g_i(x_i) \right) \\
&= \lambda x : T X. \sum_{g_1, \dots, g_n \in S X} \left( \prod_{i=1}^n f_i(g_i) \right) \cdot \lambda([g_1, \dots, g_n])(x) \\
&= \lambda x : T X. \begin{cases} \sum_{g_1, \dots, g_n \in S X} \left( \prod_{i=1}^n f_i(g_i) \right) \cdot \left( \prod_{i=1}^n g_i(x_i) \right) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Therefore, we have  $(\lambda \circ T\mu_S)(l) = (\mu_S T \circ S\lambda \circ \lambda S)(l)$ .

- $\lambda \circ \mu_T S = S\mu_T \circ \lambda T \circ T\lambda$ : let  $l = [[f_{11}, \dots, f_{1m_1}], \dots, [f_{n1}, \dots, f_{nm_n}]] \in T T S X$ . We have:

$$\begin{aligned}
(\lambda \circ \mu_T S)(l) &= \lambda(\mu_T S(l)) \\
&= \lambda(\#_{i=1}^n [f_{i1}, \dots, f_{im_i}]) \\
&= \lambda x : T X. \begin{cases} \prod_{i=1}^n \prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = \#_{i=1}^n [x_{i1}, \dots, x_{im_i}] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

$$\begin{aligned}
(S\mu_T \circ \lambda T \circ T\lambda)(l) &= S\mu_T(\lambda T(T\lambda(l))) \\
&= S\mu_T(\lambda T([\lambda([f_{11}, \dots, f_{1m_1}]), \dots, \lambda([f_{n1}, \dots, f_{nm_n}])])) \\
&= S\mu_T \left( \lambda T \left( \left[ \lambda y : T X. \begin{cases} \prod_{j=1}^{m_1} f_{1j}(y_j) & y = [y_1, \dots, y_{m_1}] \\ 0 & \text{otherwise} \end{cases}, \dots, \lambda y : T X. \begin{cases} \prod_{j=1}^{m_n} f_{nj}(y_j) & y = [y_1, \dots, y_{m_n}] \\ 0 & \text{otherwise} \end{cases} \right] \right) \right) \\
&= S\mu_T \left( \lambda h : S T X. \sum_{i=1}^n \sum_{y_1, \dots, y_{m_i} \in T X} \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & h = \lambda y : T X. \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & y = [y_1, \dots, y_{m_i}] \\ 0 & \text{otherwise} \end{cases} \\ 0 & \text{otherwise} \end{cases} \right) \\
&= \lambda x : T X. \begin{cases} \sum_{i=1}^n \sum_{y_1, \dots, y_{m_i} \in T X} \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & y_i = [y_{i1}, \dots, y_{im_i}] \\ 0 & \text{otherwise} \end{cases} & x = \#_{i=1}^n [y_1, \dots, y_{m_i}] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : T X. \begin{cases} \prod_{i=1}^n \prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = \#_{i=1}^n [x_{i1}, \dots, x_{im_i}] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Therefore, we have  $(\lambda \circ \mu_T S)(l) = (S\mu_T \circ \lambda T \circ T\lambda)(l)$ .

□

**Corollary 86**

$(ST, \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}, \mu_{\mathbf{SG} \triangleleft \mathbf{CSG}})$  is a monad on **Set**, where

$$\mu_{\mathbf{SG} \triangleleft \mathbf{CSG}} : STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu S \mu T} ST$$

Another approach to get a monad structure on  $ST$  is to use monads in extension form.

[MW10] shows<sup>1</sup> that we can define an **SG**-algebra  $(STX, (-)^\lambda)$  where for all  $f : X \rightarrow STY$ , we have  $f^\lambda : TX \rightarrow STY$  defined by  $f^\lambda([x_1, \dots, x_n]) = \lambda([f(x_1), \dots, f(x_n)])$ . It is then showed that this **SG**-algebra gives a distributive law of the monad  $T$  over the monad  $S$ , and therefore a monad structure on  $ST$ .

In extensive form, the monad is  $(ST, \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}, (-)^{(ST)})$  where for all  $f : X \rightarrow STY$ , we have  $f^{(ST)} : STX \rightarrow STY$  defined by  $f^{(ST)} = (f^\lambda)^T$ .

We also have that  $\lambda = (S\eta_T)^\lambda$  as expected.

We now may want to see whether the monad we got from the distributive law is the same as the one we get from the free **SG**  $\triangleleft$  **CSG**-model construction.

Informally, we want to check whether the following diagram commutes (also see Definition 79):

$$\begin{array}{ccc}
 \mathbf{SG}, \mathbf{CSG} & \xrightarrow{\text{combining theory}} & \mathbf{SG} \triangleleft \mathbf{CSG} \\
 \downarrow \text{adding monad structure} & \text{?} & \downarrow \text{adding monad structure} \\
 \mathbf{Mon}(\mathbf{SG}), \mathbf{Mon}(\mathbf{CSG}) & \xrightarrow{\text{distributive law}} & \mathbf{Mon}(\mathbf{SG} \triangleleft \mathbf{CSG})
 \end{array}$$

The classical monadic structure on  $\mathbf{SG} \triangleleft \mathbf{CSG}$  comes from the adjunction between **Set** and  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models given by the free  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model construction.

**Lemma 87**

$F_{\mathbf{SG} \triangleleft \mathbf{CSG}} \dashv U$  is an adjunction, where  $U : \mathbf{SG} \triangleleft \mathbf{CSG}\text{-Mod} \rightarrow \mathbf{Set}$  is the forgetful functor.

**Proof**

We need to show that for all  $X \in \mathbf{Set}$  and all  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model  $A$ , there is a natural bijection between  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model morphisms from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}}X$  to  $A$  and functions from  $X$  to  $U A$ .

Let  $X \in \mathbf{Set}$  and let  $A = (\underline{A}, \cdot_A, +_A)$  be a  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model. We have already shown that there is a unique morphism of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}}X$  to  $A$  such that the diagram below commutes:

$$\begin{array}{ccc}
 X & \xrightarrow{\eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}} & STX \\
 \searrow \varphi_A & & \downarrow \Sigma \varphi_A \\
 & & \underline{A}
 \end{array}$$

Therefore, we have a function  $\Phi_{X,A}$  from  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model morphisms from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}}X$  to  $A$  to functions from  $X$  to  $U A$  defined by  $\Phi_{X,A}(h) = h \circ \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}$ .

Conversely, we have a function  $\Psi_{X,A}$  from functions from  $X$  to  $U A$  to  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model morphisms from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}}X$  to  $A$  defined by  $\Psi_{X,A}(\varphi_A) = \Sigma \varphi_A$ .

We need to show that  $\Phi_{X,A}$  and  $\Psi_{X,A}$  are inverses of each other.

<sup>1</sup>example 8.5



Let  $h : F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X \rightarrow A$  be a morphism of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models. We have:

$$\begin{aligned} (\Psi_{X,A} \circ \Phi_{X,A})(h) &= \Psi_{X,A}(h \circ \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}) \\ &= \Sigma(h \circ \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}) \\ &= h \end{aligned}$$

Let  $\varphi_A : X \rightarrow U A$  be a function. We have:

$$\begin{aligned} (\Phi_{X,A} \circ \Psi_{X,A})(\varphi_A) &= \Phi_{X,A}(\Sigma \varphi_A) \\ &= \Sigma \varphi_A \circ \eta_{\mathbf{SG} \triangleleft \mathbf{CSG}} \\ &= \varphi_A \end{aligned}$$

Therefore,  $\Phi_{X,A}$  and  $\Psi_{X,A}$  are inverses of each other, and we have a natural bijection between  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model morphisms from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  to  $A$  and functions from  $X$  to  $U A$ .  $\square$

This adjunction gives us a monad structure on  $ST$ .

### Theorem 88

The monad structure on  $ST$  given by the adjunction  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} \dashv U$  is the same as the one given by the distributive law  $\lambda$ .

### Proof

We'll consider the extensive form of the monads.

#### ▷ Carrier sets.

Let  $X \in \mathbf{Set}$ .  $UF_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  is the carrier set of the free  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model over  $X$ , which is  $STX$ . The carrier set of the monad given by the distributive law is also  $STX$ . Therefore, the carrier sets are the same.

#### ▷ Unit.

The unit is the same for both monads, sending  $x \in X$  to the sum of the singleton list  $[x]$  in  $STX$ .

#### ▷ Extension.

Let  $f : X \rightarrow STY$ . Since  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} Y$  is a model of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ , freeness of  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  gives us a unique morphism of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models from  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} X$  to  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} Y$  such that the diagram below commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\mathbf{SG} \triangleleft \mathbf{CSG}}} & STX \\ & \searrow f & \downarrow \Sigma f \\ & & STY \end{array}$$

Therefore, the extension of  $f$  in the monad given by the adjunction is  $\Sigma f$ . The extension of  $f$  in the monad given by the distributive law is  $f^{(ST)} = (f^\lambda)^T$ . We need to show that  $\Sigma f = (f^\lambda)^T$ .

Let  $f : X \rightarrow STY$ . Suppose that

$$f(x_k) = \sum_j u_{k,j}$$

for some  $u_{k,j} \in TY$ . Then

$$\Sigma f([x_1, \dots, x_n]) = \sum_{j_1, \dots, j_n} u_{1,j_1} \cdots u_{n,j_n}$$

On the other hand, we have:

$$\begin{aligned} f^\lambda([x_1, \dots, x_n]) &= \lambda([f(x_1), \dots, f(x_n)]) \\ &= \lambda([\sum_{j_1} u_{1,j_1}, \dots, \sum_{j_n} u_{n,j_n}]) \\ &= \sum_{j_1, \dots, j_n} u_{1,j_1} \cdots u_{n,j_n} \end{aligned}$$

Therefore, we have  $f^\lambda([x_1, \dots, x_n]) = \Sigma f([x_1, \dots, x_n])$  for all  $[x_1, \dots, x_n] \in TX$ .

We can apply  $(-)^T$  to both sides of the equation  $f^\lambda = \Sigma f$  to get  $(f^\lambda)^T = (\Sigma f)^T$ . We have  $(\Sigma f)^T = \Sigma f$  because  $\Sigma f$  is a morphism of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -models, and therefore preserves the operators of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ . Therefore, we have  $f^{(ST)} = (f^\lambda)^T = \Sigma f$ .

This shows that the monad structure on  $ST$  given by the adjunction  $F_{\mathbf{SG} \triangleleft \mathbf{CSG}} \dashv U$  is the same as the one given by the distributive law  $\lambda$ .  $\square$

## 2.3 Monoids

Now that we have the free semigroup, we can easily construct the free monoid by adding a new constant (arity 0 operator) to the signature, and adding the corresponding axioms.

### 2.3.1 Free commutative monoids

Let  $X$  be a set. We consider the theory of commutative monoids  $\mathbf{CM}$ , with a signature  $\Sigma$  with two operators:  $(+)$  of arity 2, and  $0$  of arity 0, and with the three axioms

$$\begin{aligned} a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \\ a &\vdash a + 0 = a = 0 + a \end{aligned}$$

We pose<sup>2</sup>

$$SX = \{f : X \rightarrow \mathbb{N} \mid f \text{ has finite support}\}$$

There is no non-empty support condition as in the free commutative semigroup, which means that the function  $f$  can be the constant 0 function. This corresponds to the new constant 0 in the signature.

We pose

$$\mathbf{0} = \lambda x. 0 \in SX$$

.

We keep the same definition for the operator  $+$  and for  $\eta_{\mathbf{CM}}$  as in the free commutative semigroup.

We pose  $F_{\mathbf{CM}}X = (SX, (+), \mathbf{0})$  where  $+$  and  $\eta_{\mathbf{CM}}$  are defined as above.

We want to show that  $F_{\mathbf{CM}}X$  is a free  $\mathbf{CM}$ -model over  $X$ .

- ▷  $F_{\mathbf{CM}}X$  is a  $\mathbf{CM}$ -model: the axioms of  $\mathbf{CM}$  are valid in  $F_{\mathbf{CM}}X$  because of the commutativity and associativity of addition in  $\mathbb{N}$ , and because of the existence of the constant function  $\mathbf{0}$  which acts as the identity for  $+$ .
- ▷ Let  $A = (\underline{A}, +_A, 0_A)$  be a  $\mathbf{CM}$ -model over  $X$  with environment  $\varphi_A : X \rightarrow U A$ . We define  $\Sigma \varphi_A : SX \rightarrow \underline{A}$  for all  $f \in SX$  by

$$\Sigma \varphi_A(f) = \sum_{x \in X} f(x) \cdot \varphi_A(x)$$

---

<sup>2</sup>we may want to find a different name, as it may be confused with the one for the commutative semigroup

We have  $\Sigma\varphi_A(\mathbf{0}) = 0_A$ .

For the sake of doing the proof at least once, we'll prove that  $\Sigma\varphi_A$  is well defined on  $SX \setminus \{\mathbf{0}\}$ .

### Proposition 89

For all  $f \in SX$ , the sum  $\sum_{x \in X} f \ x \cdot \varphi_A(x)$  is well defined.

#### Proof

Let  $f \in SX \setminus \{\mathbf{0}\}$ . We note  $\text{Supp } f = \{x \in X \mid f(x) \neq 0\}$ .

Since  $f$  has finite support,  $\text{Supp } f$  is a finite set. Moreover, since  $f \neq \mathbf{0}$ ,  $\text{Supp } f$  is non-empty. Therefore, we can write  $\text{Supp } f = \{x_1, x_2, \dots, x_n\}$  for some  $n \geq 1$  and we have a bijection  $\sigma$  between  $\{1, 2, \dots, n\}$  and  $\text{Supp } f$  given by  $i \mapsto x_i$ .

Therefore, we can rewrite the sum as follows:

$$\begin{aligned} \sum_{x \in X} f \ x \cdot \varphi_A(x) &= \sum_{x \in \text{Supp } f} f \ x \cdot \varphi_A(x) \\ &= \sum_{i=1}^n f \ \sigma(i) \cdot \varphi_A(\sigma(i)) \end{aligned}$$

The definition is valid because  $+_A$  is associative.

Let  $\sigma'$  be another bijection between  $\{1, 2, \dots, n\}$  and  $\text{Supp } f$ . Let's show that we get the same value for the sum if we use  $\sigma'$  instead of  $\sigma$ . This is true because  $\sigma$  and  $\sigma'$  are bijections, so there exists a permutation  $\pi$  of  $\{1, 2, \dots, n\}$  such that  $\sigma' = \sigma \circ \pi$ . Therefore, we have:

$$\begin{aligned} \sum_{i=1}^n f \ \sigma'(i) \cdot \varphi_A(\sigma'(i)) &= \sum_{i=1}^n f \ \sigma(\pi(i)) \cdot \varphi_A(\sigma(\pi(i))) \\ &= \sum_{i=1}^n f \ \sigma(i) \cdot \varphi_A(\sigma(i)) \quad \text{by commutativity of } +_A \end{aligned}$$

Therefore, the definition of  $\Sigma\varphi_A(f)$  does not depend on the choice of the bijection between  $\{1, 2, \dots, n\}$  and  $\text{Supp } f$ , and is well defined.

Now if  $f = \mathbf{0}$ , we have  $\text{Supp } f = \emptyset$ , and the sum is the empty sum, which is defined to be  $0_A$ .  $\square$

We can now prove that  $\Sigma\varphi_A$  is the unique morphism  $h$  of **CM**-models from  $F_{\mathbf{CM}}X$  to  $A$  such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\mathbf{CM}}} & SX \\ & \searrow \varphi_A & \downarrow h \\ & & U A \end{array}$$

– Preservation of the operator  $+$ : let  $f, g \in SX$ . We have:

$$\begin{aligned} \Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda x. f \ x + g \ x) \\ &= \sum_{x \in X} (f \ x + g \ x) \cdot \varphi_A(x) && \text{by definition} \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) + \sum_{x \in X} g \ x \cdot \varphi_A(x) && \text{since both } f \text{ and } g \text{ have finite support} \\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g) \end{aligned}$$

- Commutation of the diagram:  $\eta_{\mathbf{CM}} = \eta_{\mathbf{CSG}}$ , so the proof is the same as for the free commutative semigroup.
- Uniqueness: let  $h : F_{\mathbf{CM}}X \rightarrow A$  be a morphism of **CM**-models such that the diagram above commutes. We want to show that  $h = \Sigma\varphi_A$ . Let  $f \in SX$ . We'll show this lemma first:

**Lemma 90**

For all  $f \in SX$ , we have  $f = \sum_{x \in X} f \cdot x \cdot \eta_{\mathbf{CM}}(x)$ .

**Proof**

Let  $f \in SX$ . We have  $\text{Supp } f = \{x \in X \mid f(x) \neq 0\}$ . Since  $f$  has finite support,  $\text{Supp } f$  is a finite set. Moreover, since  $f$  is not necessarily non-zero,  $\text{Supp } f$  can be empty.

If  $\text{Supp } f$  is empty, then  $f = \mathbf{0}$  and we have:

$$\sum_{x \in X} f \cdot x \cdot \eta_{\mathbf{CM}}(x) = \sum_{x \in X} 0 \cdot \eta_{\mathbf{CM}}(x) = 0 = f$$

If  $\text{Supp } f$  is non-empty, we can write  $\text{Supp } f = \{x_1, x_2, \dots, x_n\}$  for some  $n \geq 1$ . We'll prove by induction on  $n$  that  $f = \sum_{x \in X} f \cdot x \cdot \eta_{\mathbf{CM}}(x)$ .

- \* If  $n = 1$ , then  $\text{Supp } f = \{x_1\}$  and we have  $f = f \cdot x_1 \cdot \eta_{\mathbf{CM}}(x_1) = \sum_{x \in X} f \cdot x \cdot \eta_{\mathbf{CM}}(x)$ .
- \* If  $n > 1$ , we have  $\text{Supp } f = \{x_1, x_2, \dots, x_n\}$ .

We can write  $f = f \cdot x_1 \cdot \eta_{\mathbf{CM}}(x_1) + f'$  where  $f' = \lambda x. \begin{cases} 0 & \text{if } x = x_1 \\ f \cdot x & \text{otherwise} \end{cases}$

We have  $\text{Supp } f' = \{x_2, \dots, x_n\}$ , so by the induction hypothesis, we have  $f' = \sum_{x \in X} f' \cdot x \cdot \eta_{\mathbf{CM}}(x)$ . Therefore, we have:

$$\begin{aligned} f &= f \cdot x_1 \cdot \eta_{\mathbf{CM}}(x_1) + f' \\ &= f \cdot x_1 \cdot \eta_{\mathbf{CM}}(x_1) + \sum_{x \in X} f' \cdot x \cdot \eta_{\mathbf{CM}}(x) \\ &= f \cdot x_1 \cdot \eta_{\mathbf{CM}}(x_1) + \sum_{x \in X} f \cdot x \cdot \eta_{\mathbf{CM}}(x) - f \cdot x_1 \cdot \eta_{\mathbf{CM}}(x_1) \quad \text{by definition of } f' \\ &= \sum_{x \in X} f \cdot x \cdot \eta_{\mathbf{CM}}(x) \end{aligned}$$

□

We then have:

$$\begin{aligned} h(f) &= h\left(\sum_{x \in X} f \cdot x \cdot \eta_{\mathbf{CM}}(x)\right) \\ &= \sum_{x \in X} f \cdot x \cdot h(\eta_{\mathbf{CM}}(x)) \\ &= \sum_{x \in X} f \cdot x \cdot \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma \varphi_A(f) \end{aligned}$$

Therefore we proved that  $F_{\mathbf{CM}}X$  is a free  $\mathbf{CM}$ -model over  $X$ .

## 2.4 Distributive combination of semigroups and commutative semigroups

Let's first start with a case of study. We'll take  $\mathcal{A}$  to be commutative semigroups and  $\mathcal{M}$  to be semigroups. We denote by  $(+)$  the operator on  $\mathcal{A}$ , and  $(\cdot)$  the operator on  $\mathcal{M}$ .

We define  $\mathcal{O}$  by  $\mathcal{O}_0 = \emptyset$  and  $\mathcal{O}_n = \{x_1, \dots, x_n \vdash x_1 \cdots x_n\}$  for  $n \geq 1$ . We will simply write  $\mathcal{O}_n = \{\mu_n\}$ .

**Proposition 91**

$\mathcal{O}$  is the smallest operad containing<sup>a</sup>  $(\cdot)$ .

<sup>a</sup>we may say that  $\mathcal{O}$  is the operad spanned by  $(\cdot)$

**Proof**▷  **$\mathcal{O}$  is an operad.**

associativity comes from the associativity of  $(\cdot)$  and the neutrality of the identity is trivial.

▷ **minimality condition.**

Let  $\mathcal{O}'$  be an operad containing  $(\cdot)$ . We want to show that  $\mathcal{O}$  is a sub-operad of  $\mathcal{O}'$ . We have  $\mu_2 \in \mathcal{O}'_2$  since  $\mathcal{O}'$  contains  $(\cdot)$ . We can then construct  $\mu_n$  for all  $n \geq 1$  by composition of  $\mu_2$  with itself, and using the identity for the case  $n = 1$ . Therefore, we have  $\mu_n \in \mathcal{O}'_n$  for all  $n \geq 1$ , which means that  $\mathcal{O}$  is a sub-operad of  $\mathcal{O}'$ .  $\square$

We then define a translation  $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$  by interpreting each operator as the  $n$ -ary multiplication, i.e  $\mathfrak{T}\mu_n(x_1, \dots, x_n) = x_1 \cdots x_n$ .

**Proposition 92**

$\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$  is a surjective translation.

**Proof**▷  **$\mathfrak{T}$  is a translation.**

- the identity condition is trivially true.
- the translation respects composition thanks to the associativity of  $(\cdot)$ .

▷  **$\mathfrak{T}$  is surjective.**

Let  $t \in \mathcal{M}_n$ . Since  $\mathcal{M}$  is the algebraic theory of semigroups,  $t$  is represented by a nonempty word in the variables  $x_1, \dots, x_n$ . Thus we may write

$$t = x_{i_1} x_{i_2} \cdots x_{i_m}$$

for some  $m \geq 1$  and indices  $i_j \in \{1, \dots, n\}$ .

Consider the unique  $m$ -ary operation  $\mu_m \in \mathcal{O}_m$ . By definition of the translation,

$$\mathfrak{T}(\mu_m) = x_1 x_2 \cdots x_m.$$

We define a renaming  $\rho : m \rightarrow n$  by

$$\rho(j) = i_j.$$

Applying this renaming to  $\mathfrak{T}(\mu_m)$  gives

$$\mathfrak{T}(\mu_m)[\rho] = (x_1 x_2 \cdots x_m)[\rho] = x_{\rho(1)} x_{\rho(2)} \cdots x_{\rho(m)} = x_{i_1} x_{i_2} \cdots x_{i_m} = t.$$

Therefore every term  $t \in \mathcal{M}_n$  is of the form  $t = \mathfrak{T}\mu_m[\rho]$  for some renaming  $\rho : m \rightarrow n$ .

Hence the translation  $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$  is surjective.  $\square$

Let's give an explicit description of the theory  $\mathfrak{T} \triangleleft \mathcal{A}$ .

- **Signature.** The signature  $\Sigma_{\mathfrak{T} \triangleleft \mathcal{A}}$  contains the operators of  $\mathcal{M}$  and the operators of  $\mathcal{A}$ . Therefore, it contains the operator  $(\cdot)$  of arity 2 and the operator  $(+)$  of arity 2.

- **Axioms.**

$$\begin{aligned}
& \mathcal{M} \text{ axioms} \\
& x, y, z \vdash x \cdot (y \cdot z) = (x \cdot y) \cdot z \\
& \mathcal{A} \text{ axioms} \\
& x, y \vdash x + y = y + x \\
& x, y, z \vdash x + (y + z) = (x + y) + z \\
& \text{distributivity axioms} \\
& x, y, z \vdash x \cdot (y + z) = x \cdot y + x \cdot z \\
& x, y, z \vdash (x + y) \cdot z = x \cdot z + y \cdot z
\end{aligned}$$

We define the multi-functor  $\_ : \mathbf{Multi}(\mathcal{T}, \mathcal{U}) \rightarrow \langle \mathcal{U}, \Pi \rangle$  such that the following diagram commutes:

$$\begin{array}{ccc}
\mathbf{Multi}(\mathcal{T}, \mathcal{U}) & \xrightarrow{\_} & \langle \mathcal{U}, \Pi \rangle \\
\downarrow \scriptstyle \_ \circ & & \downarrow \scriptstyle \_ \circ \\
\mathbf{Mod}(\mathcal{T}, \mathcal{U}) & \xrightarrow{U} & \mathcal{U}
\end{array}$$

This is the anaology in **Multicat** of the forgetful functor in **Cat**, i.e the forgetful multi-functor.

### Proposition 93

Let  $\mathcal{T}$  be a theory.  $\_ : \mathbf{Multi}(\mathcal{T}, \mathbf{Set}) \rightarrow \langle \mathbf{Set}, \Pi \rangle$  has a left adjoint if and only if  $\mathcal{T}$  is commutative.

#### Proof

See section 6 of Ohad's notes. □

Now we'll try to give an explicite description of each element of the following pullback square:

$$\begin{array}{ccc}
\mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set}) & \xrightarrow{\_ \circ \triangleleft \mathcal{A}} & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \\
\downarrow \scriptstyle \_ \mathcal{M} & & \downarrow \scriptstyle \mathbf{Mod}(\mathcal{O}, \_) \\
\mathbf{Mod}(\mathcal{M}, \mathbf{Set}) & \xrightarrow{\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})} & \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \Pi \rangle)
\end{array}$$

For vertices:

- $\mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$  is the category of  $\mathfrak{T} \triangleleft \mathcal{A}$ -models in **Set** (i.e with carrier sets in **Set**), i.e the category of sem-rgs.
- $\mathbf{Mod}(\mathcal{M}, \mathbf{Set})$  is the category of  $\mathcal{M}$ -models in **Set**, i.e the category of semigroups.
- $\mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \Pi \rangle)$  is the category of  $\mathcal{O}$ -models in  $\langle \mathbf{Set}, \Pi \rangle$ . This category coincides with the category of semigroups in **Set** (cf. example 9 of Ohad's notes).
- $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$  is the category of  $\mathcal{O}$ -models in  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ .

$\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$  is the multicategory where:

- objects are  $\mathcal{A}$ -models in **Set**, i.e commutative semigroups.
- multi-morphisms are multi- $\mathcal{A}$ -homomorphisms. It can be showed<sup>3</sup> that multi- $\mathcal{A}$ -homomorphisms are multi-linear.

Similarly to the previous case,  $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$  coincides with the category of sem-rgs in **Set**. The the distributivity axioms hold in  $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$  thanks to the multi-linearity of the multi-morphisms in  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ .

---

<sup>3</sup>maybe do it?

For arrows:

- $\neg_{\mathcal{O} \triangleleft \mathcal{A}}$  is such that for  $A = (UA, (+), (\cdot)) \in \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$ ,  $\underline{A}_{\mathcal{O} \triangleleft \mathcal{A}}$  consists of:
  - $\underline{A} = (UA, (+))$  the  $\mathcal{A}$ -model given by the interpretation of the operators of  $\mathcal{A}$  in  $A$ .
  - $\underline{A}_{\mathcal{O} \triangleleft \mathcal{A}} \llbracket \mu_n \rrbracket (x_1, \dots, x_n) = x_1 \cdot \dots \cdot x_n \in \underline{A}$
- $\neg_{\mathcal{M} \triangleleft}$  is the functor sending a  $\mathfrak{T} \triangleleft \mathcal{A}$ -model  $M$  to the  $\mathcal{M}$ -model in  $\mathbf{Set}$  given by the interpretation of the operators of  $\mathcal{M}$  in  $M$ , i.e the semigroup structure of  $M$ .

For  $A = (\underline{A}, (+), (\cdot)) \in \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$ , we have  $\underline{A}_{\mathcal{M} \triangleleft} = (\underline{A}, (\cdot))$ . It is easy to check that this is indeed a  $\mathcal{M}$ -model in  $\mathbf{Set}$  because the axioms of  $\mathcal{M}$  are included in the axioms of  $\mathfrak{T} \triangleleft \mathcal{A}$ .

- $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$  is defined in Proposition 60. We have

$$(\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) A) \llbracket f \rrbracket := A \llbracket \mathfrak{T} f \rrbracket \quad \mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) h := h$$

- $\mathbf{Mod}(\mathcal{O}, \neg)$  is defined in Proposition 62. It forgets about the model structure for  $\mathcal{A}$ , keeping only its carrier set, therefore we are left with a  $\mathcal{O}$ -model in  $\mathbf{Set}$ .

**Remark :**  $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$  is therefore an isomorphism of categories.

#### Proposition 94

The square above is a pullback square.

#### Proof

Informally, this is true because a model of  $\mathfrak{T} \triangleleft \mathcal{A}$  consists of a model of  $\mathcal{M}$  and a model of  $\mathcal{A}$  such that the operations of  $\mathcal{O}$  distribute over the operations of  $\mathcal{A}$ . Conversely, we can reconstruct a model for the combination with both those models.

We'll give a constructive proof anyway.

##### ▷ commutativity.

Let  $A \in \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$ .

We clearly have  $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(\underline{A}_{\mathcal{M} \triangleleft}) = \mathbf{Mod}(\mathcal{O}, \neg)(\underline{A}_{\mathcal{O} \triangleleft \mathcal{A}})$ , since both sides are equal to  $\underline{A}$ .

Moreover, for  $f \in \mathcal{O}_n$ , we have:

$$\begin{aligned} \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(\underline{A}_{\mathcal{M} \triangleleft}) \llbracket f \rrbracket &= \underline{A}_{\mathcal{M} \triangleleft} \llbracket \mathfrak{T} f \rrbracket && \text{by definition of } \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) \\ &= \underline{A}_{\mathcal{M} \triangleleft} \llbracket \mu_n \rrbracket && \text{since } \mathcal{O} \text{ is spanned by } \mu_n \\ &= \underline{A}_{\mathcal{O} \triangleleft \mathcal{A}} \llbracket \mu_n \rrbracket && \text{by definition of } \neg_{\mathcal{O} \triangleleft \mathcal{A}} \\ &= \mathbf{Mod}(\mathcal{O}, \neg)(\underline{A}_{\mathcal{O} \triangleleft \mathcal{A}}) \llbracket f \rrbracket && \text{by definition of } \mathbf{Mod}(\mathcal{O}, \neg) \end{aligned}$$

##### ▷ pullback.

Let  $\mathcal{C}$  be a category,  $F : \mathcal{C} \rightarrow \mathbf{Mod}(\mathcal{M}, \mathbf{Set})$  and  $G : \mathcal{C} \rightarrow \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$  be two functors such that  $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) \circ F = \mathbf{Mod}(\mathcal{O}, \neg) \circ G$ . We want to show that there exists a unique

functor  $H : \mathcal{C} \rightarrow \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$  such that the following diagram commutes:

$$\begin{array}{ccc}
 \mathcal{C} & \xrightarrow{G} & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \\
 \downarrow H & \searrow \cong_{\mathcal{O} \triangleleft \mathcal{A}} & \downarrow \mathbf{Mod}(\mathcal{O}, \perp) \\
 \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set}) & \xrightarrow{\quad} & \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \Pi \rangle) \\
 \downarrow \cong_{\mathcal{M}} & \searrow \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) & \\
 \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) & \xrightarrow{\quad} & 
 \end{array}$$

(Note: The diagram shows a commutative square with an additional arrow from  $\mathcal{C}$  to  $\mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \Pi \rangle)$  labeled  $F$ .)

We define  $H : \mathcal{C} \rightarrow \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$  as follows: for  $c \in \mathcal{C}$ , we have  $Hc = (UFc, (+_c), (\cdot_c))$  where:

- $UFc$  is the underlying set of the  $\mathcal{M}$ -model  $Fc$ .
- $(+_c)$  is the binary operation on  $UFc$  given by the interpretation of the operator  $(+)$  of  $\mathcal{A}$  in  $Gc$ , i.e  $(+_c) = Gc \llbracket + \rrbracket$ .
- $(\cdot_c)$  is the binary operation on  $UFc$  given by the interpretation of the operator  $(\cdot)$  of  $\mathcal{M}$  in  $Fc$ , i.e  $(\cdot_c) = Fc \llbracket \cdot \rrbracket$ .

We then have to check that  $Hc$  is indeed a  $\mathfrak{T} \triangleleft \mathcal{A}$ -model, i.e that the axioms of  $\mathfrak{T} \triangleleft \mathcal{A}$  hold in  $Hc$ . This is true because:

- the axioms of  $\mathcal{M}$  hold in  $Hc$  since  $Fc$  is a  $\mathcal{M}$ -model and the interpretation of the operators of  $\mathcal{M}$  in  $Hc$  is the same as the interpretation of the operators of  $\mathcal{M}$  in  $Fc$ .
- the axioms of  $\mathcal{A}$  hold in  $Hc$  since  $Gc$  is a  $\mathcal{O}$ -model in  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$  and the interpretation of the operators of  $\mathcal{A}$  in  $Hc$  is the same as the interpretation of the operators of  $\mathcal{A}$  in  $Gc$ .
- the distributivity axioms hold in  $Hc$  since  $Gc$  is a  $\mathcal{O}$ -model in  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$  and the interpretation of the operators of  $\mathcal{M}$  in  $Hc$  is the same as the interpretation of the operators of  $\mathcal{M}$  in  $Fc$ , and the interpretation of the operators of  $\mathcal{A}$  in  $Hc$  is the same as the interpretation of the operators of  $\mathcal{A}$  in  $Gc$ , and the distributivity axioms hold in  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ .

□

We can construct another pullback using the generic construction of Example 21. We can deduce an isomorphism

$$\mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{T}, \mathbf{Set}) \cong \langle \mathbf{Mod}(\mathcal{M}, \mathbf{Set}), \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \rangle$$

As in [HP06], we can obtain an universal characterisation of  $\mathfrak{T} \triangleleft \mathcal{A}$  as a pushout of theories:

$$\begin{array}{ccc}
 \mathfrak{T} \triangleleft \mathcal{A} & \longleftarrow & \mathcal{O} * \mathcal{A} \\
 \uparrow & \lrcorner & \uparrow \\
 \mathcal{M} & \longleftarrow & F\mathcal{O}
 \end{array}$$

## 2.5 Free algebras for the combination

We continue with the example of the previous section and we now try to construct the free algebra for  $\mathfrak{T} \triangleleft \mathcal{A}$  using the free algebras for  $\mathcal{M}$  and  $\mathcal{A}$ , similarly to Section 2.2, but in a more abstract way. We fix a set  $X$ .

We'll use the results of Section 2.1, and we write<sup>4</sup>  $F_{\mathcal{M}} = F_{\mathbf{SG}}$  and  $F_{\mathcal{A}} = F_{\mathbf{CSG}}$ .

<sup>4</sup>acutally that's not correct since we need to define functors



**Step 1.**

Let  $A_0 := F_{\mathcal{M}}X \models \mathcal{M}$  be the free  $\mathcal{M}$ -algebra over  $X$ .

$$\underline{A_0} = TX \quad A_0 \llbracket (\cdot) \rrbracket = (+) \quad \eta^{A_0}(x) = [x]$$

**Step 2.**

We apply the functor

$$\underline{\quad} := \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \rightarrow \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$$

to  $A_0$ . This gives  $A_1 := \underline{A_0}_{\mathfrak{T}} \models \mathcal{O}$ .

$$\underline{A_1} = \underline{A_0} = TX \quad A_1 \llbracket \mu_n \rrbracket (x_1, \dots, x_n) = x_1 + \dots + x_n \quad \eta_{A_0}^{A_1} = \text{id}$$

**Step 3.**

By Proposition 93, we have an adjunction

$$F_{\mathcal{A}} \dashv \underline{\quad} : \mathbf{Multi}(\mathcal{A}, \mathbf{Set}) \rightarrow \langle \mathbf{Set}, \prod \rangle$$

in **Multicat** since  $\mathcal{A}$  is commutative. Applying the 2-functor  $\mathbf{Mod}(\mathcal{O}, -) : \mathbf{Multicat} \rightarrow \mathbf{Cat}$  gives a left adjoint

$$F_{\mathcal{O} \rightarrow \mathcal{A}} : \mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) : \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle) \rightarrow \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$$

We pose  $A_2 := F_{\mathcal{O} \rightarrow \mathcal{A}} A_1 \models \mathcal{O} * \mathcal{A}$ .

$$\underline{A_2} = STX \quad \eta_{A_1}^{A_2}(x) = \sum_{w \in TX} \delta_x^w \cdot w$$

$F_{\mathcal{A}}$  extends the morphisms in  $\langle \mathbf{Set}, \prod \rangle$  multi-linearly. Therefore if  $a_i = \sum_{w \in TX} n_w^{(i)} \cdot w \in STX$ , then

$$\begin{aligned} A_2 \llbracket \mu_n \rrbracket (a_1, \dots, a_n) &= \sum_{w_1, \dots, w_n \in TX} (n_{w_1}^{(1)} \dots n_{w_n}^{(n)}) \cdot (A_1 \llbracket \mu_n \rrbracket (w_1, \dots, w_n)) \\ &= \sum_{w_1, \dots, w_n \in TX} (n_{w_1}^{(1)} \dots n_{w_n}^{(n)}) \cdot (w_1 + \dots + w_n) \end{aligned}$$

Lemma 33 of Ohad's notes then is just saying that any  $x \in STX$  can be written as

$$x = \sum_{i=1}^n w_i$$

for some  $(w_i) \in (TX)^n$ , which is no surprise.

**Step 4.**

Let  $A_3 := \mathbf{Mod}(\mathcal{O}, \underline{\quad}) A_2 \models \mathcal{O}$  be the underlying operadic algebra.

$$\underline{A_3} = \underline{A_2} = STX \quad A_3 \llbracket \mu_n \rrbracket = A_2 \llbracket \mu_n \rrbracket \quad \eta_{A_2}^{A_3} = \text{id}$$

For each  $t \in \mathcal{M}_n$ , we use surjectivity to get some  $f \in \mathcal{O}_m$  and  $\theta : m \rightarrow n$  such that  $t = \mathfrak{T}f[\theta]$ . Then  $A_{\mathcal{M}} \llbracket t \rrbracket e := A_3 \llbracket f \rrbracket e \circ \theta$ .

**Lemma 95**

With the preceding notation:

- the interpretation  $A_{\mathcal{M}} \llbracket t \rrbracket e$  is independent of the decomposition  $t = \mathfrak{T}f[\theta]$
- the resulting algebra validates  $\mathcal{M}$ , i.e.  $A_{\mathcal{M}} \models \mathcal{M}$

**Proof**

We will show that, given  $f_i \in \mathcal{O}_{n_i}, \theta_i : n_i \rightarrow m$ , if  $m \vdash_{\mathcal{M}} \mathfrak{T}f_1[\theta_1] = \mathfrak{T}f_2[\theta_2]$ , then for all  $e : [m] \rightarrow \underline{A}_{\mathcal{M}} = \underline{A}_3 = \underline{A}_2$ , we have

$$A_2 \llbracket f_1 \rrbracket (e \circ \theta_1) = A_2 \llbracket f_2 \rrbracket (e \circ \theta_2)$$

For  $r \in [m]$ , let  $e(r) = \sum_{w \in TX} n_w^r \cdot w$ . Then

$$\begin{aligned} A_2 \llbracket f_1 \rrbracket (e \circ \theta_1) &= A_2 \llbracket f_1 \rrbracket \langle e(\theta_1(j)) \rangle_{j=1}^m && \text{evaluation in an operadic algebra} \\ &= A_2 \llbracket f_1 \rrbracket \left\langle \sum_{w \in TX} n_w^{\theta_1(j)} \cdot w \right\rangle_{j=1}^m && \text{substitution} \\ &= \sum_{w_1, \dots, w_{n_1} \in TX} (n_{w_1}^{\theta_1(1)} \dots n_{w_{n_1}}^{\theta_1(m)}) \cdot (A_1 \llbracket f_1 \rrbracket (w_1, \dots, w_{n_1})) && \mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}})\text{-def} \\ &= \sum_{w_1, \dots, w_{n_1} \in TX} (n_{w_1}^{\theta_1(1)} \dots n_{w_{n_1}}^{\theta_1(m)}) \cdot (A_0 \llbracket \mathfrak{T}f_1 \rrbracket (w_1, \dots, w_{n_1})) && \neg\mathfrak{T}\text{-def} \\ &= \sum_{w_1, \dots, w_{n_2} \in TX} (n_{w_1}^{\theta_2(1)} \dots n_{w_{n_2}}^{\theta_2(m)}) \cdot (A_0 \llbracket \mathfrak{T}f_2 \rrbracket (w_1, \dots, w_{n_2})) && \text{by hypothesis} \\ &= \dots && \text{reverse steps} \\ &= A_2 \llbracket f_2 \rrbracket (e \circ \theta_2) \end{aligned}$$

This proves the first point. The second point is then a direct consequence of the first one and the fact that  $A_3 \models \mathcal{O}$ .  $\square$

At this point we managed to go from a model of  $\mathcal{M}$  with carrier set  $TX$  to a model of  $\mathcal{M}$  with carrier set  $STX$ .

$$\underline{A}_{\mathcal{M}} = \underline{A}_3 = \underline{A}_2 = STX \quad A_{\mathcal{M}} \llbracket m \vdash x_{i_1} \dots x_{i_n} \rrbracket e = A_3 \llbracket \mu_n \rrbracket (e \circ (j \mapsto i_j)) \quad \eta_{A_2}^{A_{\mathcal{M}}} = \text{id}$$

**Step 5.**

We have  $A_{\mathcal{M}} \models \mathcal{M}$  and  $A_2 \models \mathcal{O} * \mathcal{A}$ . We can then use the pullback square to get a model  $A := A_{\mathcal{M}} \triangleleft A_2$  of  $\mathfrak{T} \triangleleft \mathcal{A}$ :

- $\underline{A} = \underline{A}_{\mathcal{M}} = \underline{A}_2 = STX$
- $A \llbracket (\cdot) \rrbracket (x, y) = A_{\mathcal{M}} \llbracket (\cdot) \rrbracket (x, y) = A_2 \llbracket \mu_2 \rrbracket (x, y)$
- $A \llbracket (+) \rrbracket (x, y) = A_2 \llbracket (+) \rrbracket (x, y)$

All axioms hold since they hold either in  $A_{\mathcal{M}}$  or in  $A_2$ .

We define  $\eta : X \xrightarrow{\eta^{A_0}} \underline{A}_0 = \underline{A}_1 \xrightarrow{\eta_{A_1}^{A_2}} \underline{A}_2 = STX = \underline{A}$ .

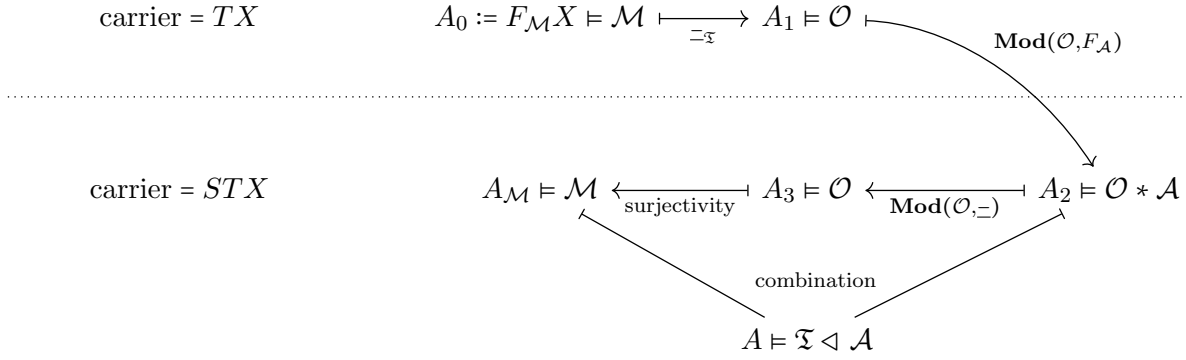
**Proposition 96**

$A$  is the free  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebra over  $X$ .

**Proof**

We need to show that  $(A, \eta)$  is an universal arrow from  $X$  to the forgetful functor  $\mathcal{A} \rightarrow \mathbf{Set}$ . Since the definitions for  $(+)$  and  $(\cdot)$  in  $A$  are the same as we defined in Section 2.2, the same proof as in Section 84 works here.  $\square$

Here is a little diagram that summarizes the steps:



## 2.6 Counter example of Lemma 95

### working point

Let  $\mathcal{M} = \mathbf{BA}$  be the theory of barycentric algebras,  $\mathcal{A} = \mathbf{SL}$  be the theory of join-semilattices,  $\mathcal{O}$  be the simplex operad and  $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$  be the usual translation sending a weight vector to the corresponding convex-combination term.

Formaly:

- $\mathbf{BA}$  consists of a binary operator  $\oplus_p$  for every  $p \in [0, 1]$  and the axioms

$$\begin{array}{ll}
 x \vdash x \oplus_p x = x & \text{idempotence} \\
 x, y \vdash x \oplus_p y = y \oplus_{1-p} x & \text{skew commutativity} \\
 x, y, z \vdash x \oplus_p (y \oplus_r z) = \left( x \oplus_{\frac{p}{p+(1-p)r}} y \right) \oplus_{p+(1-p)r} z & \text{skew associativity}
 \end{array}$$

- $\mathbf{SL}$  consists of a binary operator  $\vee$  and a constant  $\perp$ , along with the axioms

$$\begin{array}{ll}
 x \vdash x \vee \perp = x & \text{unit} \\
 x, y, z \vdash (x \vee y) \vee z = x \vee (y \vee z) & \text{associativity} \\
 x, y \vdash x \vee y = y \vee x & \text{commutativity} \\
 x \vdash x \vee x = x & \text{idempotence}
 \end{array}$$

- $\mathcal{O}$  is the operad where

$$\mathcal{O}_n = \left\{ \lambda = (\lambda_1, \dots, \lambda_n) \in [0, 1]^n \mid \sum_{i=1}^n \lambda_i = 1 \right\}$$

Composition is substitution of convex combinations, and the identity is  $(1) \in \mathcal{O}_1$ .

- $\mathfrak{T} : \mathcal{O} \rightarrow \mathbf{BA}$  sends a weighted vector  $\lambda \in \mathcal{O}_n$  to the corresponding barycentric term  $\sum_i \lambda_i x_i$ .

The free semilattice on a set  $X$  is  $F_{\mathbf{SL}} X = \mathcal{P}_f^+(X)$  the set of non-empty finite subsets of  $X$ , with union as join. to prove

The free barycentric algebra on  $X$  is  $F_{\mathbf{BA}} X = \mathcal{D}_f(X)$  the set of finitely-supported probability distributions on  $X$ . to prove

Let  $X = \{x, y\}$ . Following the pipeline of the previous section, let

$$\begin{aligned}
 A_0 &= F_{\mathbf{BA}}(X) = \mathcal{D}_f(X) \\
 A_1 &= \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})A_0 \\
 A_2 &= \mathbf{Mod}(\mathcal{O}, F_{\mathbf{SL}})A_1
 \end{aligned}$$

The carrier set of  $A_2$  is

$$\underline{A_2} = \mathcal{P}_f^+(\mathcal{D}_f(X))$$

the set of non-empty finite subsets of finitely supported probability distributions on  $X$ .

For  $p \in ]0, 1[$ , let  $f_p \in \mathcal{O}_2$  be the binary convex-combination operation of weight  $p$ .

The lifted operation on  $A_2$  is

$$A_2 \llbracket f_p \rrbracket \langle S, T \rangle = \{p\mu + (1-p)\nu \mid \mu \in S, \nu \in T\}$$

Considering the barycentric axiom  $x \oplus_p x = x$ , it is of the form

$$1 \vdash_{\mathbf{BA}} \mathfrak{T}f_p[\theta] = \mathfrak{T}\text{id}$$

where

- $\theta : 2 \rightarrow 1$  is the constant map
- $\text{id} \in \mathcal{O}_1$  is the unary identity operation

Take  $e := \{\delta_x, \delta_y\} \in \mathcal{P}_f^+(\mathcal{D}_f(X))$ .

$$A_2 \llbracket f_p \rrbracket \langle e, e \rangle = \{\delta_x, p\delta_x + (1-p)\delta_y, p\delta_y + (1-p)\delta_x, \delta_y\}$$

whereas

$$A_2 \llbracket \text{id} \rrbracket e = e = \{\delta_x, \delta_y\}$$

Since  $0 < p < 1$ ,  $p\delta_x + (1-p)\delta_y$  and  $p\delta_y + (1-p)\delta_x$  do not belong to  $e$ . Therefore

$$A_2 \llbracket f_p \rrbracket \langle e, e \rangle \neq A_2 \llbracket \text{id} \rrbracket e$$

and Lemma 95 does not hold.

It follows from this that the construction presented above does not yield the free algebra for the distributive combination of **BA** over **SL**.

## 2.7 General case

Steps 1 to 4 of the previous section can be adapted to the general case.

$\eta^{A_0}$  comes from the adjunction  $F_{\mathcal{M}} \dashv U : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \rightarrow \mathbf{Set}$  and  $\eta_{A_1}^{A_2}$  comes from the adjunction  $\mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) \dashv \mathbf{Mod}(\mathcal{O}, \_ ) : \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \rightarrow \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \Pi \rangle)$ .

We encounter a problem when trying to generalize Lemma 95: it doesn't hold anymore (counter example of Section 2.6). We need to replace it with another weaker lemma which limits which theories we can apply the construction to.

### Definition 97 ( $\mathfrak{T}$ -affine presentation)

Let  $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$  be a surjective translation.

A presentation of  $\mathcal{M}$  over  $\mathfrak{T}$  is  $\mathfrak{T}$ -affine if:

- for every operation symbol  $g : n$  of the chosen presentation of  $\mathcal{M}$ , there is a chosen operation  $\tilde{g} \in \mathcal{O}_n$  such that

$$n \vdash_{\mathcal{M}} \mathfrak{T}\tilde{g} = g(x_1, \dots, x_n),$$

- the chosen operations  $\tilde{g}$  generate  $\mathcal{O}$  under identities and operadic composition,
- every axiom of the presentation is of the form

$$x_1, \dots, x_n \vdash_{\mathcal{M}} \mathfrak{T}f_1[\theta_1] = \mathfrak{T}f_2[\theta_2],$$

for some  $f_1, f_2 \in \mathcal{O}_n$ , and injective renamings

$$\theta_1 : [n_1] \hookrightarrow [m] \quad \theta_2 : [n_2] \hookrightarrow [m]$$

**Lemma 98 (Step 4 in the  $\mathfrak{T}$ -affine case)**

With the preceding notation, assume that  $\mathcal{M}$  admits a  $\mathfrak{T}$ -affine presentation. Let

$$A_3 := \mathbf{Mod}(\mathcal{O}, -)(A_2).$$

For each operation symbol  $g : n$  of the chosen presentation of  $\mathcal{M}$ , define

$$A_{\mathcal{M}} \llbracket g \rrbracket := A_3 \llbracket \tilde{g} \rrbracket.$$

Then:

- for every  $f \in \mathcal{O}_n$ , the interpretation in  $A_{\mathcal{M}}$  of the  $\mathcal{M}$ -term  $\mathfrak{T}f$  is equal to  $A_3 \llbracket f \rrbracket$ ;
- $A_{\mathcal{M}} \models \mathcal{M}$ .

**Proof**

Let  $X := \underline{A_1}$  and let  $\eta_X : X \rightarrow \underline{F_{\mathcal{A}}X}$  be the unit of the adjunction  $F_{\mathcal{A}} \dashv U_{\mathcal{A}}$ .

By construction of

$$A_2 = \mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}})(A_1),$$

for every  $f \in \mathcal{O}_n$ , the operation  $A_2 \llbracket f \rrbracket$  is the unique multi- $A$ -homomorphism extending

$$X^n \xrightarrow{A_1 \llbracket f \rrbracket} X \xrightarrow{\eta_X} \underline{F_{\mathcal{A}}X}.$$

We first prove that, for every  $f \in \mathcal{O}_n$ , the interpretation in  $A_{\mathcal{M}}$  of the term  $\mathfrak{T}f$  is equal to  $A_3 \llbracket f \rrbracket$ . Since the chosen operations  $\tilde{g}$  generate  $\mathcal{O}$ , we may prove this by induction on the operadic expression of  $f$ .

- If  $f = \text{id}$ , the claim is immediate.
- If  $f = \tilde{g}$  is one of the chosen generators, the claim is exactly the definition of  $A_{\mathcal{M}} \llbracket g \rrbracket$ .
- If  $f = h \circ (f_1, \dots, f_k)$ , then, using preservation of operadic composition by  $\mathfrak{T}$ , the induction hypothesis, and the fact that  $A_3$  is an  $\mathcal{O}$ -model, we get

$$A_{\mathcal{M}} \llbracket \mathfrak{T}f \rrbracket = A_{\mathcal{M}} \llbracket \mathfrak{T}h[\mathfrak{T}f_1, \dots, \mathfrak{T}f_k] \rrbracket = A_3 \llbracket h \rrbracket \circ (A_3 \llbracket f_1 \rrbracket, \dots, A_3 \llbracket f_k \rrbracket) = A_3 \llbracket f \rrbracket.$$

Now let

$$x_1, \dots, x_m \vdash_{\mathcal{M}} \mathfrak{T}f_1[\theta_1] = \mathfrak{T}f_2[\theta_2]$$

be an axiom of the chosen  $\mathfrak{T}$ -affine presentation. For each  $i = 1, 2$ , define  $\Phi_i : X^m \rightarrow \underline{F_{\mathcal{A}}X}$  by

$$\Phi_i(e) := \eta_X(A_1 \llbracket f_i \rrbracket (e \circ \theta_i)).$$

Since  $A_0 \models \mathcal{M}$ , these two maps are equal:  $\Phi_1 = \Phi_2$ .

For each  $i = 1, 2$ , define  $\overline{\Phi_i} : \underline{F_{\mathcal{A}}X}^m \rightarrow \underline{F_{\mathcal{A}}X}$  by

$$\overline{\Phi_i}(e) := A_2 \llbracket f_i \rrbracket (e \circ \theta_i).$$

Because  $\theta_i$  is injective, each coordinate of  $e$  is used in at most one argument of  $A_2 \llbracket f_i \rrbracket$ , and therefore  $\overline{\Phi_i}$  is a multi- $\mathcal{A}$ -homomorphism in each variable. By construction, it extends  $\Phi_i$ .

By uniqueness of multi- $\mathcal{A}$ -homomorphic extension from generators, we conclude that

$$\overline{\Phi_1} = \overline{\Phi_2}.$$

Hence, for every environment  $e : [m] \rightarrow \underline{A}_2$ ,

$$A_2 \llbracket f_1 \rrbracket (e \circ \theta_1) = A_2 \llbracket f_2 \rrbracket (e \circ \theta_2).$$

Forgetting from  $A_2$  to  $A_3$  preserves this equality, so the two sides of the axiom have the same interpretation in  $A_{\mathcal{M}}$ . Hence all axioms of the chosen presentation hold in  $A_{\mathcal{M}}$ , and therefore  $A_{\mathcal{M}} \models \mathcal{M}$ .  $\square$

### Step 5.

We define  $A \models \mathfrak{T} \triangleleft \mathcal{A}$  the same way as in the example of Section 2.5, i.e as the pullback of  $A_{\mathcal{M}}$  and  $A_2$ . Let's prove that it is free.

Let  $B$  be any sem-rg algebra and  $f : X \rightarrow \underline{B}$  be a function.

$A_0$  is the free  $\mathcal{M}$ -algebra over  $X$ , so we have a unique morphism  $h_0 : A_0 \rightarrow B_{\mathcal{M}}$  of  $\mathcal{M}$ -algebras such that  $h_0 \circ \eta^{A_0} = f$ .

$$\begin{array}{ccc} X & \xrightarrow{\eta^{A_0}} & A_0 \\ & \searrow f & \downarrow h_0 \\ & & B_{\mathcal{M}} \end{array}$$

There is also an homomorphism  $h_1 : A_1 \rightarrow B_{\mathcal{O}}$  of  $\mathcal{O}$ -algebras such that  $h_1 \circ \eta_{A_0}^{A_1} = h_0$ , and  $h_1 = h_0$  since  $\eta_{A_0}^{A_1} = \text{id}$ .

$$\begin{array}{ccc} X & \xrightarrow{\text{id} \circ \eta^{A_0}} & A_1 \\ & \searrow f & \downarrow h_1 = h_0 \\ & & B_{\mathcal{O}} \end{array}$$

$B$  has an  $\mathcal{A}$ -structure that distributes over  $\mathcal{O}$ , therefore thanks to the adjunction  $\mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) \dashv \mathbf{Mod}(\mathcal{O}, \underline{\phantom{x}})$ , we have a unique morphism  $h_2 : A_2 \rightarrow B$  of  $\mathcal{O} * \mathcal{A}$ -algebras such that  $h_2 \circ \eta_{A_1}^{A_2} = h_1$ .

$$\begin{array}{ccc} A_1 & \xrightarrow{\eta_{A_1}^{A_2}} & A_2 \\ & \searrow h_1 & \downarrow h_2 \\ & & B_{\mathcal{O} * \mathcal{A}} \end{array}$$

Apply the 2-functor  $\mathbf{Mod}(\mathcal{O}, \underline{\phantom{x}})$  to the morphism  $h_2 : A_2 \rightarrow B$  gives a morphism  $h_3 : A_3 \rightarrow B_{\mathcal{O}}$  of  $\mathcal{O}$ -algebras such that  $h_3 \circ \eta_{A_2}^{A_3} = h_2$  and  $h_3 = h_2$  since  $\eta_{A_2}^{A_3} = \text{id}$ .

By the surjectivity of  $\mathfrak{T}$  and the independance of the decomposition given by Lemma 98, we now have a  $\mathcal{M}$ -morphism  $h_4 = h_3 : A_{\mathcal{M}} \rightarrow B_{\mathcal{M}}$  such that  $h_4 \circ \eta_{A_3}^{A_{\mathcal{M}}} = h_3$ .

### Proof

Let  $t \in \mathcal{M}_n$  and  $e : [n] \rightarrow \underline{A_{\mathcal{M}}}$ . By surjectivity, there is some  $f \in \mathcal{O}_m$  and  $\theta : m \rightarrow n$  such that  $t = \mathfrak{T}f[\theta]$ . We define  $h_4$  as

$$h_4(A_{\mathcal{M}} \llbracket t \rrbracket e) = h_3(A_3 \llbracket f \rrbracket e \circ \theta)$$

This is well-defined by Lemma 98. We then check that  $h_4$  is a morphism of  $\mathcal{M}$ -algebras. Since  $h_3$  is a morphism of  $\mathcal{O}$ -algebras,

$$\begin{aligned} h_3(A_3 \llbracket f \rrbracket e \circ \theta) &= B_{\mathcal{O}} \llbracket f \rrbracket (h_3 \circ e \circ \theta, \dots, h_3 \circ e \circ \theta) \\ &= B_{\mathcal{M}} \llbracket f \rrbracket (h_4 \circ e, \dots, h_4 \circ e) \end{aligned}$$

This proves that  $h_4(A_{\mathcal{M}} \llbracket t \rrbracket e) = B_{\mathcal{M}} \llbracket t \rrbracket (h_4 \circ e)$ , so  $h_4$  is a morphism of  $\mathcal{M}$ -algebras<sup>a</sup>.  $\square$

<sup>a</sup>I should really work on notations before writing the report as the current ones are really hard to read and don't really make sense sometimes.

In the end we have the following diagram:

$$\begin{array}{ccccc}
 X & \xrightarrow{\eta^{A_0}} & A_0 & \xrightarrow{\text{id}} & A_1 \\
 & \searrow f & \downarrow h_0 & \swarrow h_1 & \downarrow \eta_{A_1}^{A_2} \\
 & & B & \xleftarrow{h_2} & A_2 \\
 & & \uparrow h_4 & \swarrow h_3 & \downarrow \text{id} \\
 & & A_4 & \xleftarrow{\text{id}} & A_3
 \end{array}$$

We then consider the walking morphism  $I : \{0 \xrightarrow{u} 1\}$  and the two functors  $F_1 : I \rightarrow \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$  and  $F_2 : I \rightarrow \mathbf{Mod}(\mathcal{M}, \mathbf{Set})$  defined by  $F_1 0 = A_2$ ,  $F_1 1 = B_{\mathcal{O} \star \mathcal{A}}$ ,  $F_1 u = h_2$  and  $F_2 0 = A_{\mathcal{M}}$ ,  $F_2 1 = B_{\mathcal{M}}$ ,  $F_2 u = h_4$ . By the construction, we can then use the pullback square to get a unique  $H : I \rightarrow \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A})$  such that the following diagram commutes:

$$\begin{array}{ccc}
 I & \xrightarrow{F_1} & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \\
 \searrow H & & \downarrow \mathbf{Mod}(\mathcal{O}, \neg) \\
 & \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set}) & \xrightarrow{\neg \triangleleft \mathcal{A}} \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \\
 \searrow F_2 & \downarrow \neg_{\mathcal{M}} & \downarrow \mathbf{Mod}(\mathcal{O}, \neg) \\
 & \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) & \xrightarrow{\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})} \mathbf{Mod}(\mathcal{O}, (\mathbf{Set}, \Pi))
 \end{array}$$

This gives us a unique morphism  $h = H u : A \rightarrow B$  of  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebras.

We need to show that  $h$  extends  $f$  along  $\eta$ .  $h_2 \circ \eta = h_4 \circ \eta = f$ , therefore  $h \circ \eta = f$  since  $\eta$  comes from the pullback square.

This morphism is unique for a given  $h_2$  and  $h_4$ , but these are also uniquely determined by  $h_0$  and  $h_1$ , which are themselves uniquely determined by  $f$ . Therefore  $h$  is the unique morphism of  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebras such that the following diagram commutes:

$$\begin{array}{ccc}
 X & \xrightarrow{\eta} & A \\
 & \searrow f & \downarrow h \\
 & & B
 \end{array}$$

This proves that  $A$  is the free  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebra over  $X$ .

Now that we proved that  $A$  is the free  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebra over  $X$ , we can give an explicit description of the unique morphism  $h : A \rightarrow B$  of  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebras such that  $h \circ \eta = f$ .

Since we cannot access the internal structure anymore, we have the following lemma:

### Lemma 99

For every  $e \in A_2$ , there is some  $s \in \mathcal{A}_k$  and  $\mathbf{e} \in A_1^k$  such that

$$e = A_2 \llbracket s \rrbracket \langle \eta_{A_1}^{A_2} \mathbf{e}_i \rangle_{i=1}^k$$

### Proof

This is proved in Ohad's notes.  $\square$

We can actually prove a stronger statement:

**Lemma 100**

For all  $e \in \underline{A_2}$ , there exists  $s \in \mathcal{A}_k$  such that for all  $i \in [k]$ , there exists  $t_i \in \mathcal{M}_{k_i}$  and  $\mathbf{e}_{\langle i,j \rangle} \in X$  such that:

$$e = A_2 \llbracket s \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle} \right\rangle_{j=1}^{k_i} \right\rangle_{i=1}^k$$

**Proof**

Similar to the proof of Lemma 99. □

Let  $e \in \underline{A} = \underline{A_2}$ . By Lemma 100, we can write

$$e = A_2 \llbracket s \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle} \right\rangle_{j=1}^{k_i} \right\rangle_{i=1}^k$$

We can then define  $h : X \rightarrow \underline{B}$  by

$$h e = B \llbracket s \rrbracket \left\langle B \llbracket t_i \rrbracket \left\langle f \mathbf{e}_{\langle i,j \rangle} \right\rangle_{j=1}^{k_i} \right\rangle_{i=1}^k$$

**Lemma 101**

$h$  is well-defined, i.e. it does not depend on the choice of  $s, t_i$  and  $\mathbf{e}_{\langle i,j \rangle}$ .

**Proof**

Suppose

$$e = A_2 \llbracket s_1 \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_{1,k} \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle 1,k,j \rangle} \right\rangle_{j=1}^{k_{1,k}} \right\rangle_{k=1}^{n_1}$$

and

$$e = A_2 \llbracket s_2 \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_{2,k} \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle 2,k,j \rangle} \right\rangle_{j=1}^{k_{2,k}} \right\rangle_{k=1}^{n_2}$$

Equalities in  $A$  are exactly generated by the equations of  $\mathfrak{T} \triangleleft \mathcal{A}$ .

Since  $B$  is a model of  $\mathfrak{T} \triangleleft \mathcal{A}$ , we have that  $B$  validates the same equations, and therefore

$$\begin{aligned} & B \llbracket s_1 \rrbracket \left\langle B \llbracket t_{1,k} \rrbracket \left\langle f \mathbf{e}_{\langle 1,k,j \rangle} \right\rangle_{j=1}^{k_{1,k}} \right\rangle_{k=1}^{n_1} \\ &= B \llbracket s_2 \rrbracket \left\langle B \llbracket t_{2,k} \rrbracket \left\langle f \mathbf{e}_{\langle 2,k,j \rangle} \right\rangle_{j=1}^{k_{2,k}} \right\rangle_{k=1}^{n_2} \end{aligned}$$

□

We now prove that this is indeed a  $\mathfrak{T} \triangleleft \mathcal{A}$ -homomorphism and that it is the unique one such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta} & \underline{A} \\ & \searrow f & \downarrow h \\ & & \underline{B} \end{array}$$

▷ **preservation of  $\mathcal{A}$ -structure.**

Take any  $s \in \mathcal{A}_n$  and  $e_1, \dots, e_n \in \underline{A}$ . We can write

$$e_l = A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \left\langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle}^l \right\rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l}$$



Taking  $p := \sum_l k^l$  and  $s' := s[i \mapsto s^i]_{i=1}^n \in \mathcal{A}_p$ , we have:

$$\begin{aligned}
 h(A_2 \llbracket s \rrbracket \langle e_l \rangle_{l=1}^n) &= h \left( A_2 \llbracket s \rrbracket \left\langle A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle}^l \rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \right) \\
 &= h \left( A_2 \llbracket s' \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_m \rrbracket \langle \eta^{A_1} \mathbf{e}_j^m \rangle_{j=1}^{k_m} \right\rangle_{m=1}^p \right) \\
 &= B \llbracket s' \rrbracket \left\langle B \llbracket t_m \rrbracket \langle f \mathbf{e}_j^m \rangle_{j=1}^{k_m} \right\rangle_{m=1}^p \\
 &= B \llbracket s \rrbracket \left\langle B \llbracket s^l \rrbracket \left\langle B \llbracket t_i^l \rrbracket \langle f \mathbf{e}_{\langle i,j \rangle}^l \rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \\
 &= B \llbracket s \rrbracket \langle h e_l \rangle_{l=1}^n
 \end{aligned}$$

▷ **preservation of  $\mathcal{M}$ -structure.**

Take any  $t \in \mathcal{M}_n$  and  $e_1, \dots, e_n \in \underline{A}$ . We can write

$$e_l = A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle}^l \rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l}$$

Since the operations of  $\mathcal{M}$  distribute over the operations of  $\mathcal{A}$ , we have:

$$\begin{aligned}
 h(A_2 \llbracket t \rrbracket \langle e_l \rangle_{l=1}^n) &= h \left( A_2 \llbracket t \rrbracket \left\langle A_2 \llbracket s^l \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_i^l \rrbracket \langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle}^l \rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \right) \\
 &= h \left( A_2 \llbracket s^1 \rrbracket \left\langle \dots A_2 \llbracket s^n \rrbracket \left\langle A_2 \llbracket t \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_{i_l}^l \rrbracket \langle \eta^{A_0} \mathbf{e}_{\langle i_l, m \rangle}^l \rangle_{m=1}^{k_{i_l}^l} \right\rangle_{l=1}^n \right\rangle_{i_n=1}^{k^n} \dots \right\rangle_{i_1=1}^{k^1} \right) \\
 &= h \left( A_2 \llbracket s^1 \rrbracket \left\langle \dots s^n \langle x_{i_1}, \dots, x_{i_n} \rangle_{i_n=1}^{k^n} \dots \right\rangle_{i_1=1}^{k^1} \right) \left\langle A_2 \llbracket t \rrbracket \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_{\langle q \rangle_l^{-1}}^l \rrbracket \langle \eta^{A_0} \mathbf{e}_{\langle \langle q \rangle_l^{-1}, m \rangle}^l \rangle_{m=1}^{k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{l=1}^n \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \\
 &= h \left( A_2 \llbracket s^1 \rrbracket \left\langle \dots s^n \langle x_{i_1}, \dots, x_{i_n} \rangle_{i_n=1}^{k^n} \dots \right\rangle_{i_1=1}^{k^1} \right) \left\langle \eta_{A_1}^{A_2} A_0 \llbracket t_q' \rrbracket \langle \eta^{A_0} \mathbf{e}_m^m \rangle_{m=1}^{\sum_l k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \\
 &= B \llbracket s^1 \rrbracket \left\langle \dots s^n \langle x_{i_1}, \dots, x_{i_n} \rangle_{i_n=1}^{k^n} \dots \right\rangle_{i_1=1}^{k^1} \left\langle B \llbracket t_q' \rrbracket \langle f \mathbf{e}_m^m \rangle_{m=1}^{\sum_l k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \\
 &= B \llbracket s^1 \rrbracket \left\langle \dots s^n \langle x_{i_1}, \dots, x_{i_n} \rangle_{i_n=1}^{k^n} \dots \right\rangle_{i_1=1}^{k^1} \left\langle B \llbracket t \rrbracket \left\langle B \llbracket t_{\langle q \rangle_l^{-1}}^l \rrbracket \langle f \mathbf{e}_{\langle \langle q \rangle_l^{-1}, m \rangle}^l \rangle_{m=1}^{k_{\langle q \rangle_l^{-1}}^l} \right\rangle_{l=1}^n \right\rangle_{q=1}^{\prod_{l=1}^n k^l} \\
 &= B \llbracket s^1 \rrbracket \left\langle \dots B \llbracket s^n \rrbracket \left\langle B \llbracket t \rrbracket \left\langle B \llbracket t_{i_l}^l \rrbracket \langle f \mathbf{e}_{\langle i_l, m \rangle}^l \rangle_{m=1}^{k_{i_l}^l} \right\rangle_{l=1}^n \right\rangle_{i_n=1}^{k^n} \dots \right\rangle_{i_1=1}^{k^1} \\
 &= B \llbracket t \rrbracket \left\langle B \llbracket s^l \rrbracket \left\langle B \llbracket t_i^l \rrbracket \langle f \mathbf{e}_{\langle i,j \rangle}^l \rangle_{j=1}^{k_i^l} \right\rangle_{i=1}^{k^l} \right\rangle_{l=1}^n \\
 &= B \llbracket t \rrbracket \langle h e_l \rangle_{l=1}^n
 \end{aligned}$$

Note to self: never again.

▷ **commutation of the diagram.**

We need to show that for all  $x \in X$ , we have  $h(\eta x) = f x$ . We have  $\eta x = \eta_{A_1}^{A_2} \eta^{A_0} x$ . We notice

that

$$\begin{aligned}\eta x &= \eta_{A_1}^{A_2} \eta^{A_0} x \\ &= \eta_{A_1}^{A_2} A_0 \llbracket \text{id} \rrbracket \langle \eta^{A_1} x \rangle \\ &= A_2 \llbracket \text{id} \rrbracket \langle \eta_{A_1}^{A_2} A_0 \llbracket \text{id} \rrbracket \langle \eta^{A_1} x \rangle \rangle\end{aligned}$$

Therefore

$$\begin{aligned}h(\eta x) &= h(A_2 \llbracket \text{id} \rrbracket \langle \eta_{A_1}^{A_2} A_0 \llbracket \text{id} \rrbracket \langle \eta^{A_1} x \rangle \rangle) \\ &= B \llbracket \text{id} \rrbracket \langle B \llbracket \text{id} \rrbracket \langle f x \rangle \rangle \\ &= f x\end{aligned}$$

▷ **uniqueness.**

Let  $h' : A \rightarrow B$  be another morphism of  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebras such that  $h' \circ \eta = f$ . We need to show that for all  $e \in \underline{A}$ , we have  $h e = h' e$ . We can write

$$e = A_2 \llbracket s \rrbracket \langle \eta_{A_1}^{A_2} A_0 \llbracket t_i \rrbracket \langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle} \rangle_{j=1}^{k_i} \rangle_{i=1}^k$$

Therefore

$$\begin{aligned}h e &= B \llbracket s \rrbracket \langle B \llbracket t_i \rrbracket \langle f \mathbf{e}_{\langle i,j \rangle} \rangle_{j=1}^{k_i} \rangle_{i=1}^k && \text{by definition} \\ &= B \llbracket s \rrbracket \langle B \llbracket t_i \rrbracket \langle h' \eta \mathbf{e}_{\langle i,j \rangle} \rangle_{j=1}^{k_i} \rangle_{i=1}^k && \text{since the diagram commutes} \\ &= h' \left( A_2 \llbracket s \rrbracket \langle \eta_{A_1}^{A_2} A_0 \llbracket t_i \rrbracket \langle \eta^{A_1} \mathbf{e}_{\langle i,j \rangle} \rangle_{j=1}^{k_i} \rangle_{i=1}^k \right) && \text{since } h' \text{ is an homomorphism} \\ &= h' e\end{aligned}$$

and  $h = h'$ .

Therefore  $h$  is the unique morphism of  $\mathfrak{T} \triangleleft \mathcal{A}$ -algebras such that  $h \circ \eta = f$ .

## 2.8 Link with distributive laws

There seem to be a strong link between the existence of a distributive law between the two monads underlying the theories we are trying to combine and the success of the construction we described.

[ZM18] gives a bunch of theorems about the non-existence of distributive laws, and it would be useful to check that our construction indeed does not apply in the cases mentioned.

### 2.8.1 Plotkin-type no-go

One of the condition for the non-existence of a distributive law between two theories given in the Plotkin-like no-go theorems 3.5 and 3.12 of [ZM18] is the existence of an operator that is idempotent in both theories. Our construction forbids such axioms as the presentation would not be  $\mathfrak{T}$ -affine for any translation  $\mathfrak{T}$ .

### 2.8.2 Too many constants

A theory  $\mathcal{A}$  is commutative iff for every  $n$ -ary operation  $f$  and every  $m$ -ary operation  $g$ ,

$$f(g(x_{11}, \dots, x_{1m}), \dots, g(x_{n1}, \dots, x_{nm})) = g(f(x_{11}, \dots, x_{n1}), \dots, f(x_{1m}, \dots, x_{nm})) \quad (1)$$

This equation still holds for constants, therefore if

$$\begin{aligned} f &= c \in \mathcal{A}_0 \\ g &= d \in \mathcal{A}_0 \end{aligned}$$

then the commutativity of  $\mathcal{A}$  states that  $c = d$ .

Therefore the no-go theorem 4.7 of [ZM18] applies then there are two distinct constants and the theory can't be commutative, meaning our construction doesn't apply.

### 2.8.3 Lacking abides

For an operator  $*$  of arity 2, abides hold if

$$(a * b) * (c * d) = (a * c) * (b * d) \quad (2)$$

Theorem 4.17 of [ZM18] states that if some conditions apply on the theories  $\mathcal{A}$  and  $\mathcal{M}$  and that additionally equation (2) does not hold then there are no distributive law of  $\mathcal{M}$  over  $\mathcal{A}$ .

However if  $\mathcal{A}$  is commutative, then (2) holds by taking  $f = g = *$  in (1). There are no contradiction here either.

### 2.8.4 Idempotence and units

Theorem 4.24 of [ZM18] states that if some conditions apply on the theories  $\mathcal{A}$  and  $\mathcal{M}$  and that moreover  $\mathcal{M}$  has a binary term that is idempotent then there are no distributive law of  $\mathcal{M}$  over  $\mathcal{A}$ .

Our construction forbids idempotent terms, therefore there are no contradiction.

## 3 CONSTRUCTING FREE EXTENSIONS

### 3.1 Semigroups

#### 3.1.1 Semigroup

Let's construct the free extension of an **SG**-model  $A = (\underline{A}, +)$  by a set of variable  $X$ .

We define a **SG**-algebra  $\text{Frex}_A(X)$  as follows:

- The carrier set of  $\text{Frex}_A(X)$  is the set of non-empty alternating lists of elements of  $A$  and  $X^+$ .
- The operator  $\cdot$  on  $\text{Frex}_A(X)$  is defined by concatenation of lists, and by using the operator  $+$  of  $A$  to combine adjacent elements of  $A$  in the resulting list and list concatenation to combine adjacent elements of  $X^+$ .

This is indeed an **SG**-algebra, as list concatenation is associative, and the operator  $+$  of  $A$  is associative.

We define  $\text{embed} : A \rightarrow \text{Frex}_A(X)$  by  $\text{embed}(n) = [n]$  for all  $n \in A$ . For all  $m, n \in A$ , we have  $\text{embed}(m + n) = [m + n] = [m] \cdot [n] = \text{embed}(m) \cdot \text{embed}(n)$ , so  $\text{embed}$  is a morphism of **SG**-algebras.

We also define  $\text{var} : X \rightarrow \text{Frex}_A(X)$  by  $\text{var}(x) = [x]$  for all  $x \in X$ .

#### Proposition 102

$\langle \text{Frex}_A(X), \text{embed}, \text{var} \rangle$  is the free extension of  $A$  by  $X$ .

**Proof**

Suppose  $\langle B, f, g \rangle$  is another extension of  $A$  by  $X$ .

We extend  $g$  to  $X^+$  by defining  $g(x_1 \cdots x_n) = g(x_1) \cdots g(x_n)$  for all  $x_1, \dots, x_n \in X$ .

Now for an element  $[b_1, \dots, b_n]$  of  $\text{Frex}_A(X)$ , we define  $h([b_1, \dots, b_n]) = \llbracket b_1 \rrbracket \cdots \llbracket b_n \rrbracket$  where:

- if  $b_i \in A$ , then  $\llbracket b_i \rrbracket = f(b_i)$
- if  $b_i \in X^+$ , then  $\llbracket b_i \rrbracket = g(b_i)$

This is well-defined and homomorphic because the only merges in the extension are:

- concrete/concrete, handled by  $f(a + b) = f(a) \cdot f(b)$
- abstract/abstract, handled by  $g(uv) = g(u) \cdot g(v)$

The diagram below commutes:

$$\begin{array}{ccccc} A & \xrightarrow{\text{embed}} & \text{Frex}_A(X) & \xleftarrow{\text{var}} & X \\ & \searrow f & \downarrow h & \swarrow g & \\ & & B & & \end{array}$$

Uniqueness of  $h$  is straightforward, as  $h$  is defined by the values of  $f$  and  $g$  on the elements of  $A$  and  $X$ , and by the homomorphic property of  $h$ .  $\square$

**3.1.2 Commutative semigroup**

Let's construct the free extension of an **CSG**-model  $A = (\underline{A}, +)$  by a set of variable  $X$ . It is going to be linear polynomials in  $A[X]$ .

We pose  $A_\perp = \underline{A} \sqcup \{\perp\}$  where  $\perp$  is a new element, designating the absence of a concrete part in the polynomial. We extend the operator  $+$  and  $\cdot$  of  $A$  to  $A_\perp$  by defining  $\perp + a = a + \perp = a$  for all  $a \in A$  and  $\perp + \perp = \perp$ .

We define a **CSG**-algebra  $\text{Frex}_A(X)$  as follows:

- The carrier set of  $\text{Frex}_A(X)$  is the set  $\{(c, a_1x_1 + \cdots + a_nx_n) \mid n \geq 0, c \in A_\perp, x_1, \dots, x_n \in X\}$ . We note  $(c, \emptyset) \in \text{Frex}_A(X)$  when  $n = 0$ .
- The operator  $+$  on  $\text{Frex}_A(X)$  is defined by

$$(c, a_1x_1 + \cdots + a_nx_n) + (d, b_1y_1 + \cdots + b_my_m) = \left( (c + d), \sum_{z_i \in \{x_1, \dots, x_n\} \cup \{y_1, \dots, y_m\}} c_i z_i \right)$$

where  $c_i$  is the sum of the coefficients of  $z_i$  in the two polynomials.

This is indeed an **CSG**-algebra, as addition of polynomials is associative and commutative, and so is addition in  $A_\perp$ .

We define  $\text{embed} : A \rightarrow \text{Frex}_A(X)$  by  $\text{embed}(n) = (n, \emptyset)$  for all  $n \in A$ . For all  $m, n \in A$ , we have  $\text{embed}(m + n) = (m + n, \emptyset) = (m, \emptyset) + (n, \emptyset) = \text{embed}(m) + \text{embed}(n)$ , so  $\text{embed}$  is a morphism of **CSG**-algebras.

We also define  $\text{var} : X \rightarrow \text{Frex}_A(X)$  by  $\text{var}(x) = (\perp, x)$  for all  $x \in X$ .

**Proposition 103**

$\langle \text{Frex}_A(X), \text{embed}, \text{var} \rangle$  is the free extension of  $A$  by  $X$ .

**Proof**

Suppose  $\langle B, f, g \rangle$  is another extension of  $A$  by  $X$ . We define  $h : \text{Frex}_A(X) \rightarrow B$  by

$$h(c, a_1x_1 + \dots + a_nx_n) = \begin{cases} f(c) + \sum_{i=1}^n a_i g(x_i) & \text{if } c \neq \perp \\ \sum_{i=1}^n a_i g(x_i) & \text{if } c = \perp \end{cases}$$

This is well-defined and homomorphic because the only merges in the extension are:

- concrete/concrete, handled by  $f(a + b) = f(a) + f(b)$

The diagram below commutes:

$$\begin{array}{ccccc} A & \xrightarrow{\text{embed}} & \text{Frex}_A(X) & \xleftarrow{\text{var}} & X \\ & \searrow f & \downarrow h & \swarrow g & \\ & & B & & \end{array}$$

Uniqueness of  $h$  is straightforward, as  $h$  is defined by the values of  $f$  and  $g$  on the elements of  $A$  and  $X$ , and by the homomorphic property of  $h$ .  $\square$

**3.2 Sem-rgs**

Let  $A = (\underline{A}, (+), (\cdot))$  be a **SG**  $\triangleleft$  **CSG**-model and  $X$  be a set.

We define

$$\mathbf{Mon}_A(X) = \{[b_1, \dots, b_n] \mid n \geq 1, b_i \in X^+ \sqcup \underline{A}, b_i b_{i+1} \text{ alternating kinds}, \exists i, b_i \in X^+\}$$

We define  $A_\perp$  as in the previous section, and extend the operations with  $\perp \cdot a = a \cdot \perp = \perp$  for all  $a \in A$ , and  $\perp \cdot \perp = \perp$ .

We define

$$\text{Frex}_A(X) = \{(c, M) \mid c \in A_\perp, M \in \text{Bag}(\mathbf{Mon}_A(X)), c \neq \perp \text{ or } M \neq \emptyset\}$$

Defining the addition is straightforward:

$$(c, M) + (d, N) = (c + d, M \uplus N)$$

We define the monomial multiplication  $m \otimes n$  for  $m, n \in \mathbf{Mon}_A(X)$  by concatenation of lists, and by using the operator  $\cdot$  of  $A$  to combine adjacent elements of  $A$  in the resulting list and list concatenation to combine adjacent elements of  $X^+$  (as in Section 3.1).

We also define multiplication of a concrete element with a monomial by treating the concrete element as a monomial of length 1:

- $r \triangleright m = [r] \otimes m$
- $m \triangleleft r = m \otimes [r]$

This is well-defined because  $m$  already contains an element of  $X^+$ .

We naturally extend the multiplication to bags, which then gives us a definition for the multiplication of two elements of  $\text{Frex}_A(X)$ :

$$(c, M) \cdot (d, N) = (c \cdot d, (c \triangleright N) \uplus (M \otimes N) \uplus (M \triangleleft d))$$

We define  $\text{embed} : A \rightarrow \text{Frex}_A(X)$  by  $\text{embed}(a) = (a, \emptyset)$  for all  $a \in A$ . For all  $a, b \in A$ , we have  $\text{embed}(a + b) = (a + b, \emptyset) = (a, \emptyset) + (b, \emptyset) = \text{embed}(a) + \text{embed}(b)$  and  $\text{embed}(a \cdot b) = (a \cdot b, \emptyset) = (a, \emptyset) \cdot (b, \emptyset) = \text{embed}(a) \cdot \text{embed}(b)$ , so  $\text{embed}$  is a morphism of **SG**  $\triangleleft$  **CSG**-algebras.

We also define  $\text{var} : X \rightarrow \text{Frex}_A(X)$  by  $\text{var}(x) = (\perp, \{[x]\})$  for all  $x \in X$ .

**Proposition 104**

$\langle \text{Frex}_A(X), \text{embed}, \text{var} \rangle$  is the free extension of  $A$  by  $X$ .

**Proof**

- **$\text{Frex}_A(X)$  is a  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model.** All axioms of  $\mathbf{SG} \triangleleft \mathbf{CSG}$  but the distributivity axioms are straightforward to check. For the distributivity axioms, we have:

$$\begin{aligned}
 (c, M) \cdot ((d, N) + (e, P)) &= (c, M) \cdot (d + e, N \uplus P) \\
 &= (c \cdot (d + e), (c \triangleright (N \uplus P)) \uplus (M \otimes (N \uplus P)) \uplus (M \triangleleft (d + e))) \\
 &= (c \cdot d + c \cdot e, (c \triangleright N) \uplus (c \triangleright P) \uplus (M \otimes N) \uplus (M \otimes P) \uplus (M \triangleleft d) \uplus (M \triangleleft e)) \\
 &= (c, M) \cdot (d, N) + (c, M) \cdot (e, P)
 \end{aligned}$$

and

$$\begin{aligned}
 ((c, M) + (d, N)) \cdot (e, P) &= (c + d, M \uplus N) \cdot (e, P) \\
 &= ((c + d) \cdot e, ((c + d) \triangleright P) \uplus ((M \uplus N) \otimes P) \uplus ((M \uplus N) \triangleleft e)) \\
 &= (c \cdot e + d \cdot e, (c \triangleright P) \uplus (d \triangleright P) \uplus (M \otimes P) \uplus (N \otimes P) \uplus (M \triangleleft e) \uplus (N \triangleleft e)) \\
 &= (c, M) \cdot (e, P) + (d, N) \cdot (e, P)
 \end{aligned}$$

- **Universality.** Let  $B$  be any  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -model and let  $f : A \rightarrow B$  and  $g : X \rightarrow B$  be morphisms of  $\mathbf{SG} \triangleleft \mathbf{CSG}$ -algebras.

We extend  $g$  to  $X^+$  by defining  $g(x_1 \cdots x_n) = g(x_1) \cdots g(x_n)$  for all  $x_1, \dots, x_n \in X$ .

We define  $h([b_1, \dots, b_n]) = \llbracket b_1 \rrbracket \cdots \llbracket b_n \rrbracket$  where:

- if  $b_i \in A$ , then  $\llbracket b_i \rrbracket = f(b_i)$
- if  $b_i \in X^+$ , then  $\llbracket b_i \rrbracket = g(b_i)$

We extend  $h$  to  $\text{Frex}_A(X)$  by defining

$$h(c, M) = \begin{cases} f(c) + \sum_{m \in M} h(m) & \text{if } c \neq \perp \\ \sum_{m \in M} h(m) & \text{if } c = \perp \end{cases}$$

This is well-defined and homomorphic<sup>a</sup> because the only merges in the extension are:

- concrete/concrete, handled by  $f(a + b) = f(a) + f(b)$  and  $f(a \cdot b) = f(a) \cdot f(b)$
- abstract/abstract, handled by  $g(uv) = g(u) \cdot g(v)$

The diagram below commutes:

$$\begin{array}{ccccc}
 A & \xrightarrow{\text{embed}} & \text{Frex}_A(X) & \xleftarrow{\text{var}} & X \\
 & \searrow f & \downarrow h & \swarrow g & \\
 & & B & & 
 \end{array}$$

Uniqueness of  $h$  is straightforward, as  $h$  is defined by the values of  $f$  and  $g$  on the elements of  $A$  and  $X$ , and by the homomorphic property of  $h$ .

□

<sup>a</sup>it is less straightforward than in the examples for the semigroups, so doing it may be a good idea

### 3.3 Distributive combination of free extensions

We'll try to adapt the construction of Section 2.5 to the case of free extensions.

Let  $B$  be a  $\mathfrak{T} \triangleleft \mathcal{A}$  model, and  $X$  be a set.

#### Step 1.

Let  $B_{\mathcal{M}}$  be the underlying  $\mathcal{M}$ -model of  $B$ . Let  $E_0 := \text{Frex}_{B_{\mathcal{M}}}(X) \models \mathcal{M}$  be the free extension of  $B_{\mathcal{M}}$  by  $X$ . From the free extension construction, we also have the canonical inclusion  $i_{\mathcal{M}} : B_{\mathcal{M}} \rightarrow E_0$ .

#### Step 2.

We apply the functor

$$\neg_{\mathfrak{T}} := \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \rightarrow \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$$

to  $E_0$ . This gives  $E_1 := \underline{E_0}_{\neg_{\mathfrak{T}}} \models \mathcal{O}$ , and the canonical inclusion  $i_{\mathcal{O}} : B_{\mathcal{O}} \rightarrow E_1$ .

#### Step 3.

Now the next part is tricky. Let's first look at this abstractly, and then find a way to actually construct the required elements.

From [All+25], we know that we can construct free extensions based on free algebras. This means that there exists  $E_1^{\text{new}}$  such that  $E_1 \cong B_{\mathcal{O}} \sqcup E_1^{\text{new}}$ . The model of  $\mathcal{O} \triangleleft \mathcal{A}$  that we want is then

$$E_2 = B_{\mathcal{O} \triangleleft \mathcal{A}} \sqcup F_{\mathcal{O} \rightarrow \mathcal{A}} E_1^{\text{new}}$$

with

$$F_{\mathcal{O} \rightarrow \mathcal{A}} : \mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) : \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle) \rightarrow \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$$

obtained as in step 3 of Section 2.5. The problem with this is that we can't get  $E_1^{\text{new}}$  from  $E_1$ , so we'll have to find another approach.

Let's note  $F_{\mathcal{O} \rightarrow \mathcal{A}} \dashv U_{\mathcal{O} \rightarrow \mathcal{A}}$  the adjunction between  $\mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$  and  $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ . We have  $B_{\mathcal{O}} = U_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O} \triangleleft \mathcal{A}}$ . Therefore the identity  $\text{id}_{B_{\mathcal{O}}} : B_{\mathcal{O}} \rightarrow U_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O} \triangleleft \mathcal{A}}$  has an adjunct  $\varepsilon_B : F_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O}} \rightarrow B_{\mathcal{O} \triangleleft \mathcal{A}}$ . We can then define  $E_2$  as the pushout of  $\varepsilon_B$  and  $F_{\mathcal{O} \rightarrow \mathcal{A}} i_{\mathcal{O}}$ :

$$\begin{array}{ccc} F_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O}} & \xrightarrow{F_{\mathcal{O} \rightarrow \mathcal{A}} i_{\mathcal{O}}} & F_{\mathcal{O} \rightarrow \mathcal{A}} E_1 \\ \varepsilon_B \downarrow & & \downarrow \\ B_{\mathcal{O} \triangleleft \mathcal{A}} & \longrightarrow & E_2 \end{array}$$

#### Lemma 105

Let  $\mathcal{C}$  be a category,  $X, Y, Z$  be objects of  $\mathcal{C}$ , and  $f : X \rightarrow Z$ .  
The pushout of  $\iota_X : X \rightarrow X \sqcup Y$  and  $f$  is isomorphic to  $Y \sqcup Z$ :

$$(X \sqcup Y) \sqcup_X Z \cong Y \sqcup Z$$

#### Proof

□

#### Lemma 106

$$E_2 := B_{\mathcal{O} \triangleleft \mathcal{A}} \sqcup_{F_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O}}} F_{\mathcal{O} \rightarrow \mathcal{A}} E_1 \cong B_{\mathcal{O} \triangleleft \mathcal{A}} \sqcup F_{\mathcal{O} \rightarrow \mathcal{A}} E_1^{\text{new}}$$

**Proof**

Because  $F_{\mathcal{O} \rightarrow \mathcal{A}}$  is a left adjoint, it preserves coproducts, so

$$F_{\mathcal{O} \rightarrow \mathcal{A}} E_1 \cong F_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O}} \sqcup F_{\mathcal{O} \rightarrow \mathcal{A}} E_1^{\text{new}}$$

$F_{\mathcal{O} \rightarrow \mathcal{A}} i_{\mathcal{O}} : F_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O}} \rightarrow F_{\mathcal{O} \rightarrow \mathcal{A}} E_1$  is then the first coproduct injection. Therefore the pushout span for  $E_2$  is

$$B_{\mathcal{O} \rightarrow \mathcal{A}} \xleftarrow{\varepsilon_B} F_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O}} \xrightarrow{i_1} F_{\mathcal{O} \rightarrow \mathcal{A}} B_{\mathcal{O}} \sqcup F_{\mathcal{O} \rightarrow \mathcal{A}} E_1^{\text{new}}$$

By Lemma 105, the pushout of this span is isomorphic to the coproduct  $B_{\mathcal{O} \triangleleft \mathcal{A}} \sqcup F_{\mathcal{O} \rightarrow \mathcal{A}} E_1^{\text{new}}$ .  $\square$

Since the pushout is taken in  $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ ,  $E_2$  is a model of  $\mathcal{O}$  in  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$  and we have the canonical inclusion  $B_{\mathcal{O} \triangleleft \mathcal{A}} \rightarrow E_2$ .

**Intuition:** The intuition behind this construction is that we can't directly apply  $F_{\mathcal{O} \rightarrow \mathcal{A}}$  to  $E_1$  because the concrete part is then duplicated. By doing the pushout, we separate the concrete part (with  $\varepsilon_B$ ) from the new part (with  $F_{\mathcal{O} \rightarrow \mathcal{A}} i_{\mathcal{O}}$ ), and we get the desired result.

**Step 4.**

Let  $E_3 := \mathbf{Mod}(\mathcal{O}, \_ ) E_2 \models \mathcal{O}$  be the underlying operadic algebra. We use surjectivity as in step 4 of Section 2.5 to get an  $\mathcal{M}$ -model  $E_{\mathcal{M}} \models \mathcal{M}$ .

**Step 5.**

We can then define  $E$  as the pullback of  $E_{\mathcal{M}}$  and  $E_2$ . This should give us the free extension of  $B$  by  $X$ .

**3.4 Practical combination**

While the construction of Section 3.3 works in theory and has showed that we can indeed construct the free extension for the distributive combination using both of the free extensions, it is not practical as it uses a pushout, which is hard to compute.

The goal of this section is to give a practical way of constructing the free extension for the combination that could be implemented in Idris2.

We call  $B \models \mathfrak{T} \triangleleft \mathcal{A}$  the model we are trying to extend. We denote by  $B_{\mathcal{M}}[-]$  (resp.  $B_{\mathcal{A}}[-]$ ) the functor that takes a set  $X$  and returns the free extension of  $B$  by  $X$  as an  $\mathcal{M}$  (resp.  $\mathcal{A}$ ) model. We write  $B[X]$  for the free extension of  $B$  by  $X$  as a  $\mathfrak{T} \triangleleft \mathcal{A}$ -model.

**Hypothesis:** We would need to have the following structure with the following property:

- a model  $C \models \mathcal{O}$
- a function  $X \xrightarrow{\nu} C$
- a function  $C \xrightarrow{u} B[X]$

such that

- for all  $D \models \mathcal{O}$
- for all function  $X \xrightarrow{f} D$
- for all homomorphism  $B_{\mathcal{O}} \xrightarrow{p} D$

there exists a unique homomorphism  $q : C \rightarrow D$  such that the following diagram commutes:

$$\begin{array}{ccc}
 & X & \\
 \nu \swarrow & & \searrow f \\
 C & \xrightarrow{\quad q \quad} & D \\
 u \searrow & & \nearrow [f, p] \\
 & B_{\mathcal{O}}[X] &
 \end{array}$$



Then we would define

$$A_2 := \mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-]) \quad C \models \mathcal{O} * \mathcal{A}$$

and continue the construction as for the case of the free algebra.

### 3.4.1 Example of the semigroups

**Remark :** In the semigroup example, let

$$\mathcal{M} = \mathbf{SG}, \quad \mathcal{A} = \mathbf{CSG},$$

let  $\mathcal{O}$  be the semigroup operad, with one operation  $\mu_n \in \mathcal{O}_n$  for each  $n \geq 1$ , and let

$$\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$$

be the surjective translation defined by

$$\mathfrak{T}(\mu_n) = x_1 \cdots x_n.$$

If  $D$  is an  $\mathcal{O}$ -model, then its underlying set carries a semigroup structure

$$x \cdot_D y := D[\llbracket \mu_2 \rrbracket](x, y),$$

and the operad laws imply that this multiplication is associative. We denote the resulting semigroup by  $D_{\mathcal{M}}$ . By construction,

$$\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(D_{\mathcal{M}}) = D.$$

#### Proposition 107 (The translated monomial object)

Let  $B \models \mathcal{M} \triangleleft \mathcal{A}$ , and write

$$B_{\mathcal{O}} := \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(B_{\mathcal{M}}).$$

Let

$$\mathbf{Mon}_B(X) = \{[b_1, \dots, b_n] \mid n \geq 1, b_i \in X^+ \sqcup \underline{B}, b_i, b_{i+1} \text{ alternate kinds}, \exists i, b_i \in X^+\}.$$

Equip  $\mathbf{Mon}_B(X)$  with the semigroup structure of Section 3.2, and write

$$C_{\mathcal{M}} := \mathbf{Mon}_B(X).$$

Set

$$C := \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(C_{\mathcal{M}}).$$

Let

$$\nu : X \rightarrow \underline{C}, \quad \nu(x) := [x],$$

and let

$$u_{\mathcal{M}} : C_{\mathcal{M}} \rightarrow B_{\mathcal{M}}[X]$$

be the inclusion of variable-containing alternating lists into the alternating-list representation of the multiplicative free extension  $B_{\mathcal{M}}[X]$ . Define

$$u := \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(u_{\mathcal{M}}) : C \rightarrow B_{\mathcal{O}}[X],$$

where

$$B_{\mathcal{O}}[X] := \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(B_{\mathcal{M}}[X]).$$

Then the triple  $(C, \nu, u)$  satisfies the hypothesis of Section 3.4.

**Proof**

By definition,  $C_{\mathcal{M}} = \mathbf{Mon}_B(X)$  is a semigroup: multiplication is given by concatenation followed by seam normalisation, and associativity follows from associativity of concatenation, associativity of multiplication in  $B_{\mathcal{M}}$ , and associativity of concatenation in  $X^+$ . Applying the translation functor

$$\mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \rightarrow \mathbf{Mod}(\mathcal{O}, \mathbf{Set})$$

therefore yields an  $\mathcal{O}$ -model

$$C = \mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(C_{\mathcal{M}}).$$

Now let  $D \models \mathcal{O}$ , let

$$f : X \rightarrow \underline{D},$$

and let

$$p : B_{\mathcal{O}} \rightarrow D$$

be an  $\mathcal{O}$ -homomorphism. By the preceding remark, the carrier of  $D$  carries a semigroup structure

$$x \cdot_D y := D \llbracket \mu_2 \rrbracket (x, y),$$

which we denote by  $D_{\mathcal{M}}$ , and we have

$$\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(D_{\mathcal{M}}) = D.$$

Since  $p$  is an  $\mathcal{O}$ -homomorphism, it is also a semigroup homomorphism

$$p_{\mathcal{M}} : B_{\mathcal{M}} \rightarrow D_{\mathcal{M}}.$$

Indeed, for all  $b, b' \in \underline{B}$ ,

$$\begin{aligned} p_{\mathcal{M}}(b \cdot b') &= p(B_{\mathcal{O}} \llbracket \mu_2 \rrbracket (b, b')) \\ &= D \llbracket \mu_2 \rrbracket (p(b), p(b')) \\ &= p_{\mathcal{M}}(b) \cdot_D p_{\mathcal{M}}(b'). \end{aligned}$$

By freeness of the multiplicative extension  $B_{\mathcal{M}}[X]$ , there exists a unique semigroup homomorphism

$$[f, p]_{\mathcal{M}} : B_{\mathcal{M}}[X] \rightarrow D_{\mathcal{M}}$$

such that

$$[f, p]_{\mathcal{M}} \circ \iota_{B_{\mathcal{M}}} = p_{\mathcal{M}} \quad \text{and} \quad [f, p]_{\mathcal{M}} \circ \eta_X = f.$$

Applying  $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$  gives an  $\mathcal{O}$ -homomorphism

$$[f, p]_{\mathcal{O}} : B_{\mathcal{O}}[X] \rightarrow D.$$

Now define

$$q := [f, p]_{\mathcal{O}} \circ u : C \rightarrow D.$$

Since  $u$  and  $[f, p]_{\mathcal{O}}$  are  $\mathcal{O}$ -homomorphisms,  $q$  is an  $\mathcal{O}$ -homomorphism. Moreover,

$$q \circ \nu = [f, p]_{\mathcal{O}} \circ u \circ \nu = f,$$

because  $u \circ \nu$  is the variable map into  $B_{\mathcal{O}}[X]$ .

Finally, if

$$q' : C \rightarrow D$$

is another  $\mathcal{O}$ -homomorphism satisfying the same condition from Section 3.4, namely

$$q' \circ \nu = f \quad \text{and} \quad q' = [f, p]_{\mathcal{O}} \circ u,$$

then clearly  $q' = q$ . Thus  $(C, \nu, u)$  satisfies the required hypothesis.  $\square$

**Theorem 108 (Applying the Section 3.4 construction in the semigroup case)**

Let  $C$  be the  $\mathcal{O}$ -model defined in the previous proposition, and let

$$E_2 := \mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-])(C).$$

Let

$$E_3 := \mathbf{Mod}(\mathcal{O}, -)(E_2).$$

Using the surjectivity of

$$\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M},$$

reconstruct from  $E_3$  a semigroup model  $E_{\mathcal{M}}$  on the same carrier, exactly as in Step 4 of the construction, and let  $E$  be the resulting model of  $\mathcal{M} \triangleleft \mathcal{A}$  obtained by Step 5.

Then  $E$  is the free extension of  $B$  by  $X$  for the distributive combination  $\mathbf{SG} \triangleleft \mathbf{CSG}$ .

**Proof**

We compare the object obtained from Section 3.4 with the explicit free extension constructed in Section 3.2.

**Underlying carrier of  $E_2$ .**

By definition,

$$E_2 = \mathbf{Mod}(\mathcal{O}, B_{\mathcal{A}}[-])(C),$$

so the underlying  $\mathcal{A}$ -model of  $E_2$  is the free extension of  $B_{\mathcal{A}}$  by the carrier  $\underline{C}$ . Since

$$\underline{C} = \mathbf{Mon}_B(X),$$

the underlying carrier of  $E_2$  is

$$\{(c, M) \mid c \in \underline{B}^{\perp}, M \in \mathbf{Bag}(\mathbf{Mon}_B(X)), c \neq \perp \text{ or } M \neq \emptyset\}.$$

This is exactly the carrier of the explicit model of Section 3.2.

**The  $\mathcal{O}$ -structure on  $E_2$ .**

$\square$

## REFERENCES

- [All+25] Guillaume Allais et al. “Frex: Dependently Typed Algebraic Simplification”. In: *Proceedings of the ACM on Programming Languages* 9.ICFP (Aug. 2025), pp. 30–65. ISSN: 2475-1421. DOI: 10.1145/3747506. URL: <http://dx.doi.org/10.1145/3747506>.
- [Bec69] Jon Beck. “Distributive laws”. In: *Seminar on Triples and Categorical Homology Theory*. Ed. by B. Eckmann. Berlin, Heidelberg: Springer Berlin Heidelberg, 1969, pp. 119–140. ISBN: 978-3-540-36091-9.
- [HP06] Martin Hyland and John Power. “Discrete Lawvere theories and computational effects”. In: *Theoretical Computer Science* 366.1 (2006). Algebra and Coalgebra in Computer Science, pp. 144–162. ISSN: 0304-3975. DOI: <https://doi.org/10.1016/j.tcs.2006.07.007>. URL: <https://www.sciencedirect.com/science/article/pii/S0304397506004026>.
- [Mac71] Saunders MacLane. *Categories for the Working Mathematician*. Graduate Texts in Mathematics, Vol. 5. New York: Springer-Verlag, 1971, pp. ix+262.
- [MW10] Francisco Marmolejo and R. J. Wood. “Monads as Extension Systems—No Iteration is Necessary”. In: *Theory and Applications of Categories* (2010). URL: <https://api.semanticscholar.org/CorpusID:18849266>.
- [YGK18] Jeremy Yallop, Tamara von Glehn, and Ohad Kammar. “Partially-static data as free extension of algebras”. In: *Proc. ACM Program. Lang.* 2.ICFP (July 2018). DOI: 10.1145/3236795. URL: <https://doi.org/10.1145/3236795>.
- [ZM18] Maaike Zwart and Dan Marsden. *Don’t Try This at Home: No-Go Theorems for Distributive Laws*. 2018. arXiv: 1811.06460 [math.CT]. URL: <https://arxiv.org/abs/1811.06460>.

## A DAILY RECAP

- 05/05/2026: first day, talk with Ohad, building tour, admin stuff, setting up the git repo and starting to read the notes of Ohad
- 06/05/2026: reading Ohad’s notes, starting to write my own notes. lots of definitions to try and understand how things work.
- 07/05/2026: read the paper on frexes and frals, wrote down the relevant definitions. Meeting with Ohad: 3 equivalent definitions of adjunctions  $\Rightarrow$  universal arrows, examples where/when they are used + equivalence of definitions of free algebras and how to construct a free algebra for (commutative) semigroups.  
 Gave examples of uses of universal arrows: defining adjunctions, defining products and coproducts, and defining free algebras.  
 Defined and proved the free algebra for commutative semigroups, started to work on the free algebra for semigroups.
- 08/05/2026: Defined and proved the free algebra for semigroups and commutative monoids. Proved the details that were missing for the free algebra for commutative semirings (in the commutative monoid part).  
 Started to work on the free algebra for the combination of semigroups and commutative semigroups, but still need to define the product operator and check that the distributivity axioms hold, and that the resulting algebra is indeed a free algebra for the combination.
- 11/05/2026: finished the definition of the product operator for the free algebra for the combination of semigroups and commutative semigroups, and proved that it is indeed a free algebra for the combination.

Defined multiplication for the free semigroup and commutative semigroup, and proved that we get a monad structure on **Set** in both cases.

Defined the distributive law of the monad for the free semigroup over the monad for the free commutative semigroup, and proved that it is indeed a distributive law  $\rightarrow$  we get a monad structure on **Set** for the free algebra for the combination of semigroups and commutative semigroups.

- 12/05/2026: definition of Kleisli category and related objects, research and definition of algebra and distributive laws of monads in extension form, and link with the previous definition of distributive laws. Example 8.5 of [MW10] is almost the same thing as Section 2.2.
- 13/05/2026: started looking into the distributive combination of semigroups and commutative semigroups. Defined the operad  $\mathcal{O}$  spanned by the multiplication operator of semigroups, and defined a translation from  $\mathcal{O}$  to the theory of semigroups. Proved that  $\mathcal{O}$  is minimal and that the translation is surjective.
- 14/05/2026: defined the monad on  $\mathbf{SG} \triangleleft \mathbf{CSG}$  with the adjunction, and proved that it is the same as the one given by the distributive law.

Copied over relevant information to start describing the pullback square for the combination of models in more details. A lot of reading Ohad's notes.

- 15/05/2026: more copying definitions to understand the pullback square. Figured out what all of the objects (both vertices and arrows) are. 2-hour meeting with Ohad to check that I got everything right, and to talk about what I have to do next.
- 18/05/2026: finished giving the full description of all objects in the pullback square.

Jesse helped me understand how to create the pullback in the category of categories, and started working on the proof that the square is a pullback square. Proved the commutativity of the square, and started working on the pullback condition. It seems like the distributivity axioms pose a problem.

- 19/05/2026: sick day, stayed home. Solved the distributivity problem thanks to Ohad. Finished proving the pullback and started working on constructing the free algebra for the example of the semigroups.
- 20/05/2026: finished constructing the free algebra for the semigroups, giving explicit description at each step, and proved that the resulting algebra is the free algebra for the combination. Starting working on completing Ohad's notes for the general case.
- 21/05/2026: finished Ohad's proof of lemma 34 in the general case. Started proving step 5 for the general case  $\rightarrow$  2 approaches: build morphisms along the steps or constructive approach (which I think might work).
- 22/05/2026: the constructive approach cannot be used to prove that we indeed have a free algebra: we need the fact that the algebra is free so that the morphism is well-defined. Almost finished the proof using the construction with the adjunctions and the pullback, still need to tie some loose ends.
- 26/05/2026: finished the proof. Proved that the homomorphism defined constructively is indeed the right one.

Started reading about free extensions.

- 27/05/2026: read about free extensions. I'm pretty sure I know how to adapt the construction of the free algebra to free extensions.
- 28/05/2026: wrote the definition of free extensions. Defined and proved the free extensions for the semigroups and commutative semigroups. Started looking into extending the result to free extensions.

- 29/05/2026: (almost) proved that you can indeed extend the result to free extensions with some modifications. Meeting with Ohad → what I came up with works but it uses a pushout which is not easy to calculate, so we started searching for another way to do this.
- 01/06/2026: Started learning some Idris2.  
Ohad gave a counter-example to the construction of the free algebra → I had overlooked something in the proof of Lemma 34 of Ohad's notes. Partially fixed things, but stronger hypothesis now.
- 02/06/2026: read [ZM18] and checked every no-go theorem they provide and proved that our construction doesn't apply in those cases.  
Worked on the construction of the free extension.
- 03/06/2026: finished learning the basics of Idris2, started looking into the current implementation of frex in Idris2 (<https://github.com/frex-project/idris-frex>).
- 04/06/2026: looked at the whole structure of the project and read every file, understanding of where and how things are defined and how they are used and work together.  
Started working on defining the distributive combination.  
Defined the theory of semigroups and commutative semigroups.
- 05/06/2026: defined some models for the semigroups.  
Came back to the drawing board of last friday with Ohad, and tried to make sense of things. Applied it to the example of the semigroups, I can't quite picture how to get the definition the the operations once you apply the lifted functor.

## B TODO

- follow Ohad's abstract picture
- write the definition of free extensions with universal arrows

## C QUESTIONS FOR OHAD

-